AUTOMAX

Multitest computerized control console

50-C20M82, 50-C20M84

MANUALE DI ISTRUZIONI
INSTRUCTION MANUAL

Index

1. INTRODUCTION

NOTE:
The present manual is updated for the product it is sold to grant an adequate reference in operating and maintaining the equipment.
The manual may not reflect changes to the product not impacting service operations.

AUTOMAX Multitest represents a step-change in versatility that allows modular upgrades from basic failure tests up to advanced displacement controlled tests.
The system offers step-by-step flexibility to build your testing system to cover possible future testing needs.
The console is supplied with DATAMANAGER software for standard failure tests to EN and ASTM, but with additional frames, accessories and dedicated software packages, your system can easily perform many tests including:

- Tension on steel rebar (with optional software package 82-SW/UTS);
- Modulus of Elasticity and Poisson ratio determinations (with optional software package 82-SW/EM);
- Displacement controlled tests for FRC/Shotcrete characterization (with optional software package 82-SW/DC).

The **AUTOMAX Multitest** control console is also the perfect solution to update, with limited investment, an existing tension/compression tester series 70-C0019, by replacing the old semi-automatic system composed by hydraulic pump and DIGIMAX PLUS readout unit. Ask our technical department for more information.

Main features:

- Suitable for any kind of test from basic failure tests (compression, flexure, splitting, tension) through cyclic tests for elastic modulus and Poisson ratio determination up to advanced displacement controlled tests;
- Suitable for any type of sample, can be connected up to 4 frames ranging from 15 kN (5,500 lbf) up to 5000 kN (1,100,000 lbf) in compression and 500 kN (112,000 lbf) in tension;
- Suitable for any budget: the machine can be upgraded in step-by-step investments by adding suitable testing frames, accessories and dedicated software packages;
- Suitable for any user: 4 easy-to-use software packages available, each one tailored for a specific test method, guiding the operator through all the test phases;
- Test cycle with closed-loop PID control automatically performed by pressing the start button via PC;
- DC-driven variable speed pump for silent operation, energy-saving, and highly accurate drop-by-drop oil flow for precise control during complex tests;
- 500 Hz high control frequency for optimum oil pressure adjustment during critical tests;
- Double frame control, expandable to four, with active frame selection via software;
- Soft platen-to-specimen contact and smooth load rate control from every beginning of the ramp.

The following models are available:

50-C20M82
Automax Multitest stand-alone power and control console for the control of 2 (expandable to 4) testing frames. PC included.
230 V, 50-60 Hz, 1 ph.

50-C20M84
Automax Multitest stand-alone power and control console for the control of 2 (expandable to 4) testing frames. PC included.
110 V, 60 Hz, 1 ph.

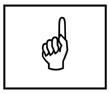
The machine must be completed with frames and relevant accessories.

The device must be used in compliance with the procedures described in this manual. Never use the device for purposes different from those herewith indicated.

NOTE:
This manual includes the hardware description of the **AUTOMAX Multitest** computerized control console.
The console must be operated via a computer connected to it and equipped with **MULTITEST SOFTWARE** that includes the **DATAMANAGER** module. Refer to the relevant documentation to operate the system and perform the tests.

Please read this manual thoroughly before you start using the equipment.

1.1 Icons appearing in the manual



This icon indicates a NOTE; please read thoroughly the items marked by this picture.

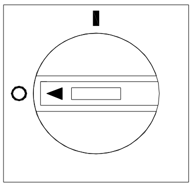
This icon indicates a WARNING message; the items marked by this icon refer to the safety aspects of the operator and/or of the service engineer.

1.2 Manual revision history

Revision/Date	Change description
Rev. 1 September 05th 2018	Manual release
Rev. 2 June 08th 2020	General updates Chapter 6.7 and 6.8 added
Rev. 3 October 29th 2020	General updates Chapter 6.7 updated

1.3 Symbols used

In this manual and on the equipment itself, apart from the symbols indicated on the control panel, the following icons are also used:

Symbol	Description
	Mains switch: O = device not connected to the mains line I = device connected to the mains line
	Emergency stop switch
	Dangerous voltage
CE	Conformity to the CE Directive

1.4 Intended use and improper use

The **AUTOMAX Multitest** control console can only be used to supply and control frames to perform building material tests according to the relevant international standards.

The unit is designed to be operated by one operator that has to install the necessary distance pieces, load-unload the sample from the frame, input the test parameters, and start the test via software.
During normal operation, the working area of the equipment is the front side where the computer is located.

The operator is responsible for starting up the equipment, performing the test sequence, shutting down the equipment, and eventually stopping it in case any emergency conditions are met.

The operator must be trained on the correct use of the machine and the relative safety aspects of its use.

The machine should be used following the procedures described in this manual.

Never use the machine for reasons other than those for which it was designed and manufactured. Any other use of the machine is to be considered improper, not foreseen, and hence dangerous.
CONTROLS will not be responsible for the improper use of the machine.

1.5 Safety information

WARNING:
Please read this chapter thoroughly.

CONTROLS designs and builds its devices complying with the related safety requirements; furthermore, it supplies all information necessary for correct use and the warnings related to the use of the equipment.

CONTROLS will not be held responsible for:

- use of the equipment different than the intended use,
- damages to the unit, to the operator, caused both by installation and maintenance procedures different than those described in this manual supplied with the unit, and by wrong operations,
- mechanical and/or electrical modifications performed during and after the installation, different than those described in this manual

The unit is not designed to be used in an explosive atmosphere.

Installation and any technical intervention must only be performed by qualified technicians authorized by CONTROLS.

Only authorized personnel can remove the covers and/or have access to the components under voltage.

During normal use, if the operator detects irregularities or damages, he/she should immediately inform the authorized technical personnel.

Maintenance and service activities can only be performed by skilled authorized technical personnel that has been properly trained on the residual risks of the equipment.

It is the responsibility of the purchaser to make sure that the operators have been properly instructed concerning the safety issues and the residual risks related to the equipment.

The machine is equipped with the following devices to limit the residual risks in using it:

- The emergency button that turns OFF the motor and discharges the hydraulic pressure once pressed, thus stopping the raising of the piston;

Fig. 1-1

- Max. pressure valve to avoid machine overloading.
- Dual input for ram travel microswitches to prevent the excessive piston travel; when triggered it disables the functioning of the pump motor;
- Panels locked with screws.

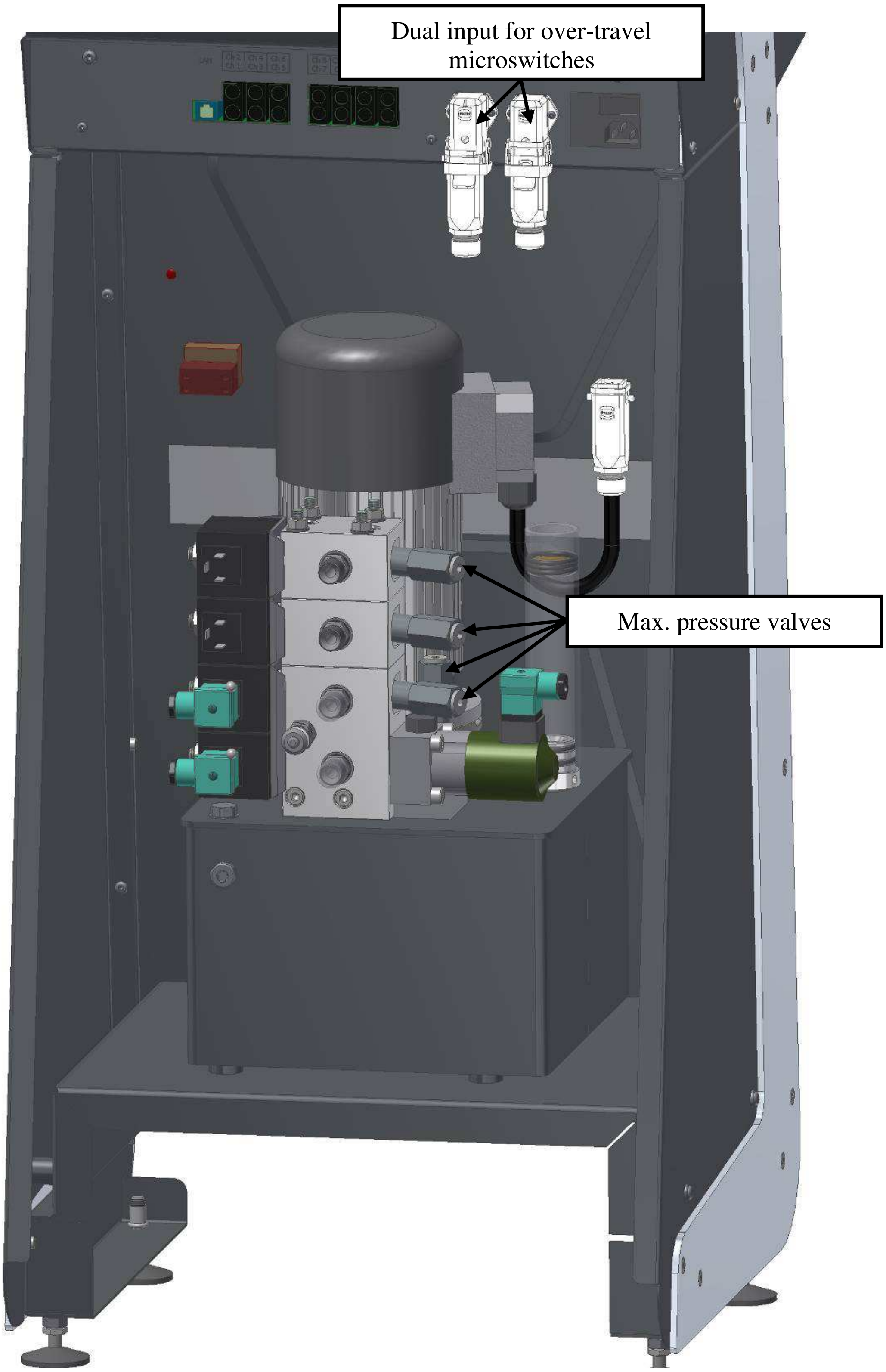


Fig. 1-2

The following table lists those parts of the equipment that may present some residual risks for the safety of the personnel if the instructions provided in this manual are not duly followed.

Personnel	An area with residual risks
Operator	The sample during the load-unloading phase from the frame, due to its weight and sharp surfaces. Use suitable personal protection devices (e.g. gloves, protective shoes, etc.).
	The distance pieces and platens and accessories during the installation phase on the frame, due to their weight. Use suitable personal protection devices (e.g. gloves, protective shoes, etc.).
	The area around the frame during the execution of the test, in case of test on explosive samples and with the front and rear fragment guards not in place.
	The gap between the sample and the upper platen on the frame during the compression phase, although the maximum speed is very low (max. 40 mm/min).
Technical personnel	Above listed areas
	The area around the hydraulic tubes, due to the high pressure of the oil, in case the tubes are damaged or not properly attached
	Areas around compartments closed by removable panels
	Areas around compartments that contain electrical parts
	For continued fire protection, replace fuses with the same type and rating. Also, in case of failure, components may only be replaced by using original spare parts. It is in the responsibility of the purchaser to ensure that fire prevention policies are properly implemented according to the CE provisions.
Personnel responsible for transportation and moving of the equipment	Raising and moving of the equipment from the wooden crate must be performed by people properly trained and equipped with suitable personal protection devices (e.g. gloves, helmet, etc.). Failure to follow the instructions above may endanger the personnel involved.

Here follows a list of all WARNINGs present in the manual; please see relevant chapters for full details on each related safety issue.

WARNING:
When operating with the covers open/removed and the unit is attached to the mains line, care must be taken as high voltage is present in some parts of the unit (e.g. electrical panel). Only authorized and qualified service technicians are allowed to open /remove covers.

WARNING:
All safety devices must be functional at all times. Damaged protective covers or devices must be replaced immediately. When safety components are replaced, the protective devices are to be properly attached and tested. Any manipulation of the safety devices endangers the operating personnel.

WARNING:
To avoid direct contact with the oil of the hydraulic circuit, always wear protective gloves while performing the checking.

WARNING:
Raising and moving of the equipment must be performed by people properly trained and equipped with suitable personal protection devices (e.g. gloves, helmet, etc.). Failure to follow the instructions above may endanger the personnel involved.

WARNING:
The wooden crate is high; be careful in balancing while raising it.

WARNING:
If the instructions to remove the equipment from the shipping box are not carefully followed, damages to the equipment cabinet may result. This will void the warranty terms of the equipment.

WARNING:
Do not stand under suspended loads

WARNING:
Considering the weight of the equipment and the footprint (see chapter 2.10), check the maximum allowed floor load before installation.

WARNING:
If the equipment is brought from a cold environment into a heated room, condensation on and in the unit can constitute a danger and lead to malfunctioning of the unit when it started. Wait to connect and operate the unit until it is at room temperature.

WARNING:
The general grounding must comply with the rules in force; a wrong quality of the grounding could be dangerous for the operator's safety and cause the bad function of the electrical devices.

WARNING:
Due to the high pressure inside the hydraulic circuit (up to 700 bars/10,150 psi), only tubes approved by CONTROLS can be used. CONTROLS will not be responsible for damages to the equipment or personnel injures in case of the use of improper material and/or operations not carried out by authorized technical personnel.

WARNING:
Covers/doors can only be removed/opened by maintenance people, by using relevant tools/buttons. These tools and buttons are for use by maintenance personnel only. Never leave the tools and buttons attached to the unit as this may endanger operator safety.

After performing maintenance/repair, make sure that all covers/doors are properly closed and locked.

WARNING:
Be careful and use proper personal protection devices (e.g. gloves, protective shoes) while loading/unloading samples from the test frame, due to the sample weight and sharp surfaces.

WARNING:
Be careful and use proper personal protection devices (e.g. gloves, protective shoes) while handling distance pieces and platens of the frame due to their weight.

WARNING:
Always close the front door and place the rear guard on the frame while executing the tests. Also, use proper personal protection devices (e.g. glasses) to avoid injures due to fragments of samples ejected during the test (e.g. in case of test on explosive samples).

WARNING:
Although the maximum speed of the loading piston of the frame is very low (max. 40 mm/min), to avoid injures during the compression phase, always close the front door and place the rear guard while executing the tests. Also, avoid putting hands/fingers in the gap between the sample and the upper platen.

WARNING:
Failing to perform the recommended maintenance actions or maintenance performed by unauthorized people can void the warranty.
CONTROLS will not be responsible for maintenance and service actions performed by unauthorized people.

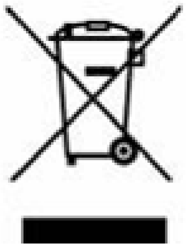
WARNING:
Before opening/removing the covers, disconnect the mains supply to the device, and wait at least 5 minutes.

WARNING:
For continued fire protection, replace fuses with the same type and rating. Also, in case of failure, components may only be replaced by using original spare parts. It is in the responsibility of the purchaser to ensure that fire prevention policies are properly implemented according to the CE provisions.

WARNING:
Avoid pouring water, even accidentally, or other liquids into the device, as this could cause short circuits. Before cleaning the device, disconnect it from the mains line.

WARNING:
Refer to qualified service organization authorized by CONTROLS to carry out the maintenance actions described in the chapter “Authorized service engineer maintenance action”. CONTROLS has not to be held responsible for damages to the equipment and/or injuries to personnel in case the above is not strictly followed.

1.6 Environmental risks and disposal



INFORMATION TO THE OWNER OF THE EQUIPMENT

The above symbol, when attached to the equipment or the relevant packaging, indicates that the product must be disposed of separately from other rubbish at the end of its useful life.

Therefore, at the end of its useful life, the owner should dispose of the product in a suitable collection point for electrical and electronic products provided by the local authorities.

The correct disposal of this product and the subsequent treatment encourages the manufacture of products using recycled materials and limits the environmental impact of the product caused by improper disposal.

Improper disposal of the product is subject to penalties as foreseen by the local regulations.

1.7 CE declaration

This page shows a copy of the CE declaration. The original is supplied with the equipment as a separate document.

REV.19

DECLARATION OF CONFORMITY
DICHIARAZIONE DI CONFORMITÀ
Directive 2006/42/CE, Annex II, sub A) - Direttiva 2006/42/CE, Allegato II, parte A)

CE

Manufacturer *Fabbricante*	**CONTROLS S.P.A.**
Address *Indirizzo*	Via Salvo D'Acquisto 2/4 20060 Liscate, (MI) Italy
	Herewith declares that the machine *Dichiara che la macchina*
Model *Modello*	**50-C20M82**
Serial number *Matricola*	18006722
Description *Descrizione*	AUTOMAX MULTITEST computerized control console for up to 2 frames (expandable up to 4) capable to perform the following tests: *AUTOMAX MULTITEST, consolle di comando controllata da PC per sistemi composti da 2 telai di prova (espandibili a 4) per l'esecuzione automatica di: - Prove di compressione, flessione, trazione indiretta (con software Datamanager, incluso) - Prove di Modulo Elastico e di Poisson (con*

is in conformity with the provisions of the following EC directives:
è conforme a quanto previsto dalle seguenti direttive CE:

2006/42/CE (Machinery Directive)
(Direttiva Macchine)

2014/35/UE (Low Voltage Directive)
(Direttiva Bassa Tensione)

2014/30/UE (Electromagnetic Compatibility Directive)
(Direttiva Compatibilità Elettromagnetica)

Date of issue
Data di emissione

02 August 2018

Nicola Lezzerini
Technical Director - Person authorized to compile the technical file
Direttore tecnico - Persona autorizzata alla costituzione del fascicolo tecn

CONTROLS S.p.A. Via Salvo D'Acquisto 2/4, 20060 Liscate (MI) Tel. +39- 02921841 fax +39- 0292103333 e-mail: controls@controls.it www.controls.it

2. DESCRIPTION

Refer to the following figures for main components identification.

Fig. 2-1

Ref.	Description
1	Computer with display
2	Keyboard
3	***POWER ON*** lamp
4	***EMERGENCY BUTTON***
5	Electronic drawer
6	DC motor dual-stage hydraulic pump
7	Distribution block (up to 4 frames)
8	Oil tank

2.1 Hydraulics

- Dual-stage pump: centrifugal low pressure for fast approach and automatic switching to radial multi-piston high pressure for loading
- DC motor. 720 W, 50-60 Hz
- Maximum working pressure 700 bar (10,150 psi)
- Third and fourth frame option, active frame selection by software
- Flow-sharing technology to perform loading and unloading cycles
- ES Energy Saving technology to reduce power consumption and silent operation

2.2 Hardware

- 131.000 points effective resolution
- High-frequency closed-loop P.I.D. control
- Control frequency 500 Hz
- Sampling rate 500 Hz
- 4 channels for load sensors (pressure transducers and load cells)
- 6 channels to measure rebars elongation with extensometers (see Accessories) or strain/displacement under compression testing with LVDT and magnetostrictive transducers (see Accessories)
- 4 channels for strain measurement with strain gauges
- Memorization of the calibration curve enables sensors to be connected and used immediately
- Digital linearization of the calibration curve (multi-coefficient)

2.3 User interface

Main software MULTITEST allows:

- Remote control of the whole system and automatic execution of tests including fast approaching, zeroing, test execution, numerical and graphical management of test results, etc.
- Active frame selection via software
- Printing and saving of customized test reports both for single and batch tests in Excel format
- Multi-language selection, customizable with a further local language (only Latin characters).
- Automatic load measurement verification procedure including data acquisition and printing of traceable calibration certificates, when the PC is connected to the digital readout unit mod. 82-P0801/E and the suitable load cells
- The use of 4 software packages each one tailored for a specific test method

DATAMANAGER software package (included) for compression, indirect tensile, 3 points and 4 points flexural tests on different types of specimens

UTS software package (optional) for steel tensile testing allowing:

- load/stress control
- crosshead separation control
- simultaneous display of: stress/elongation [mm], stress/time; stress/elongation [%] and elongation [mm]/time, with possibility to display multi diagrams
- elaboration of tension test results: ReH, ReL or Rp, final elongation, etc. in conformity to EN ISO 6892-1 (method B) and EN 15630-1 (for steel rebars)

An E-MODULE software package (optional) for Elastic modulus and Poisson ratio determination allowing:

- Free unlimited programmable load/stress cycles to fulfill any kind of test procedure
- Real-time monitoring of test data, stress/time, stress/axial strain, stress/lateral strain graphs
- Automatic verification of sample positioning and sensors functionality, as per standards requirements
- Automatic calculation of test results as per Standards requirements

D-Control software package (optional) for displacement controlled tests allowing:

- Automatic calculation of test results according to EN 14651, 14488-3, 14488-5, UNI 11039-2, ASTM C1550, C1609, C947, UNE 83515
- 8 test pre-set testing procedures according to the above Standards
- Customizable test procedure allowing desired loading history
- Possibility to change in real-time the test parameters: target load/displacement, control variable, test speed.

The software comprises a base program – the **MULTITEST** - where calibrations and system settings such as language and passwords are managed, and the **DATAMANAGER** test module which contains all the functions necessary for running compression, flexure, and indirect tensile tests.
The software operates in MS Windows; the PC connects to the Controls control console via a LAN port. The next chapters explain the main features of the software.

2.4 MULTITEST main features

- Multi-language, with the possibility to create multiple customized languages
- Selection of unit of measure: kN, Ton, and lbf
- Up to three testing frames can be managed by the software (not simultaneously and depending on the testing device)
- Transducer calibrations can be performed and uploaded to the control console
- Multiple calibrations can be stored for each channel
- The pump is controlled via the software during calibration
- Calibration sequence can be performed automatically

2.5 DataManager main features

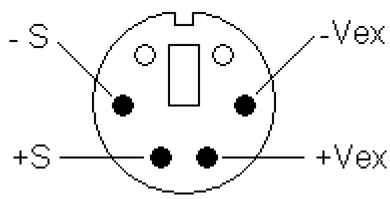
- Allows the user to perform compression, flexure and indirect tensile tests to EN and ASTM standards when connected to a suitable testing device
- Test and sample report fields can be customized
- Data are shown numerically and graphically in real-time
- Historical test results are managed through a database
- Data can be outputted to MS Excel
- Tuning/control settings can be managed via the software
- Active frame selection

2.6 Input channel specifications

The following table provides the specifications of the input channel of the **AUTOMAX Multitest**.

Channel number	Function	V excitation	V input	Acquisition point
CH 1, 2, 3, 4	Loadcells, pressure transducers	10V	0÷30mV	131072 points
CH 5, 6, 11, 12, 13, 14	Potentiometric transducers, Amplified LVDTs, compressometers	10 V	0÷10 V	131072 points
CH 7, 8, 9, 10	Strain gauges, LDTs, clip gauges	2 V	-19.5÷19.5 mV	±65535 points

The layout of the connectors (same for all channels) is as follows (seen facing the rear panel of the equipment):



-S	- Signal
+S	+ Signal
-Vex	- Excitation voltage
+Vex	+ Excitation voltage

2.7 Connections

The next figure shows the position of the connectors/ports used to connect the different modules composing the equipment and equipment itself with external devices.

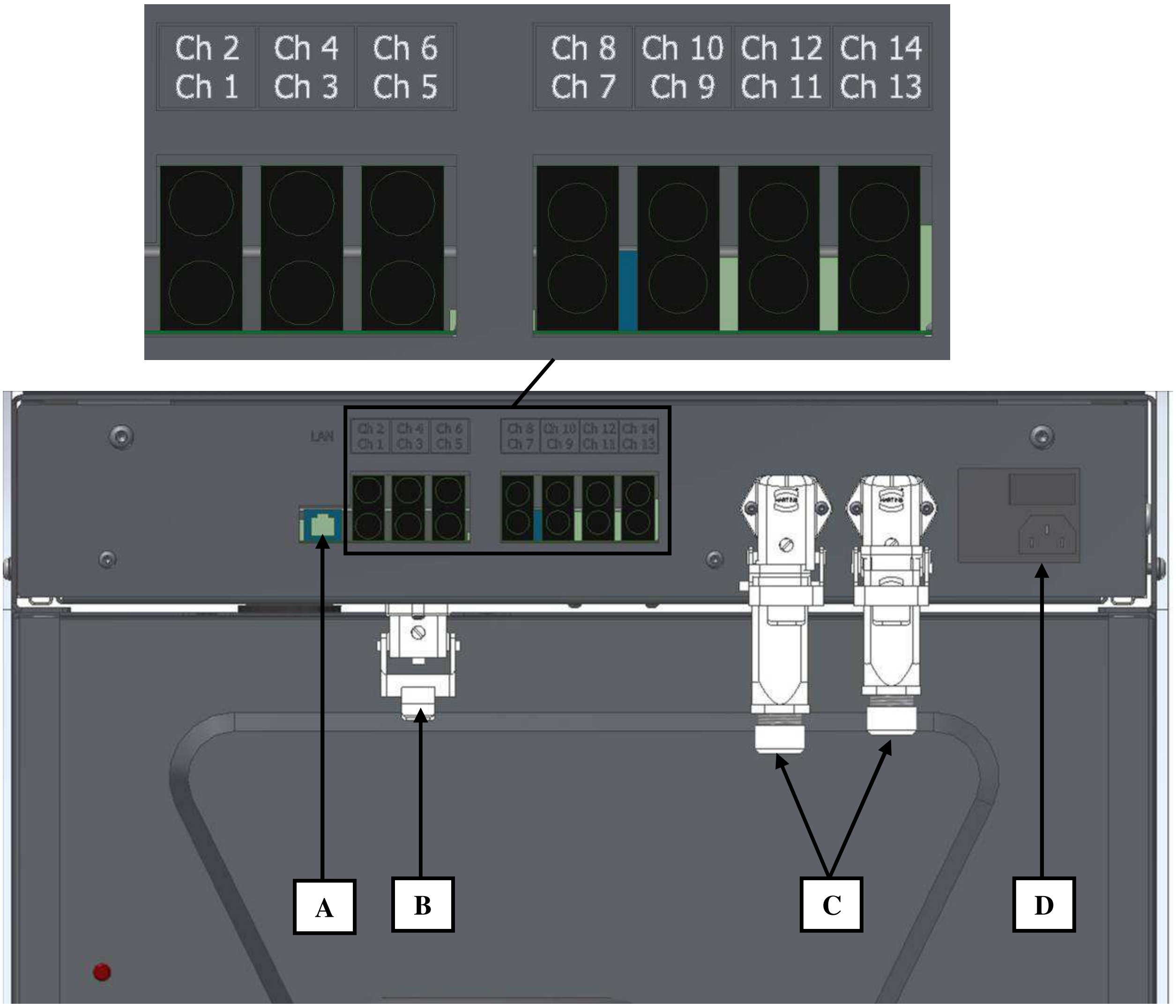


Fig. 2-2

Ref.	Description
CH1,2,3,4	Input for load cells and/or pressure transducers of the frames
CH5,6,11,12,13,14	Input for potentiometric transducers and/or LVDT applied to the sample (e.g. codes P0331/C, P0331/D1, C0222/F, etc.
CH7,8,9,10	Input for strain gauges applied to the samples (e.g. codes P0390, P0391, P0392, P0393, etc.)
A	LAN connector for PC connection
B	Connector for hydraulic pump DC motor
C	Dual connection for frames over-travel microswitches
D	Power cord receptacle

2.8 Identification plate

The identification plate is located on the rear side of the equipment, near the power cable.

Fig. 2-3

2.9 Commands and controls

This chapter describes the commands and controls of the equipment.

NOTE:
This manual includes the hardware description of the **AUTOMAX Multitest** computerized control console.
The console must be operated via a computer connected to it and equipped with **MULTITEST SOFTWARE** that includes the **DATAMANAGER** module. Refer to the relevant documentation to operate the system and perform the tests.

Fig. 2-4

Ref.	Description
1	Computer with display
2	Computer keyboard and mouse
3	***POWER ON*** lamp
4	***EMERGENCY BUTTON***

2.10 Technical specifications

Main characteristics	
Product	**AUTOMAX Multitest** computerized control console
Manufacturer	CONTROLS Liscate (MI) Italy
Product code 230VAC version Product code 110VAC version	50-C20M82 50-C20M84
Single phase line voltage	110 VAC/230 VAC +10%, -10%
Frequency	50/60 Hz
Power	750 W
Method of power entry	Power cable with plug
Computer connection	LAN port
Fast approach speed	40 mm/min (1.57 inch/min) approx.
Max. working pressure	700 bar (10,150 psi)
Oil output at max. pressure	0.3 l/min (10 fl oz/min) approx.
Oil tank capacity (approx.)	9 liters (2.5 Gal)
Acoustic pressure	< 70db (A)
Net weight (approx.)	120 Kg (265 lbs)
Overall dimensions	See the next drawings

Environmental conditions	
Operating temperature	+ 10 ÷ + 40°C (50 to 105 F)
Operating humidity	≤ 50%RH @ 40°C

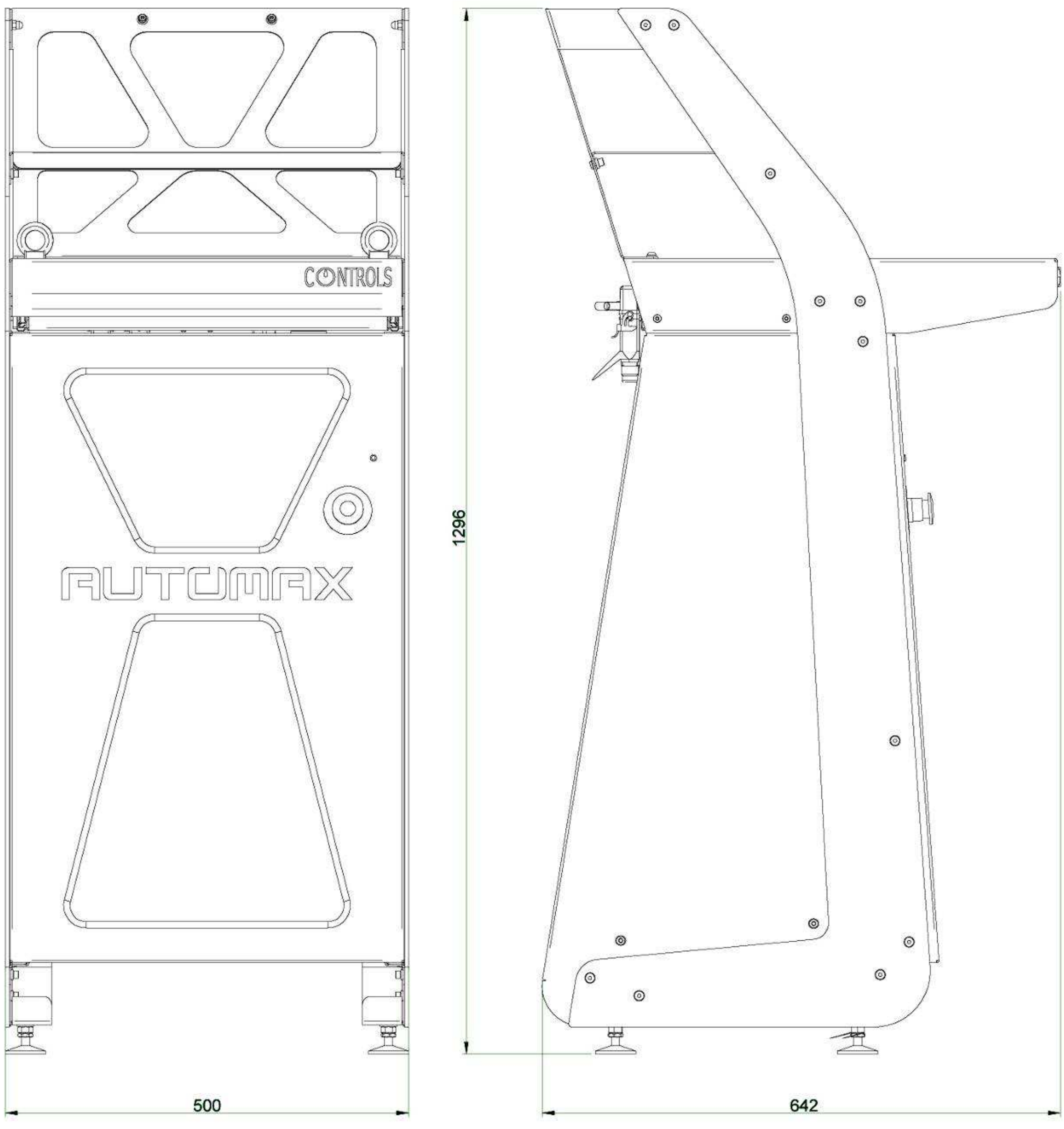


Fig. 2-5

3. INSTALLATION

The instructions indicated in this chapter enable you to perform a correct installation and interfacing to grant a regular operation of the equipment.
Included are initial inspection procedures, power requirements, and instructions for installing the unit.

The information in this chapter is intended for authorized service-trained personnel.

CONTROLS can supply the assistance and the necessary technical advice for pre-installation, all the pre-installation phases are at the purchaser's charge and must be performed complying with the indications given below.

WARNING:
When operating with the covers open/removed and the unit is attached to the mains line, care must be taken as high voltage is present in some parts of the unit (e.g. electrical panel). Only authorized and qualified service technicians are allowed to open /remove covers.

WARNING:
All safety devices must be functional at all times. Damaged protective covers or devices must be replaced immediately. When safety components are replaced, the protective devices are to be properly attached and tested. Any manipulation of the safety devices endangers the operating personnel.

WARNING:
To avoid direct contact with the oil of the hydraulic circuit, always wear protective gloves while performing the checking.

3.1 Shipment

The equipment is normally shipped in a wooden crate having the following dimensions and gross weight depending on the model:

Model	Gross weight	Overall dimensions	Minimum forks-length
C20M8X	180 Kg (400 lbs) Approx.	800 x 800 x 1550 mm (31.5” x 31.5” x 61”)	1200 mm (47”)

Use a proper forklift with minimum fork length as above indicated to raise and move the wooden crate (see figure below).

Fig. 3-1

WARNING:
Raising and moving of the equipment must be performed by people properly trained and equipped with suitable personal protection devices (e.g. gloves, helmet, etc.). Failure to follow the instructions above may endanger the personnel involved.

WARNING:
The wooden crate is high; be careful in balancing while raising it.

3.2 Unpacking and inspection

The unit was carefully checked both mechanically and electrically before shipment; it should be inspected for any damage that may have occurred in transit.

NOTE:
If the shipping container or packaging material is damaged, it should be kept until the unit has been mechanically and electrically checked.
If there is mechanical damage and/or the contents are incomplete (see the shipping list), please notify the local CONTROLS representative.
If the shipping container is damaged or shows signs of stress, notify the carrier as well as the CONTROLS representative. Save the shipping material for the carrier's inspection.
Also, take some pictures.

Here follows the procedure for the unpacking of the equipment.

3.2.1 How to remove the equipment from the shipping crate

The following table shows the net weight of the different models of the equipment:

Model	Net weight
C20M8X	120 Kg (265 lbs) approx.

Follow the instructions below to remove it from the shipping crate:
1. Inspect shipping crate (see NOTES above);
2. Open the shipping crate;
3. Cut and remove vacuum bag (if present) to expose the equipment;
4. The equipment must only be lifted by the eyebolts screwed in on the top of the console. Do not lift it by any other part to avoid damaging the equipment (see Fig. 3-2).

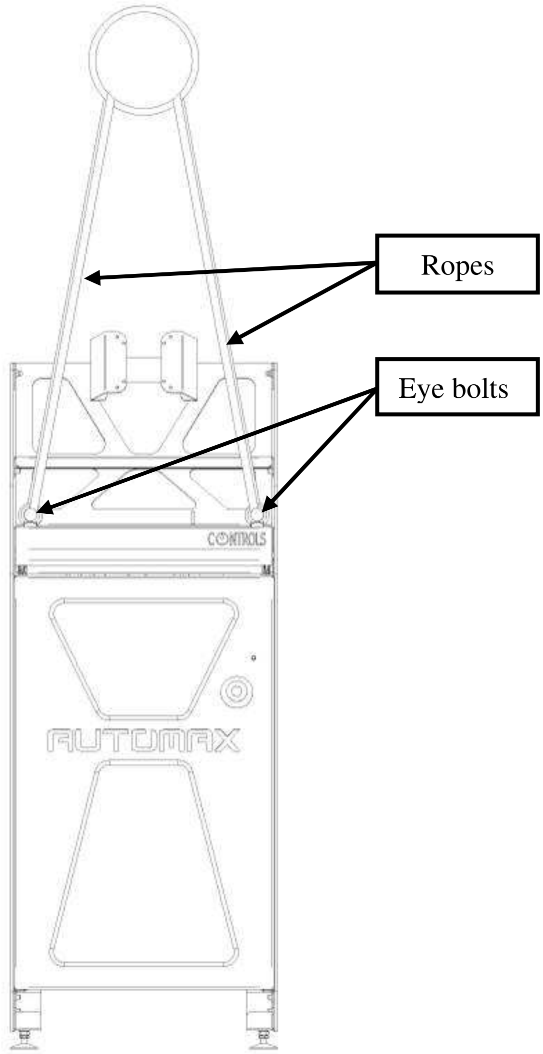


Fig. 3-2

WARNING:
If the instructions to remove the equipment from the shipping box are not carefully followed, damages to the equipment cabinet may result. This will void the warranty terms of the equipment.

WARNING:
Raising and moving of the equipment must be performed by people properly trained and equipped with suitable personal protection devices (e.g. gloves, helmet, etc.). Failure to follow the instructions above may endanger the personnel involved.

WARNING:
Do not stand under suspended loads

5. Before powering on the unit, check for loose connections of cables inside the equipment.

3.3 Positioning of the unit and space requirements

WARNING:
Considering the weight of the equipment and the footprint (see chapter 2.10), check the maximum allowed floor load before installation.

The unit must be placed on a flat and level surface.

The area around the equipment must be kept free from obstacles and from substances that may cause sliding of the personnel.

It is the responsibility of the purchaser to provide an installation place with adequate lighting.

Parts used to build the machine and electrical/electronics components are fire-proof.
It is anyway the responsibility of the owner of the equipment to make sure that the installation place is equipped with proper fire extinguishing systems and that the operators have been instructed on the relevant procedures.

Use a proper means to bring the unit to the installation place.

Place the unit in its final location, taking into account the free space around the unit: the equipment requires at least 100 cm (40”) of clearance on the front and at least 50 cm (20”) on the 3 sides for maintenance and service purposes (next drawing shows the console and a compression frame as an example).

Also, consider that the mains connection is achieved through the mains cable socket on the rear side of the equipment cabinet.

The unit requires adequate air circulation around it to assure proper cooling of the internal devices.

WARNING:
If the equipment is brought from a cold environment into a heated room, condensation on and in the unit can constitute a danger and lead to malfunctioning of the unit when it started. Wait to connect and operate the unit until it is at room temperature.

3.4 Placing the computer and accessories

The **AUTOMAX Multitest** console is equipped with a tray for the keyboard, and mouse.

Arrange the above components on the trays depending on the models provided.

Fig. 3-3

The PC should be mounted with a small frame provided with the system, on the rear side of the display.

Fix the display to the frame of the console by tightening the 4 screws with their supports (marked in red in the following figure), in the relevant holes.

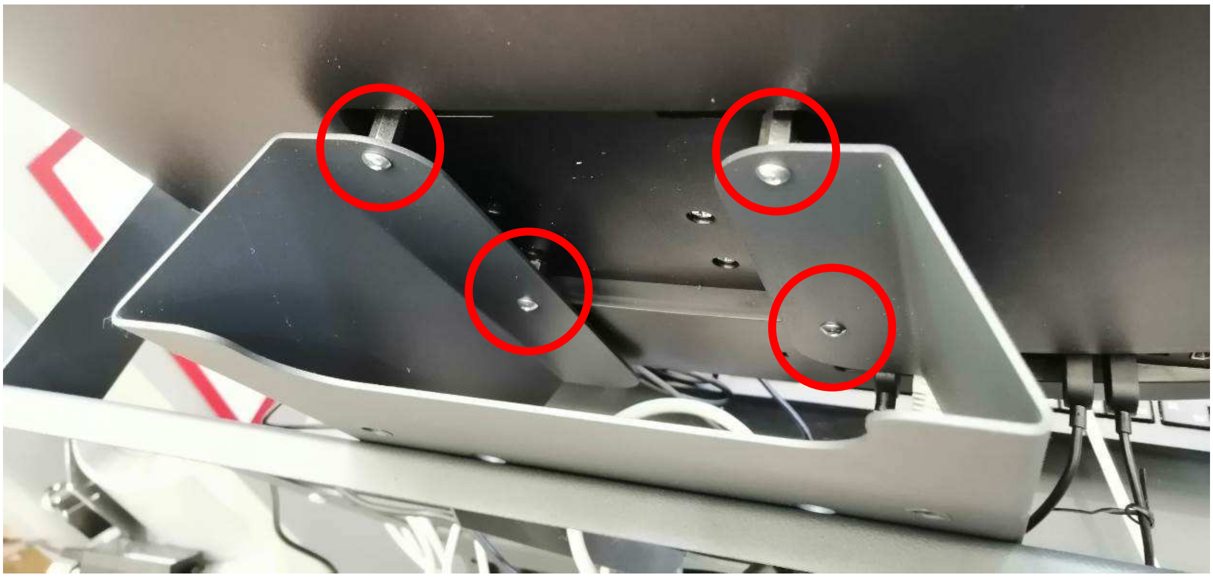


Fig. 3-4

3.5 Electrical requirements

The next table shows the electrical specifications of the equipment:

Item	Specifications
Mono phase voltage	230VAC / 110VAC +6%, -10%
Frequency	50Hz/60Hz
Maximum power	750 W

The equipment shall be connected to a proper earth system, the efficiency of which shall be checked by qualified personnel. The earth shall be via the power cable, as specified above.

WARNING:
The general grounding must comply with the rules in force; a wrong quality of the grounding could be dangerous for the operator's safety and cause the bad function of the electrical devices.

The power supply line shall be equipped with a safety device (breaker and ground fault switch) properly sized with respect to the electrical specifications provided above.

3.6 Electrical connections

The next figure shows the position of the connectors/ports used to connect the different modules composing the equipment and equipment itself with external devices.

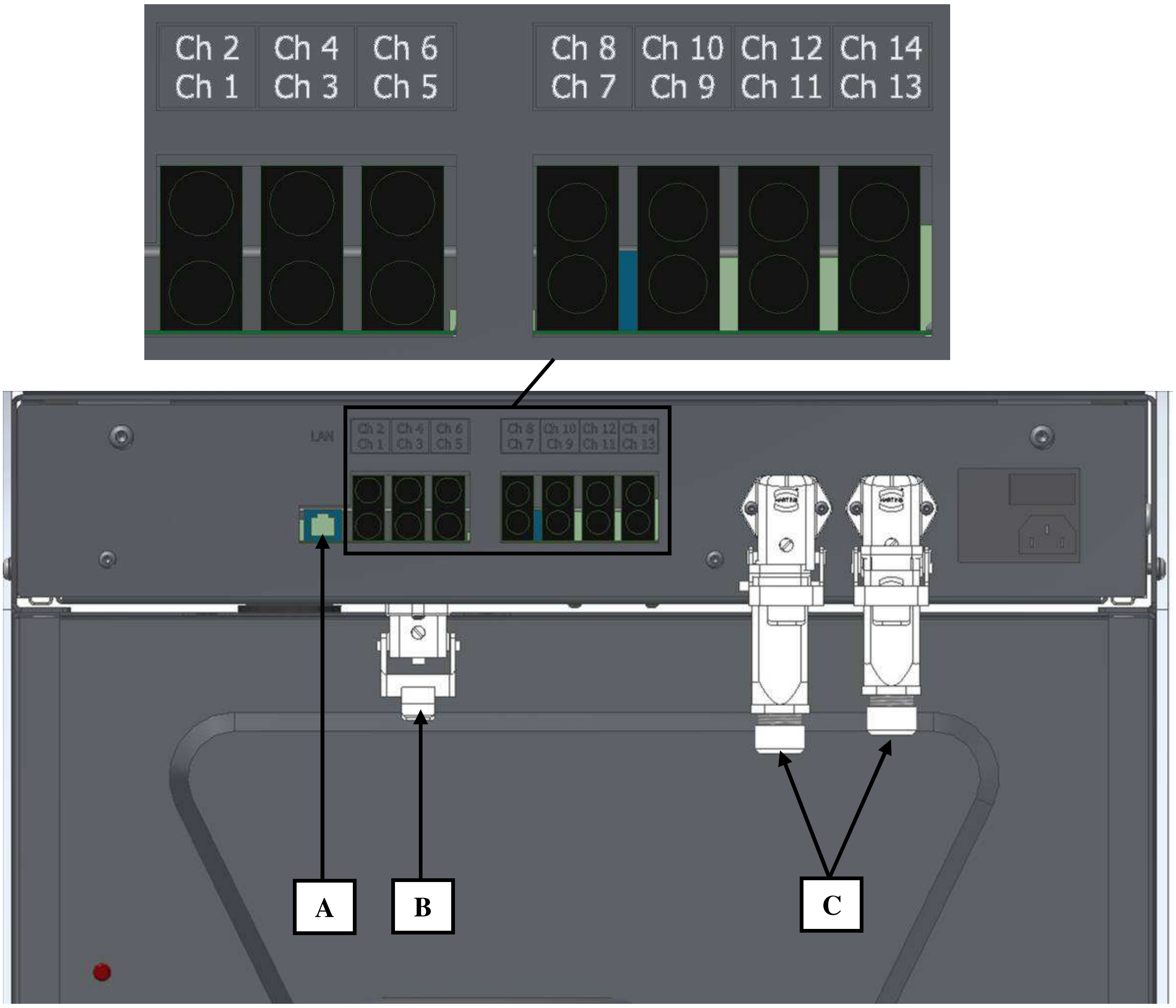


Fig. 3-5

Ref.	Description
CH1,2,3,4	Input for load cells and/or pressure transducers of the frames
CH5,6,11,12,13,14	Input for potentiometric transducers and/or LVDT applied to the sample (e.g. codes P0331/C, P0331/D1, C0222/F, etc.
CH7,8,9,10	Input for strain gauges applied to the samples (e.g. codes P0390, P0391, P0392, P0393, etc.)
A	LAN connector for PC connection
B	Connector for hydraulic pump DC motor (inside the console)
C	Dual connection for frames over-travel microswitches
D	Power cord receptacle

3.7 Hydraulic connections

This chapter explains where to connect the hydraulic hose to supply the frame. In case the device is equipped with the triple or quadruple output hydraulic block (code 50-C10D/3F and 50-C20E/4F will be installed in the factory) it will be possible to connect the third and fourth frames.

WARNING:
To avoid direct contact with the oil of the hydraulic circuit, always wear protective gloves while performing the checking.

NOTE:
The system can control the functioning of one frame per time.

1. With the unit OFF and unplugged from the mains, connect the flexible tube supplied to the selector block and tighten properly the relevant fixing nuts;

WARNING:
Due to the high pressure inside the hydraulic circuit (up to 700 bars/10,150 psi), only tubes approved by CONTROLS can be used. CONTROLS will not be responsible for damages to the equipment or personnel injures in case of the use of improper material and/or operations not carried out by authorized technical personnel.

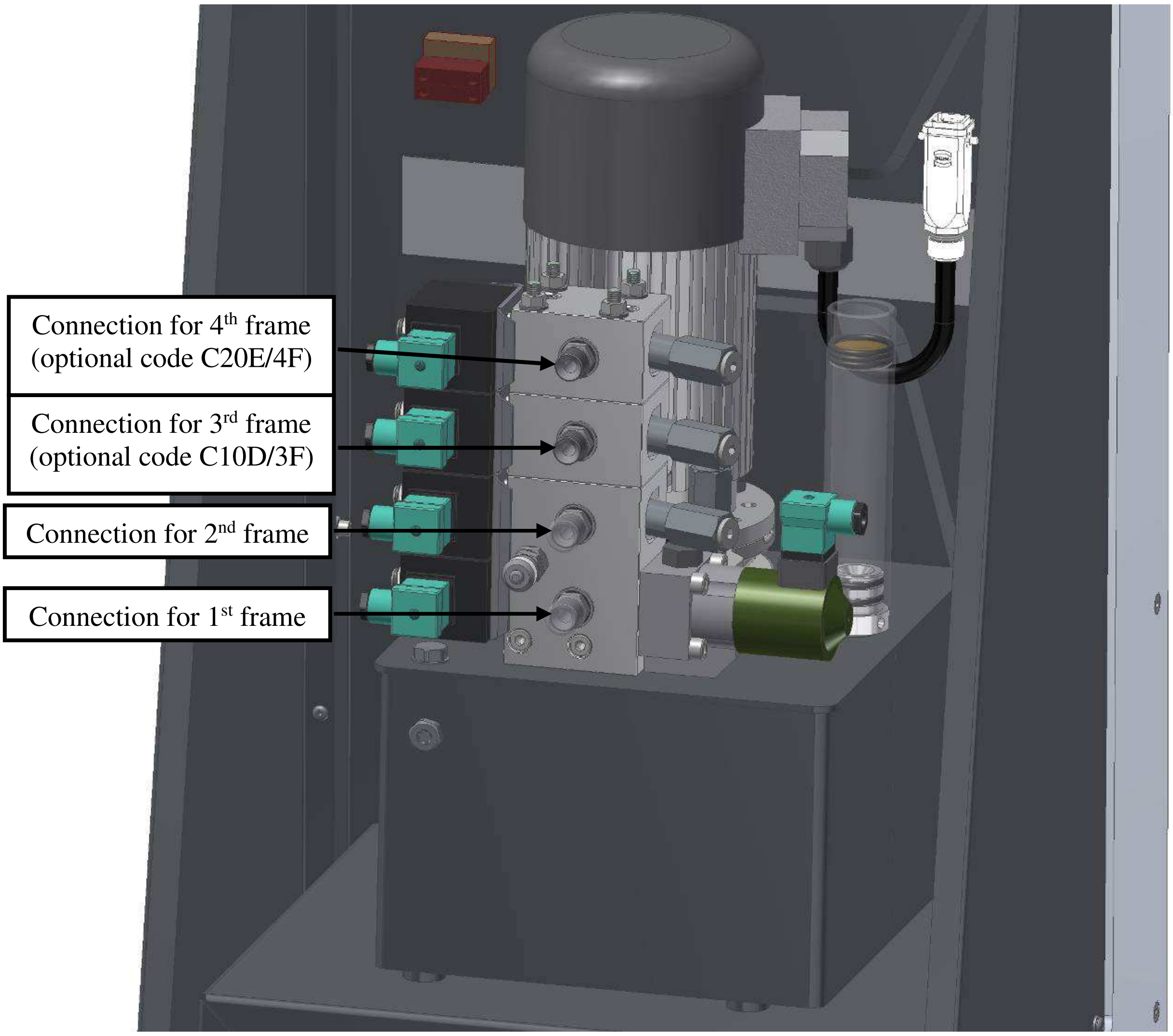


Fig. 3-6

2. After making the hydraulic connections, check the oil level in the tank via the transparent level indicator (the oil should be max. at half of the indicator); if necessary refill the oil tank (refer to chapter 7 of this manual);

3. If the capacity of the additional frames is higher than the one of the frames already connected to the equipment, the max. pressure valve might need to be re-adjusted:
    a. Locate the max. pressure adjustment valve;
    b. Use a 6mm Allen key to remove the cap
    c. Use a flat screwdriver to turn counter-clockwise the valve to reduce the max. pressure;
    d. Adjust the valve by turning the screw clockwise until it intervenes at the full scale of the frame;
    e. Turn the screw still ¼ of a turn clockwise;
    f. Re-install the cap.

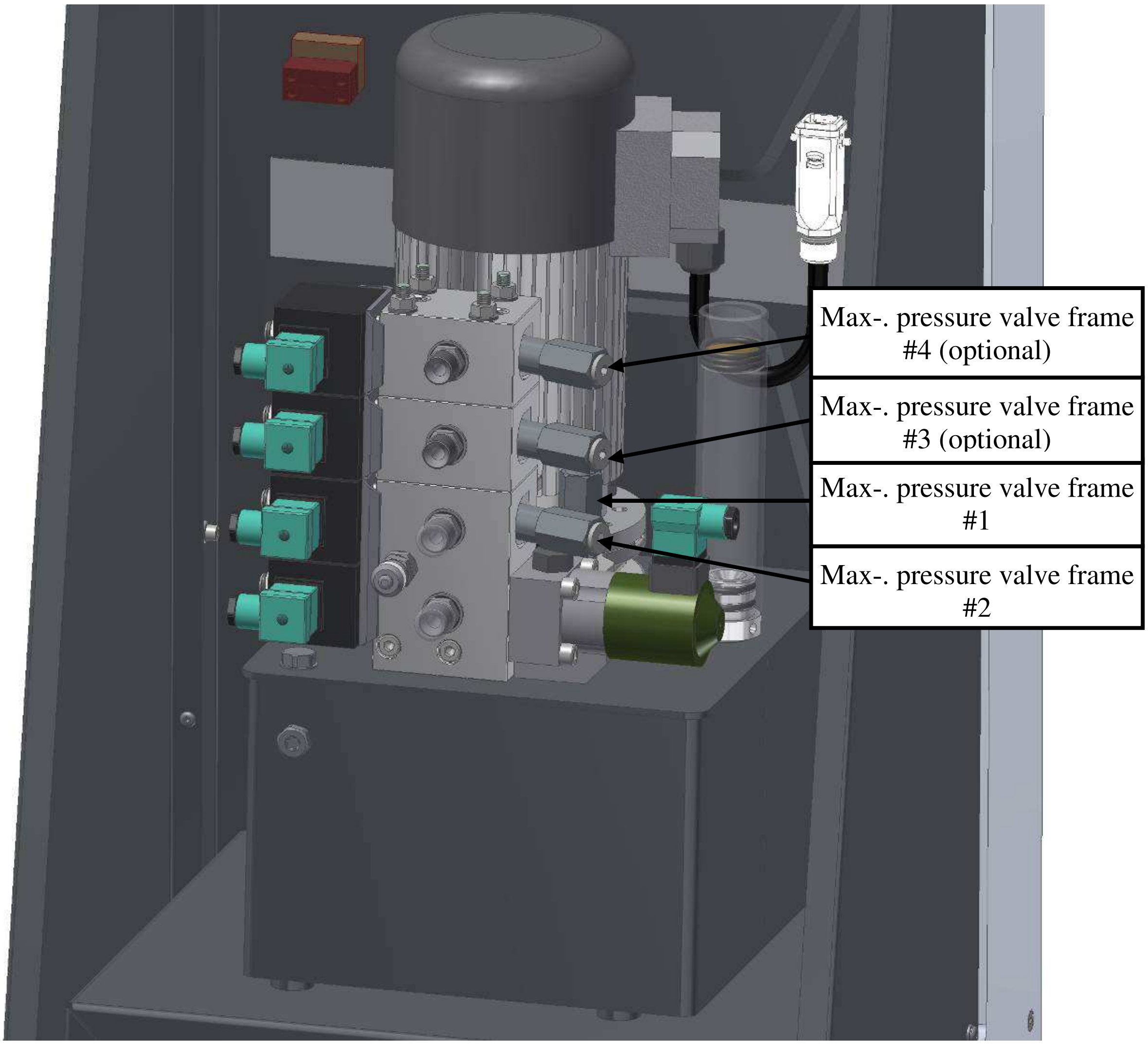


Fig. 3-7

4. USE OF THE EQUIPMENT

This chapter describes the operator's interface and the execution of a test.

WARNING:
All safety devices must be functional at all times. Damaged protective covers or devices must be replaced immediately. When safety components are replaced, the protective devices are to be properly attached and tested. Any manipulation of the safety devices endangers the operating personnel.

WARNING:
Covers/doors can only be removed/opened by maintenance people, by using relevant tools/buttons. These tools and buttons are for use by maintenance personnel only. Never leave the tools and buttons attached to the unit as this may endanger operator safety.

After performing maintenance/repair, make sure that all covers/doors are properly closed and locked.

NOTE:
This manual includes the hardware description of the **AUTOMAX Multitest** computerized control console.
The console must be operated via a computer connected to it and equipped with **MULTITEST SOFTWARE** that includes the **DATAMANAGER** module. Refer to the relevant documentation to operate the system and perform the tests.

4.1 Switching on the equipment

1. Once all the connections have been made correctly, check that the ***EMERGENCY BUTTON*** is not pressed (otherwise release it by rotating clockwise) and switch ON the console via the power switch on the rear panel: the ***POWER ON*** lamp;

2. Turn ON the computer;

Fig. 4-1

3. On the computer activate the **MULTITEST** software and then the **DATAMANAGER** module;

4. Refer to the **MULTITEST** software manual for full details on the programs and test execution.

NOTE:
Each time the unit is turned OFF or ON, repeat the above procedure in the same order (power up the control console first and start the software after) to ensure correct communication between the software and the MULTITEST electronics.

4.2 Carry out a test

The present chapter provides some safety ***WARNINGS*** to take into account to carry out a test.

WARNING:
Be careful and use proper personal protection devices (e.g. gloves, protective shoes) while loading/unloading samples from the test frame, due to the sample weight and sharp surfaces.

WARNING:
Be careful and use proper personal protection devices (e.g. gloves, protective shoes) while handling distance pieces and platens of the frame due to their weight.

WARNING:
Always close the front door and place the rear guard on the frame while executing the tests. Also, use proper personal protection devices (e.g. glasses) to avoid injures due to fragments of samples ejected during the test (e.g. in case of test on explosive samples).

WARNING:
Although the maximum speed of the loading piston of the frame is very low (max. 40 mm/min), to avoid injures during the compression phase, always close the front door and place the rear guard while executing the tests. Also, avoid putting hands/fingers in the gap between the sample and the upper platen.

Before starting any test check that:

- The test frame is free from residual of the previous sample and that the surface of the platens is clean;
- Depending on the size of the sample, use proper distance pieces to reduce the daylight on the press (leave a gap of at least 3-4mm/0.15”), also considering that the piston total travel is 50mm (2”).

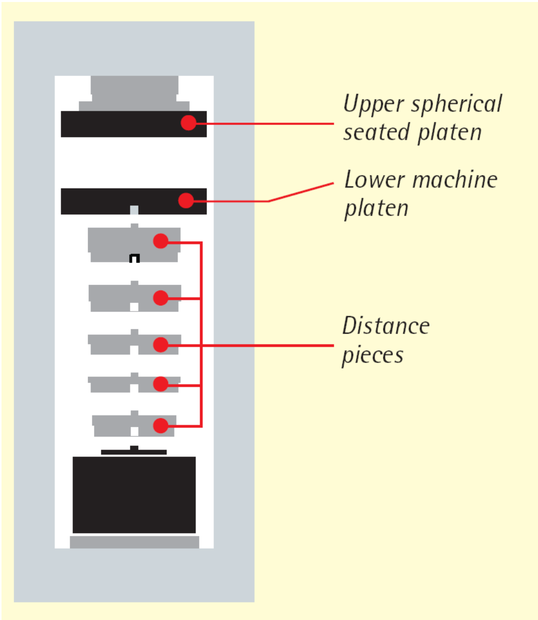


Fig. 4-2

4.3 Stopping the cycle via the Emergency button

In case dangerous conditions are encountered during the execution of a test cycle, it is possible to stop the machine by pressing the ***EMERGENCY BUTTON***.

Fig. 4-3

Pressing this button will stop the movement of the piston (pump motor turned OFF) releasing the pressure of the hydraulic circuit.
The ***POWER ON*** lamp will remain ON to indicated that the console is still supplied by the electricity.

On the MULTITEST program computer display, the warning field will turn red (normally it should be green).

Fig. 4-4

Once the dangerous condition is cleared, release the ***EMERGENCY BUTTON*** (turn it clockwise) and re-activate the function previously in progress by acting on the software program of the computer.

4.4 Switching off the unit

At the end of a test session, remove residuals of the samples from the frame.

Close the program running, shut down the computer, and then switch the equipment OFF via the mains switch on the control unit.

NOTE:
Each time the unit is turned OFF or ON, repeat the above procedure in the same order (power up the control console first and start the software after) to ensure correct communication between the software and the MULTITEST electronics.

Fig. 4-5

5. SOFTWARE COMPONENTS

5.1 Overview of the software

The software comprises two parts; the MULTITEST base software and the DataManager test module.

The **MULTITEST** base software communicates with the testing device and is used for managing all the system settings, options, and calibrations.
The **DataManager** test module is used to run, view, and manage current and previous compression, indirect tensile, and flexure tests.

The DataManager test module is located and opened from within the MULTITEST.
The components of both parts are described in the following sections.

5.2 MULTITEST Main panel

This panel contains functions for setting and editing the language, changing passwords, loading or saving system parameters, running the DataManager test module, and managing calibrations. An information bar at the bottom of the panel displays the status of the system. See Fig. 5-1. ***It is possible to choose other modules available as an upgrade. For further information refer to the relevant instruction manuals.***
The panel is displayed when the MULTITEST software is first started and when there is no test module running. It cannot be accessed while a test module is open.

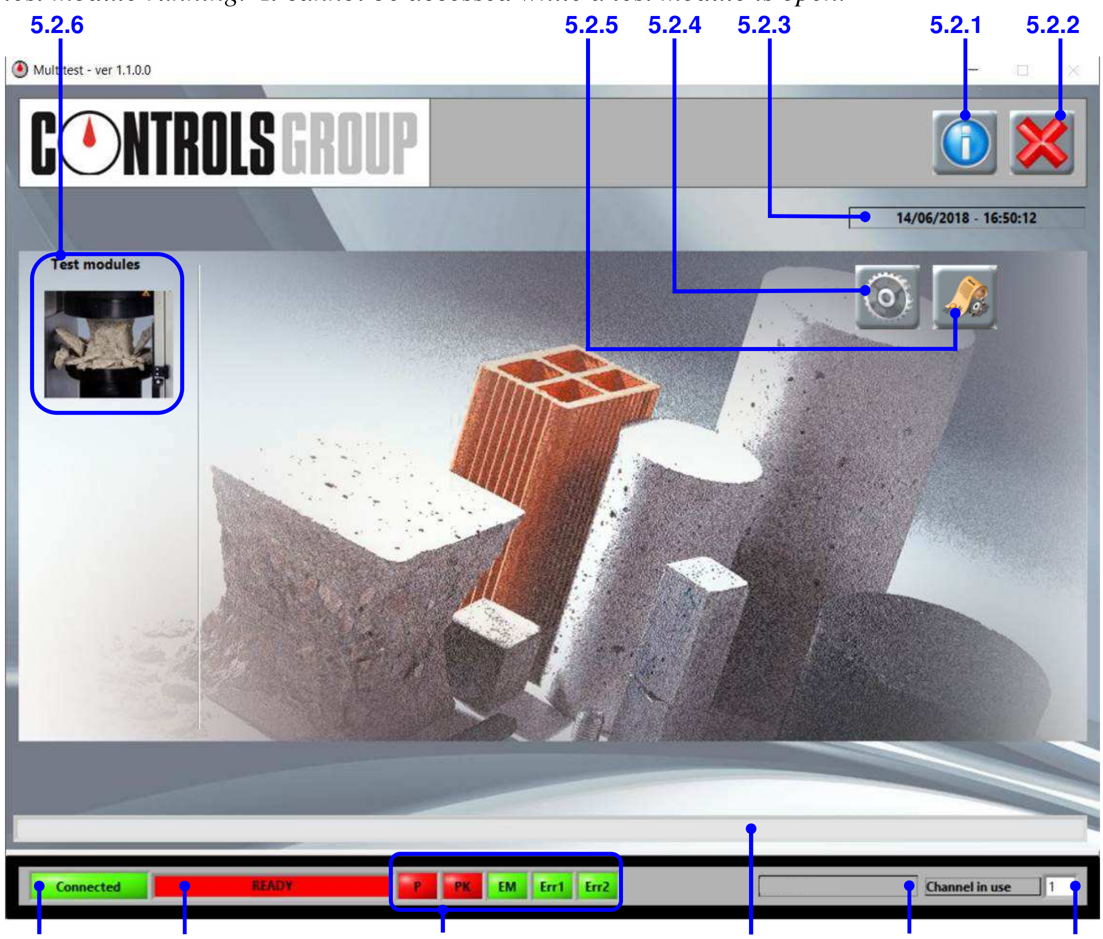


Fig. 5-1

5.2.1 Info button

Opens a window showing company information and version numbers.
To close the window, click the close button in the top right corner of the window.

5.2.2 Close button

Closes the software.

5.2.3 Date and time

Displays the current date and time.
The information is taken from the system clock of the PC.

5.2.4 Options button

Opens the **System options panel** for managing the language, changing the passwords, and for loading and saving system parameters. See Section 5.3.

5.2.5 Channel configuration button

Opens a list of available channels with information about what is currently connected to the console device. By double-clicking on the row, the calibration data for the channel can be opened in the **Calibration panel** (Section 5.4).

5.2.6 Test modules

This area of the MULTITEST contains links for running the available test modules.
Click on a test module icon to open the test module.
It is possible to choose other modules available as an upgrade. For further information refer to the relevant instruction manuals.

5.2.7 Message bar

Displays any messages from the software with the date and time that the accident occurred.

5.2.8 Connection status

Indicates the status of the connection between the PC and the console.

5.2.9 System status

Indicates the status of the testing system:
'READY' means that the system is ready to start loading;
'START COMMAND DISABLED' indicates that loading cannot be started because the pump is not ready.
'TEST IN PROGRESS' is shown when a test is running.

5.2.10 Warning indicators

When the indicators are green, they warn the user that certain events have occurred in the system:

- P - The hydraulic pump is working (green) or not (red)PK – The peak load has been reached during a test and the specimen has failed
- EM - The emergency button has been pressed. If so, the current test is stopped and it's not possible to start a new one

5.2.11 Software version

Displays the version of the software.
*The software version is automatically detected. With **AUTOMAX Multitest** control console, the software version is set to 'GOLD'.*

5.2.12 Machine channel number

Displays the channel number of the testing machine that is currently selected.
*The number can be from 1 – 4. See also **Channel number control** (Item 5.7.3) in the **Test settings panel**.*

5.3 MULTITEST System options panel

This panel contains functions for setting and editing the language, changing the passwords, and loading or saving system parameters. See Fig. 5-2.
*To open the panel, click the **Options button** on the **MULTITEST Main panel**.*
You will have to enter the level 1 password ("controls1") to gain access to the panel.

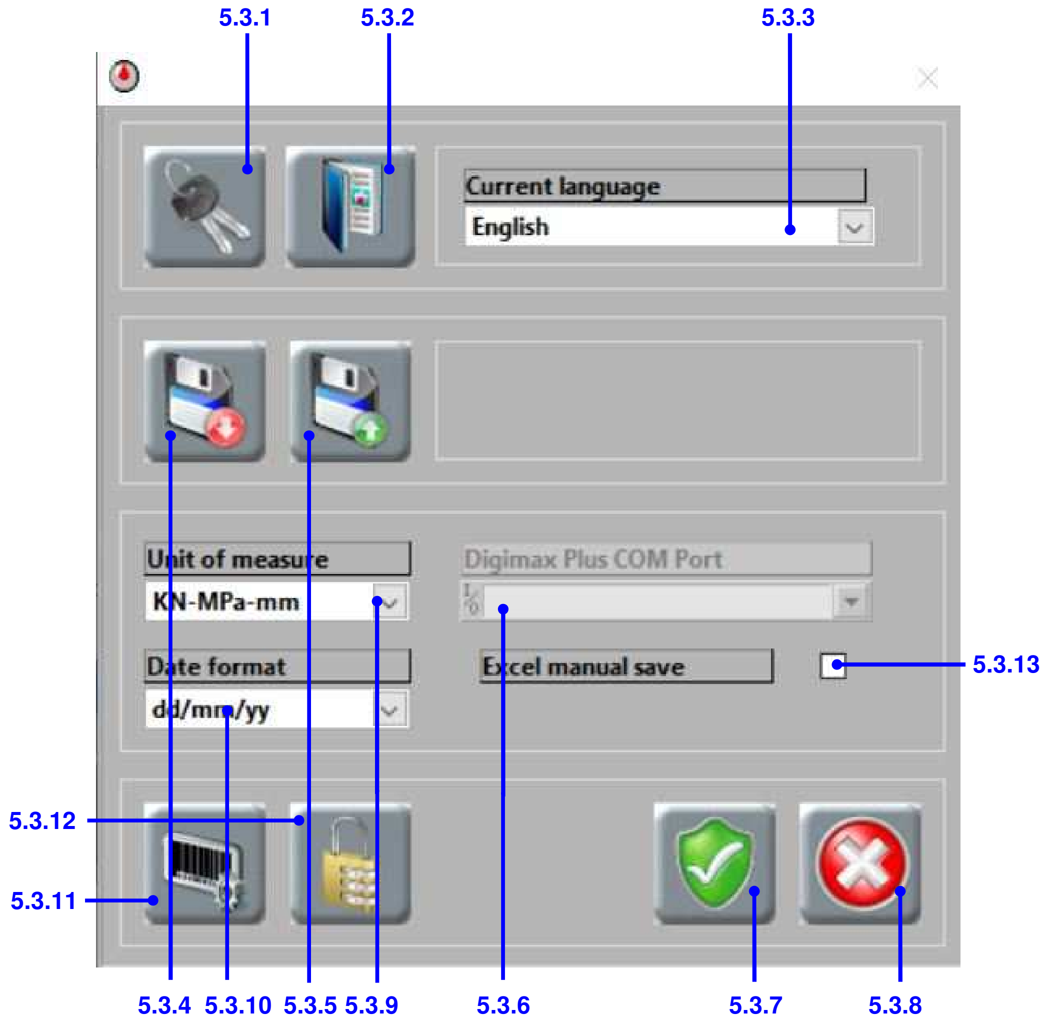


Fig. 5-2

5.3.1 Password button

Opens a panel for changing the passwords. See Section 6.2 for the procedures.
There are three levels of password:
Level 1 ("controls1") is for the operator and must be known to access the system options, edit calibrations and customize test types;
Level 2 ("controls2") is for the technical personnel authorized by Controls and must be known to access the ADC and PID settings;
Level 3 is for the Controls service engineer.
The default level 1 password is "controls1"; for level 2 the default is "controls2".

NOTE:
If you lose or forget the new password it cannot be recovered. It is recommended that you keep a list of the passwords in a safe place.

5.3.2 Dictionary button

Opens a panel for editing the text used in the software. See Section 6.1 for more information and instructions on editing the language.

5.3.3 Language selection cell

For selecting the language to be used in the software.
The software comes with English and Italian languages. A custom language may also be entered – see Section 6.1.3.
After changing the language, the software must be restarted for the change to be effective.

5.3.4 Save parameters button

Saves the system 'data' folder (*C:\Program Files\MULTITEST\data*) and all its contents in a zip file so that they can be reloaded, if required, at a later date.
A standard Windows save file dialogue box will open to allow you to choose a location for the file.

5.3.5 Load parameters button

Loads in all the system files that were previously saved using the **Save parameters button**.
A standard Windows open file dialogue box will open for you to find the required file.

5.3.6 Digimax plus COM port

For Controls installation and service engineers only.
This cell is used for selecting the PC serial port that the Digimax Plus calibration device is connected to when carrying out a traceable calibration/verification on the frame channels (1 to 4).

5.3.7 Save button

Saves any changes made to the settings in the panel.

5.3.8 Close button

Closes the panel.
If you close the panel and have not saved any changes, you will be asked to confirm that you want to close without saving.

5.3.9 Measurement units

For selecting the units that will be used for force, strength, and deformation measurements. You can choose from:

- kN - MPa - mm
- Ton - kg/cm^2 - mm
- Lbf - psi - inch

IMPORTANT:
Remember to perform the following procedures, each time the measurement units will be changed. For a complete explanation refer to sections 5.4 and 6.3:

Fig. 5-3

1. Choose the measurement units from the list in the **MULTITEST System options panel**. Confirm and close the window;

2. Open **Calibration panel** (see section 5.4);
3. Double-click on the row of the channel that you want to work with;
4. Click **Edit calibration** button (**A**) and enter the level 1 password "*controls1*" to unlock settings;
5. Click the **Load calibration** button (**B**) to open previous calibrations performed on the current channel to allow one of them to be loaded into the panel. Calibration files can also be deleted in this panel;
6. Click **Save** button (**C**) or **Save as** button (**D**) to save the calibration loaded;
7. Click **Send to device** button (**E**) to send the calibration to the console;
8. Click **Edit calibration** button (**A**) to lock settings;
9. Click the **Close** button (**F**).

5.3.10 Date format

Choose the format of the date for the test.

5.3.11 KLAB/E settings

Allows the user to change the settings for the devices connected to the machine, if the module KLAB/E is purchased. Refer to chapter 6.7 for more details.

5.3.12 Modules activation menu

In case a test module is purchased after the installation of the Multitest console, use this panel to see "YOUR CODE" and then contact Controls SPA to receive the activation code needed.

5.3.13 Excel manual save

When using certain versions of MS Excel, an error such as the one in the next figure can appear when saving a .xls test report. If so, ticking this option allows the software to create the test report but not to save it, leaving this to the user. In this way, the error should not appear anymore.

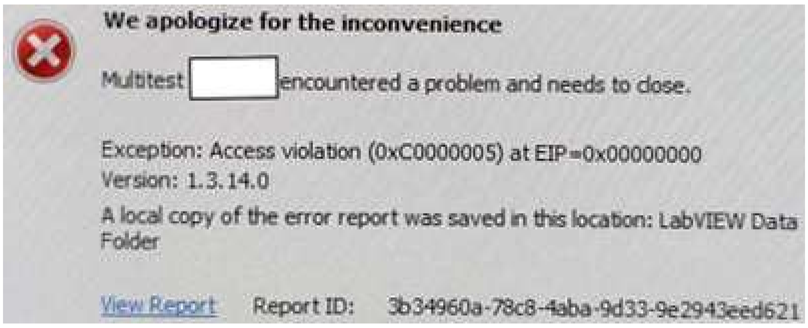


Fig. 5-4

5.4 MULTITEST Calibration panel

This panel displays calibration data in tabulated and graphical form, and contains functions for performing, viewing, editing, and reporting calibrations. See Fig. 5-5.

*To open the panel, click the **Channel configuration button** (Item 5.2.5) on the **MULTITEST Main panel** and then double-click on the row of the channel that you want to work with (see Fig. 5-7).*

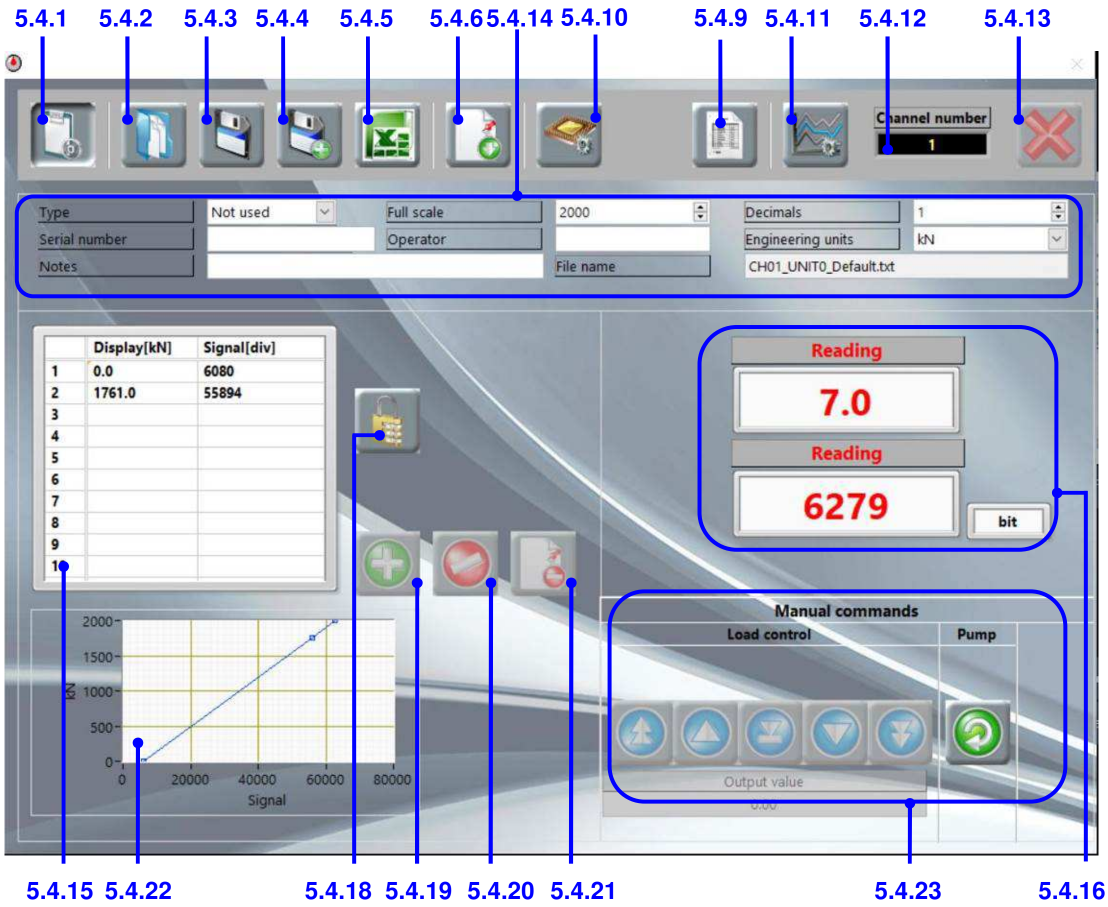


Fig. 5-5 Calibration unlocked

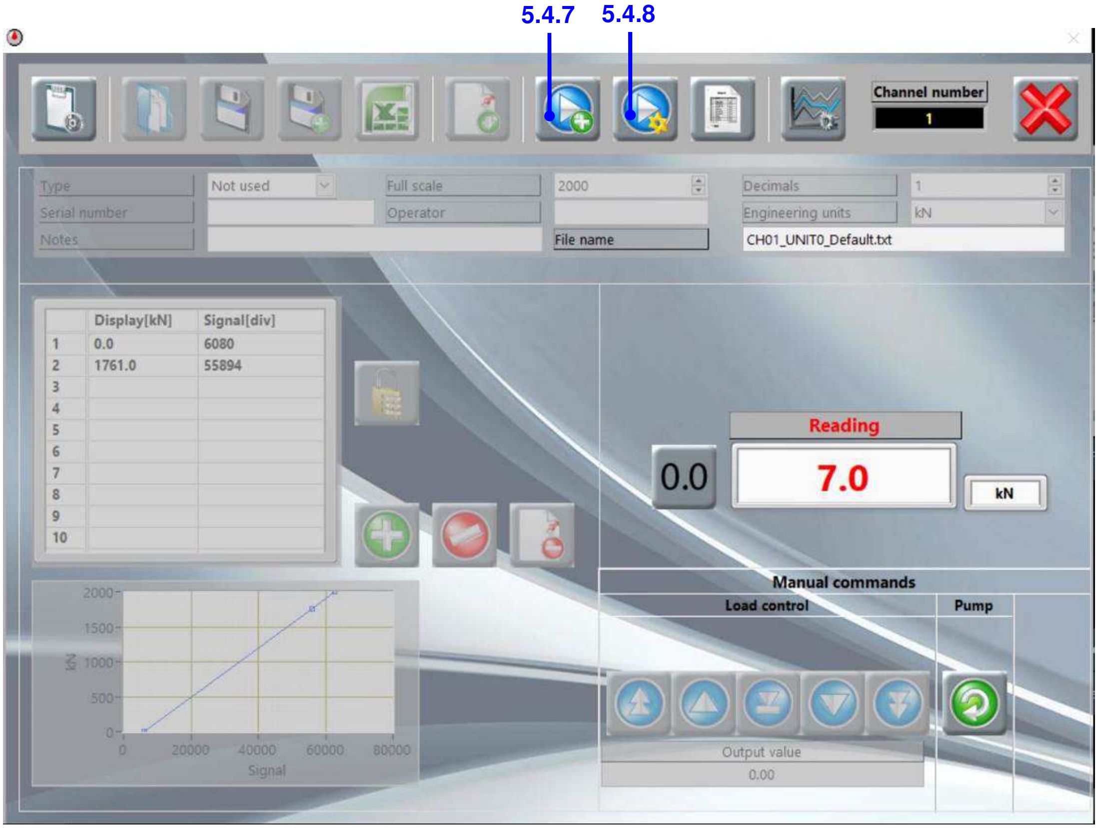


Fig. 5-6 Calibration locked

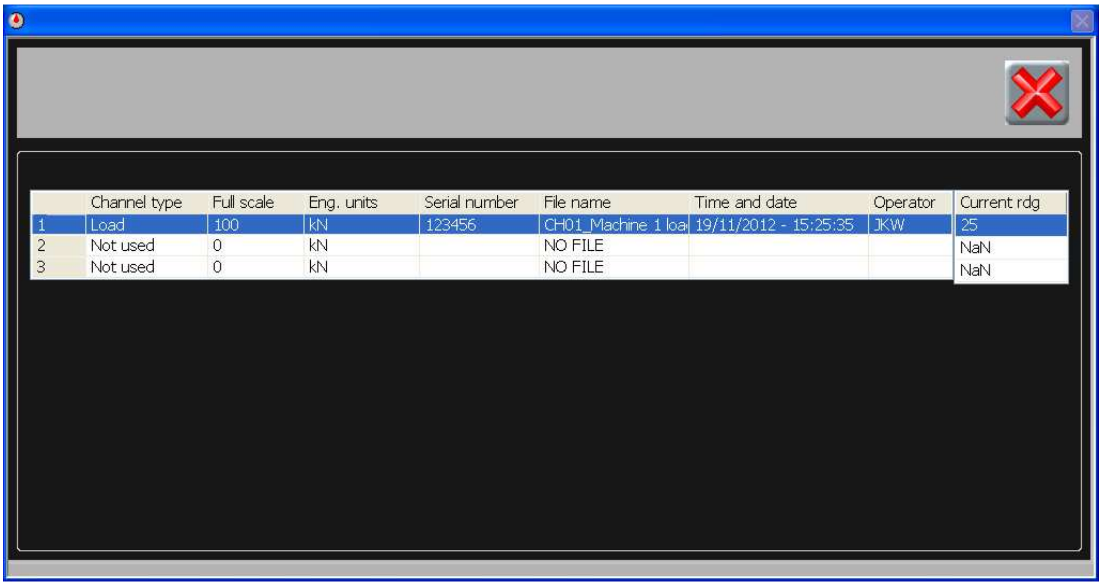


Fig. 5-7

5.4.1 Edit calibration button

Unlocks the panel to allow the calibration to be edited.
You will have to enter the level 1 password ("controls1") to gain access to the panel.
When the panel is unlocked, the button will appear depressed and the other buttons along the top of the panel will be disabled. To enable the other buttons, the ***Edit calibration button*** *must be clicked again to lock the sheet. The pump must be off before locking the sheet (see Item 5.4.23).*

5.4.2 Load calibration button

Opens a panel containing a list of all the calibrations performed on the current channel to allow a previous calibration to be loaded into the panel. Calibration files can also be deleted in this panel.

5.4.3 Save button

Saves the current calibration over the existing calibration.
You will be prompted to confirm the command before the file is overwritten. If you don't want to overwrite the existing calibration, use the ***Save as*** *button (Item 5.4.4) instead.*

5.4.4 Save as button

Saves the current calibration as a new calibration file.
You will be prompted to enter a filename for the calibration. The software automatically prefixes the file name you enter with the channel number (for example, channel 1 would be saved as 'CH01_yourfilename.txt').

NOTE:
The default location for the saved calibration files is C:\Users\...\Documents\MULTITEST\data\Channels.

5.4.5 Save to Excel button

Creates a report for the current calibration in Excel and automatically saves it as a workbook.
A standard Windows save file dialogue box will open to allow you to choose a location for the file. The report contains all the ***Transducer details*** *(Item 5.4.14) and the data from the* ***Calibration table*** *(5.4.15). An example report is included in the Appendix, Section 8.*

NOTE:
The system PC must have MS Excel installed and closed, to use the Save to Excel function.

5.4.6 Send to device button

Sends the calibration data to the device to ensure that the transducer readings are synchronized between the software and the device.

5.4.7 Manual verification button

For verifying a frame calibration one point at a time.
When this button is pressed, a panel appears (Fig. 5-8) where the required load (setpoint) can be entered.

Fig. 5-8

5.4.8 Automatic verification button

For verifying a frame calibration with a sequence of loading points automatically applied by the software.
When this button is pressed, a panel appears (Fig. 5-9) where the required loading sequence can be entered with a hold time at each point to allow the user to compare (and record, if required) the load reading in the software and the reference load readout.

Fig. 5-9

5.4.9 Traceable calibration button

For Controls installation and service engineers only.
This button is used for carrying out a traceable calibration/verification on channels 1 to 3 with a Digimax Plus calibration device.

5.4.10 Hardware calibration button

For use by Controls installation and service engineers only.
This button is for setting the zero and gain for the frame channels (1 to 4) when they are first calibrated. It is only available when the panel is unlocked for editing (see Item 5.4.1).

5.4.11 PID settings button

This function has the same function as Item 5.7.9 on the **Test settings panel** in the DataManager test module.

5.4.12 Channel number indicator

Displays the machine channel number of the calibration that is currently open.

5.4.13 Close button

Closes the panel.
This button is disabled when the sheet is unlocked for editing. Before closing the panel, it must be locked – see Item 5.4.1.
If you close the panel and have not saved any changes or sent the new calibration data to the device, you will be asked to confirm that you want to close without saving or sending.

5.4.14 Transducer details

For entering the following information about the transducer that is connected to the selected channel:

- Type – the type of transducer.

In this version of the software, you can only select load (load cell).

- Serial number - the manufacturer or supplier's serial number for the transducer.
- Full scale - the maximum capacity of the test frame.

This value should appear automatically according to the test frame that's connected.

- Operator - the name or initials of the person who performed the calibration.
- Decimals - the number of decimal places that you want to see displayed for the engineering units reading.
- Engineering units - the measurement units of the transducer.
- Notes - any notes or comments you may have about the calibration or the transducer.
- Filename – this cell displays the filename of the existing calibration.

5.4.15 Calibration table

Displays the following calibration data:

- Column 1 - Calibration point number.
- Column 2 - Reference data ('Display') – the reading entered for the reference equipment at each calibration point, in engineering units.
- Column 3 - Measured data ('Signal') – the measured transducer reading, in bits, that was recorded at each calibration point.

5.4.16 Transducer reading

Displays the transducer reading in bits and engineering units.
When the panel is locked, only the reading in engineering units is displayed.

5.4.17 Zero button

See Fig. 5-10. Applies a zero offset to the transducer reading in engineering units to negate the effect of friction and weight of the piston, spacers, and load cell.
The button is only available when the panel is locked.

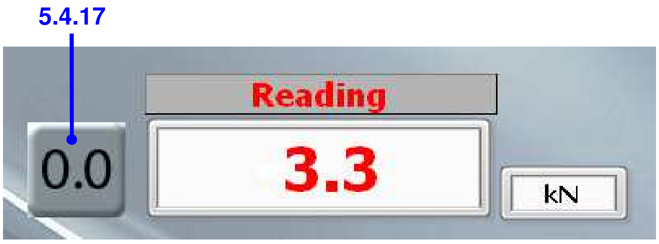


Fig. 5-10

5.4.18 Unlock table button

Unlocks the calibration table so that data can be added or deleted.

5.4.19 Add point button

Displays a cell for entering the reference reading and a button to acquire the transducer reading and add both values to the calibration table.

5.4.20 Delete point button

Deletes the selected point from the calibration table.

5.4.21 Clear table button

Clears all the data from the calibration table.

5.4.22 Calibration graph

Displays the measured transducer data from column 3 of the **Calibration table** as abscissa (x) with the reference data from column 2 as ordinates (y).
*The final point on the graph is plotted using the **Full scale** of the transducer as the ordinate, with the abscissa value extrapolated using the gain and offset calculated from the calibration data.*

5.4.23 Pump controls

For controlling the loading speed of the test machine so that the calibration points can be set. Available only for channels 1-4.

Pump on / off button

Increase speed fast button
Increases the pump speed by 1%

Increase speed slow button
Increases the pump speed by 0.1%

Half-speed button
Halves the pump speed

Decrease speed slow button
Decreases the pump speed by 0.1%

Decrease speed fast button
Decreases the pump speed by 1%

The **Output value** beneath the buttons displays the speed of the machine as a percentage of the maximum machine speed.

5.4.24 Acquire point button

Captures the current reading of the transducer and enters it in the next blank row of the **Calibration table** along with the reference reading entered in the **Reference reading cell**.

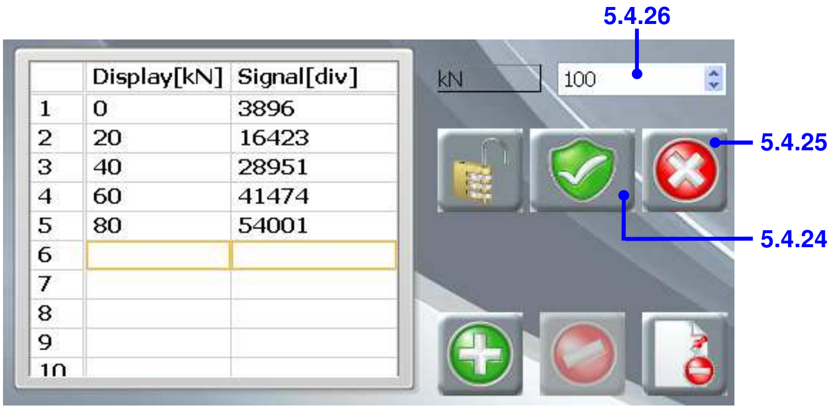


Fig. 5-11

The following items are only displayed after the ***Add point button*** *has been clicked. See Fig. 5-11.*

5.4.25 Cancel point button

Cancels acquisition of the calibration point.

5.4.26 Reference reading cell

For entering the reading on the reference transducer at the current calibration point.

5.5 DataManager Main panel

This panel contains buttons for opening the results database, saving and creating reports of the current test, customizing test types, saving test parameters, and uploading stored tests. See Fig. 5-13 and Fig. 5-15. Also on the panel are tabs for viewing live test results (the **Graph window**, Section 5.6) and entering test information (the **Report window**, Section 5.8).

The panel is always visible when the DataManager test module is running. (To run the test module, click on the DataManager icon in the ***Test modules*** *area of the* ***MULTITEST Main panel*** *(see Item 5.2.6 and Fig. 5-12).)*

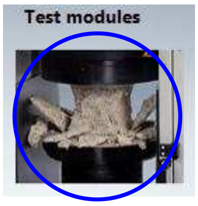


Fig. 5-12

5.5.1 Results button

Opens the **Results panel** for managing, reviewing, and printing previous tests.

5.5.2 Save test button

Saves the current test to the database.

This button is not available if the ***Autosaves*** *in the* ***Results panel*** *is ticked.*

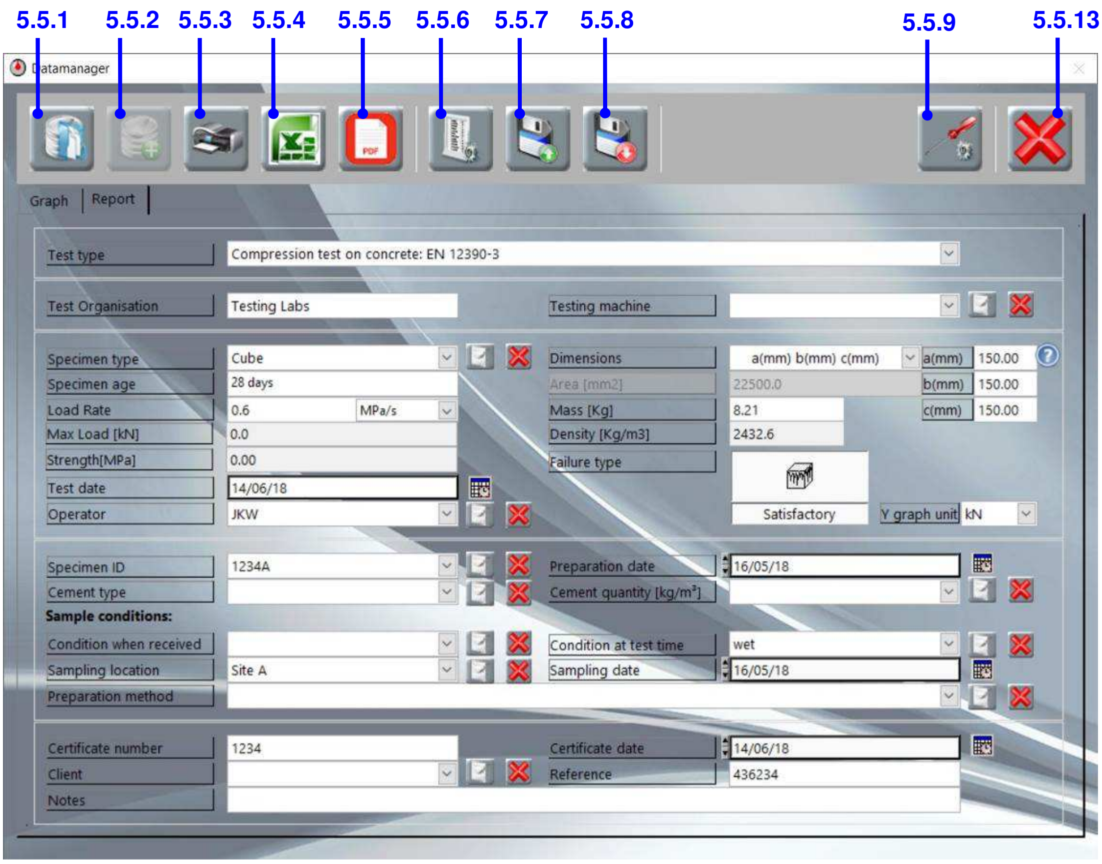


Fig. 5-13

5.5.3 Print button

Prints a test report for the current test on the default printer.
The report presents the test information from the ***Report window*** *and a copy of the graph on the* ***Graph window****. An example report is included in the Appendix, Section 8.*

5.5.4 Save to Excel button

Creates a report for the current test in Excel and automatically saves it as a workbook.
You will be prompted to choose a name and location for the file before it is created.
The report presents the test information from the ***Report window*** *and a copy of the graph on the* ***Graph window****. The second sheet of the Excel workbook contains the graph data, with time (in seconds) in the first column and strength (in megapascals) or load (in kilonewtons) in the second column. An example report is included in the Appendix, Section 8.*

NOTE:
The system PC must have MS Excel installed and closed, to use the Save to Excel function.

5.5.5 Save PDF report button

Creates a report for the current test in PDF format automatically saved in the *Documents\Controls* folder. The file name is automatically assigned as "Certificate number – Specimen ID".
The PDF report presents the test information from the Report window and a copy of the graph on the Graph window.
If enabled, a digital signature will be added to the report.
Remember to save in *Documents\Controls\MULTITEST\Operator signatures* the digital signature file in PNG format, before the operator creates the PDF report.
If enabled, the calculated loading rate will be printed instead of the set one.
The second sheet of the report contains the graph data with an acquisition interval of one second.

5.5.6 Customize button

Opens a panel for editing, adding, or deleting test types.
The level 1 password ("controls1") must be entered to access these functions.
This button is only available when the ***Report window*** *is being viewed.*

5.5.7 Save report data button

Saves all the entries from the **Report window** into a text file that can be loaded back in when needed using the **Load report data button**.
*This button is only available when the **Report window** is being viewed.*

5.5.8 Load report data button

Loads data that has been stored in a text file into the cells of the **Report window**.
*A text file can be created after entering information in the cells of the **Report window** using the **Save report data button**. A file can also be created manually using a program such as Excel or Notepad. This could be useful where, for example, a file is created when the samples are received in the laboratory so most of the information can be loaded in before the start of the test without the operator knowing all the information about the client or the history of the sample.*
*This button is only available when the **Report window** is being viewed.*

5.5.9 Settings

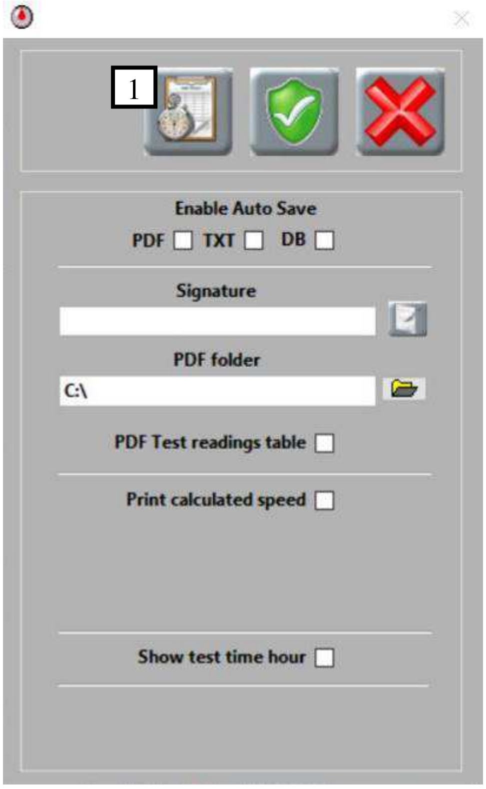


Fig. 5-14

This menu allows the management of automatic saving and test reports in PDF format.

- Graph buffer and Sampling rate button (1): allows the choice of the graph buffer and sampling rate.
- AutoSave PDF: enables the automatic saving of each test in PDF format. A report of the current test is created in PDF format and automatically saved in the *Documents\Controls* folder. The file name is automatically assigned as "Certificate number - specimen ID". When this function is not enabled, the PDF button must be clicked at the end of each test.
- AutoSave TXT: enables automatic saving of the last test in TXT format, which is saved in a file called "last_test.txt" which is automatically overwritten each time.
- AutoSave DB: enables automatic saving of the test in the test archive (database). When this function is not enabled, the SAVE button must be clicked at the end of each test.

NOTE:
If you are carrying out a 'Compression test on concrete' and you want to be able to select the failure type after the test has finished, do not tick the **AUTO Save DB** (it automatically saves the test in the archive of the software folder) and/or **AUTO Save TXT** (it automatically saves a txt file named "last_test.txt" in which the last test will be overwritten each time). This will allow you to select the failure and then save the test using the **Save test button** (Item 5.5.2).

- Signature: allows the selection of an electronic signature that will be automatically added to the PDF report. The digital signature file must be in PNG format. Files containing electronic signatures to be saved in the folder named *Documents\Controls\MULTITEST\Operator signatures*.
- PDF Folder: allows the selection of the destination folder of PDF reports.
- Print Calculated Speed: Allows the automatic calculation of the loading rate applied during the test, in a selectable time interval (normally 20 - 80% of the test duration), which will be printed on the PDF report.
- Show test time hour: when enabled, it allows you to set the test execution hour in addition to the date. This information is also presented on the print-out report.

5.5.10 Cement test button

See Fig. 5-15. Starts / stops a cement test.
When starting a test, a message will be shown to allow you to select whether to resume an incomplete cement test or start a new one.
After a test has started, the image and function of this button will change for stopping the test.
*This button will only be displayed when a cement test has been selected for the **Test type control** (Item 5.8.1).*

5.5.11 Cement test summary button

See Fig. 5-15. Opens a panel containing summary data for the completed cement test, with a button to create an Excel report of the test (see Fig. 5-16).
This button is only available after a complete sequence of nine tests has finished.

NOTE:
The system PC must have MS Excel installed and closed, to use the Save to Excel function.

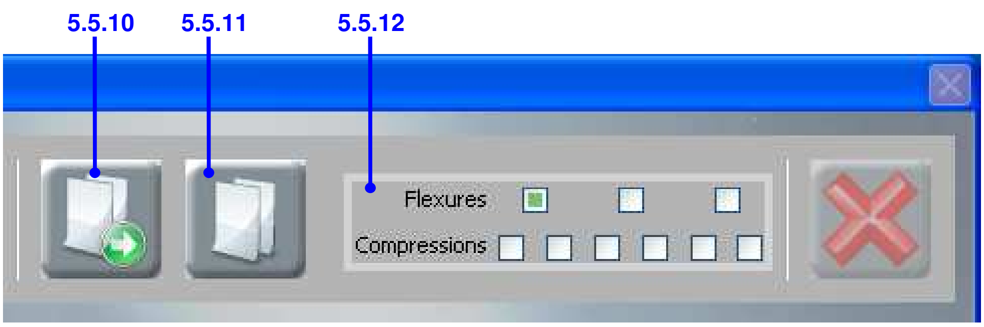


Fig. 5-15

5.5.12 Cement test status indicator

See Fig. 5-15. Indicates which stages of the current cement test have been completed, and which one will be performed next.
*This indicator will only be displayed when a cement test has been selected for the **Test type control** (Item 5.8.1).*

A tick in the box indicates that the test has been completed;
A green square in the box indicates that the test is the next to be performed;
A white box shows that the test is yet to be performed.

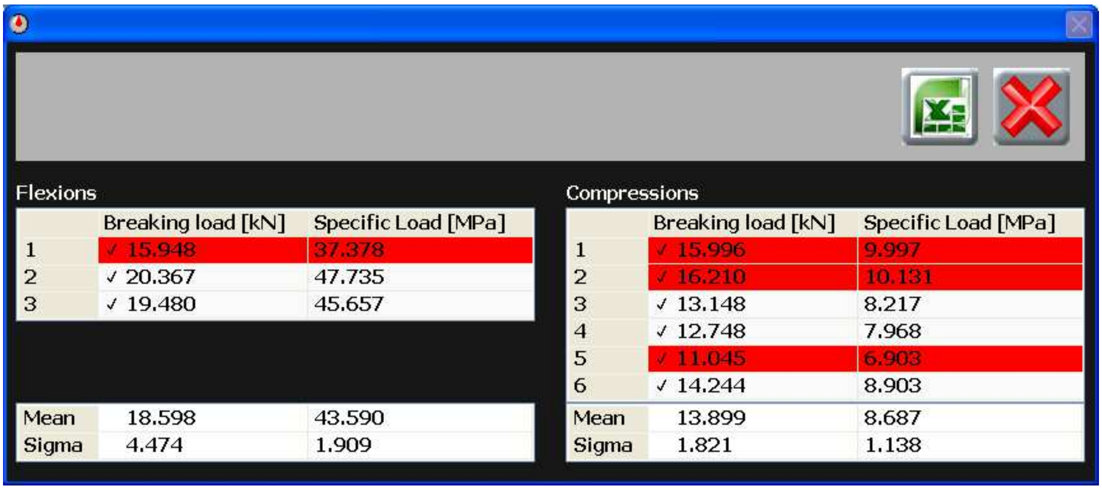


Fig. 5-16

5.5.13 Close button

Closes the DataManager test module.

5.6 DataManager Graph window

This window displays live test data. See Fig. 5-17.
To view the window, click the 'Graph' tab.

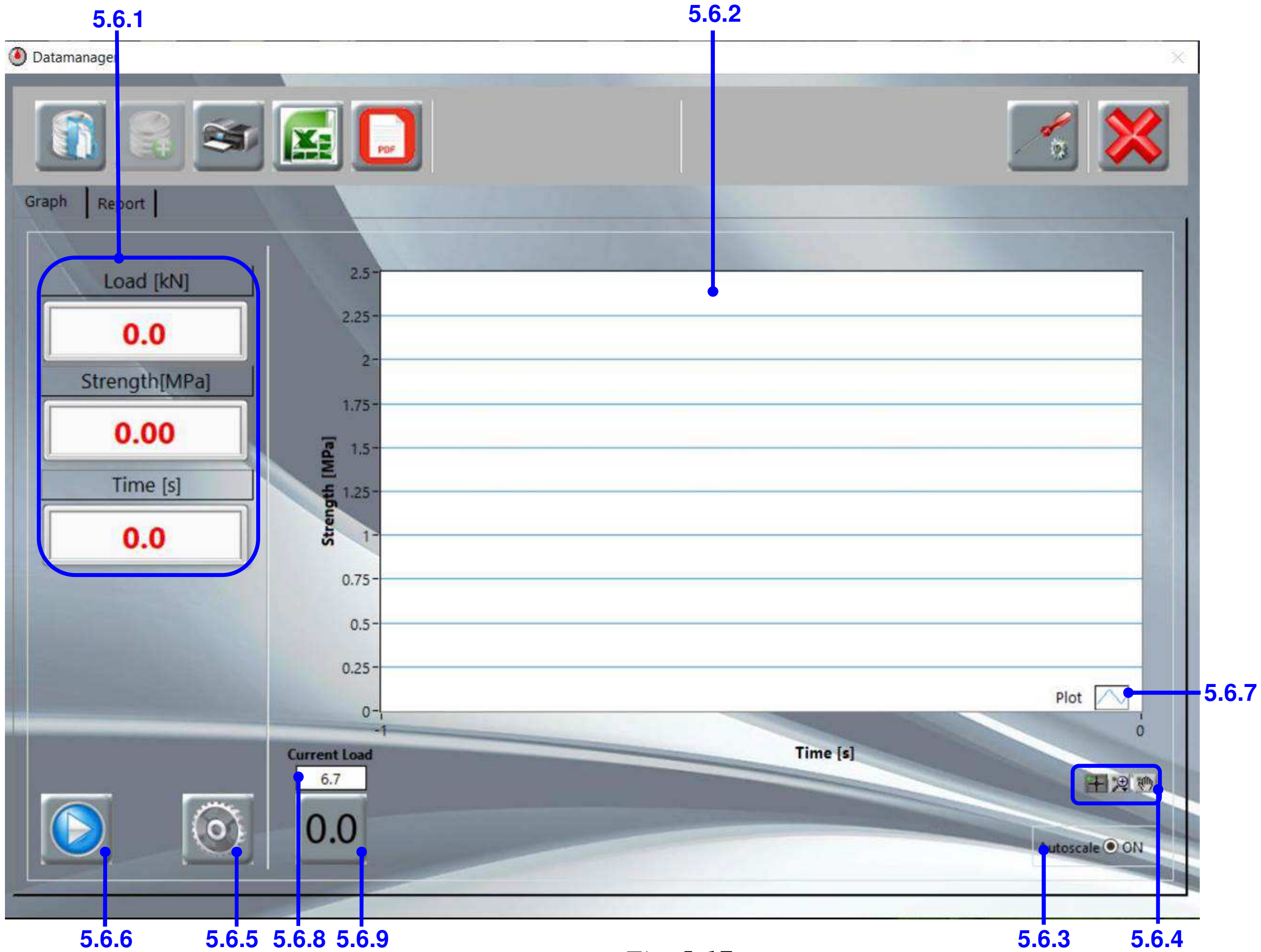


Fig. 5-17

5.6.1 Readings

Displays real-time values of:
Load - the load cell reading, in kilonewtons;
Strength - the calculated strength of the test sample, in megapascals;
Time – the elapsed time of the test, in seconds.

5.6.2 Graph

Displays a live plot of time against strength or load.
You can select which value is plotted on the y-axis in the ***Test settings panel*** *(Item 5.7.7) or the* ***Report window*** *(Item 5.8.2).*

5.6.3 Autoscale option

Turns the autoscaling of both graph axes on or off.
This function is only active when a test is running.

5.6.4 Graph tools

Three tools are available for manipulating what can be seen on the graph:

 Default – for viewing the graph normally with zoom and panning off

 Zoom –for zooming in and out on the graph or a selected area of the graph

 Panning – drags the contents of the graph around within the axes

5.6.5 Test settings button

Opens the **Test settings panel** for setting the y-axis units and setting the peak sensitivity and peak load. See Section 5.7.

5.6.6 Start test button

When the machine is connected the software tells the device to start the test and starts logging data as soon as the preload* is reached. The pump is started automatically.
After the test has started, it can be stopped manually using the same button.

*The preload value is specified in the PID settings and should be entered depending on the type of test (see Section 7.3).

5.6.7 Plot option

Click to modify the properties of the graph, for example, line color, line thickness, type of line, etc.

5.6.8 Current reading

Shows the current load if the selected channel is a load cell.

5.6.9 Zero button

Click to set the load to zero.

5.7 DataManager Test settings panel

It contains functions for selecting the channel number for testing, setting PID values, and entering parameters for automatically stopping a test. See Fig. 5-18.
*To open the panel, click the **Test settings button** (Item 5.6.5) on the **Graph window**.*

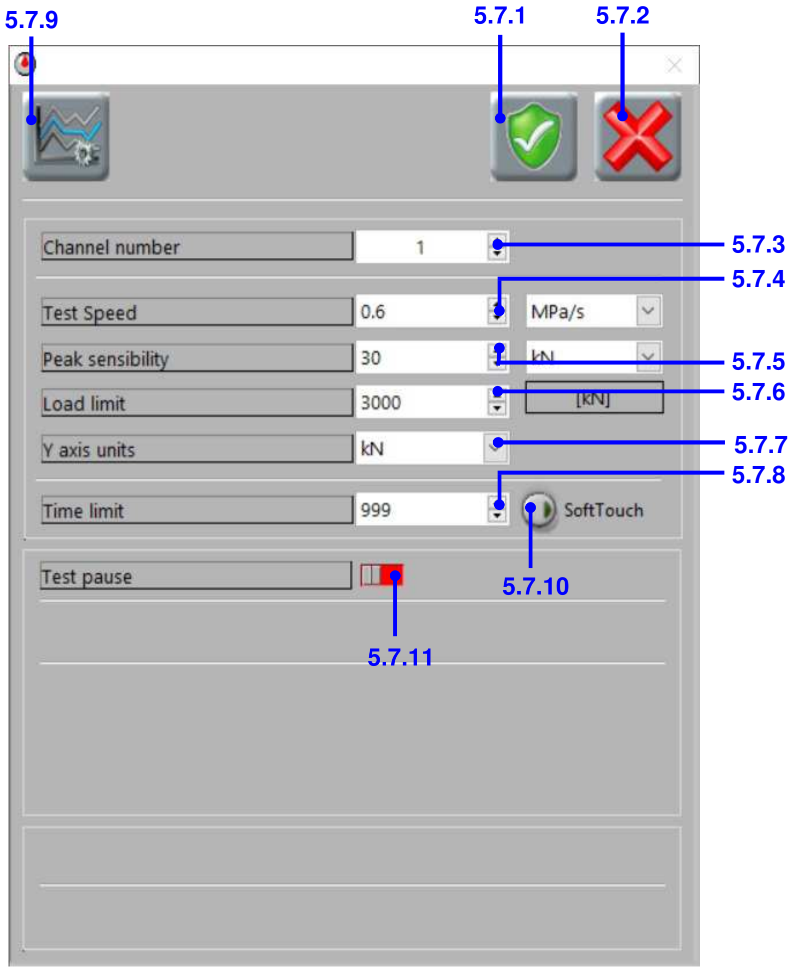


Fig. 5-18

5.7.1 Save button

Saves any changes made to the settings and closes the panel.

5.7.2 Close button

Closes the panel.

5.7.3 Channel number

For selecting the channel number of the testing frame that you want to use.
Channels 1 to 3 can be selected. This active frame selection is only available when using an AUTOMAX Multitest control unit.

5.7.4 Test speed

For entering the load rate for the test, in newtons per second or megapascals per second.
Select the units required from the drop-down list adjacent to the control.
*This control has the same function as the load rate cell in the **Report window** (see Item 5.8.2). If you change the value of this control, the value in the **Report window** will be automatically updated, and vice versa.*

5.7.5 Peak sensitivity

For entering the value, in kilonewtons, that corresponds to the decrease in load during the test at which the machine will consider the sample to have failed, and will stop the test.
The recommended values are 10 to 20 kN for compression tests on concrete and indirect tensile tests, 1 to 3 kN for compression tests on cement, and flexure tests on concrete and 0.2 to 0.4 kN for flexure tests on cement. As a general rule, the value should be proportional to and always lower than the failure load.
The peak sensitivity can also be expressed as a percentage of the maximum load reached during the test execution. For example, by setting sensitivity of 90%, the test will be automatically stopped after specimen failure when the load drops below 90% of the peak. ASTM C39 requires a peak sensitivity of 95% for all tests.

NOTE:
The peak sensitivity value should be decreased to 5kN (in any case not more than 10kN) when testing high-performance concrete, to avoid the possibility of explosive failures. ASTM C39 does not allow this kind of reduction – 95% is mandatory.

5.7.6 Load limit

For entering a maximum load for the machine or a required fixed load for the specimen, in kilonewtons, which you do not want to be exceeded during the test. If this load is reached the test will be stopped.

5.7.7 Y-axis units

For entering the units to use for the y-axis of the test graph (Item 5.6.2).
Click on the arrow to the right of the select to select from kN or MPa (lbf or psi).
*This control has the same function as the y-axis cell in the **Report window** (see Item 5.8.2). If you change the value of this control, the value in the **Report window** will be automatically updated, and vice versa. The change will be effective when the next test is started.*

5.7.8 Time limit

For entering the maximum number of seconds that you want the test to last. If this time limit is reached before sample failure is achieved or any other limit is reached, the test will be stopped.

5.7.9 PID settings button

Opens a panel for setting the control parameters for each machine connected to the control unit, according to the machine type (compression, flexural, or indirect tensile). See Fig. 5-19.
The level 2 password ("controls2") must be entered to access the panel. The values should only be changed by technical personnel authorized and trained by Controls. The values to be used are given in the instruction manual of the machine.

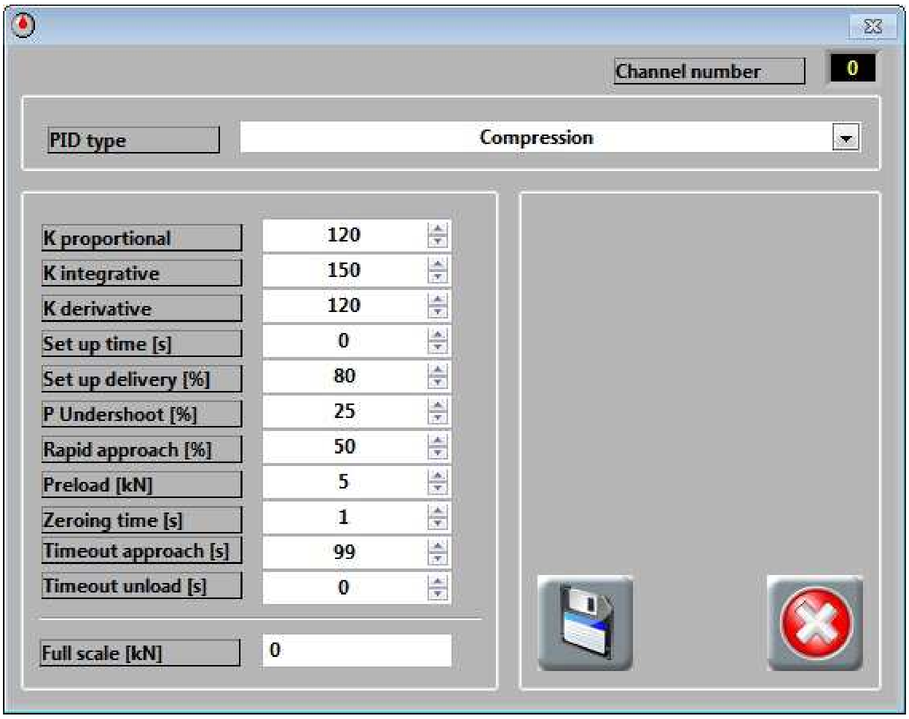


Fig. 5-19

5.7.10 Soft-touch function

Enable this function to lower the pre-load limit and reduce the approach speed. It is activated for low strength tests and low-speed tests.

5.7.11 Test pause (required for ASTM C39)

It is possible to activate this function using the dedicated switch (red = pause disabled; green = pause enabled).

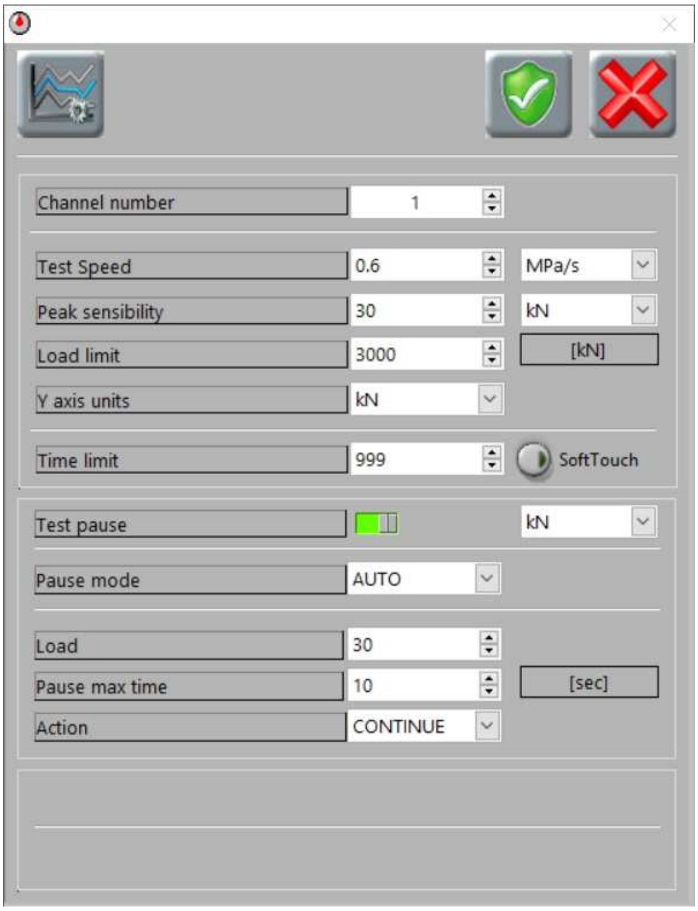


Fig. 5-20

- PAUSE MODE: choose between *AUTO* or *MAN*: test pause can be *MANUAL* or *AUTOMATIC*. In the first case, during the execution of the test, the operator can activate a pause by pushing the button "*PAUSE*".
- LOAD: In the second case, the operator can be set the load or stress at which the machine will pause.

  ATTENTION: it is not possible to set a load for pause less than pre-load.

- PAUSE MAX TIME: to set max test time of the pause in seconds.
- ACTION: to set the action that the machine will be started after the pause phase. It is possible to select between "CONTINUE" or "STOP".

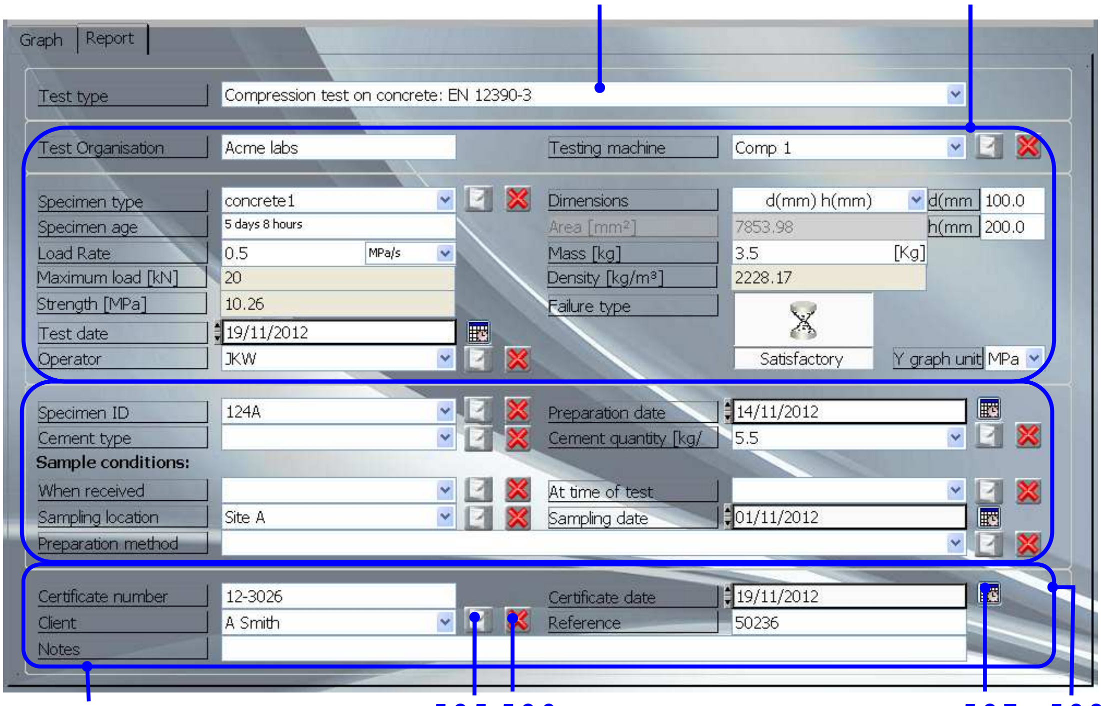
5.7.12 Dual Load Rate

This function is only available for the compression test according to **ASTM C39**.
It allows us to get a loading rate higher by a factor set by the user, up to 50% of the estimated failure load of the specimen under test. Once this function is triggered, you must enter the estimated max load/stress value for your cylinder in the report page, so the increased load rate will be applied to 50% of this value.
ATTENTION: if the PAUSE function is active, the DUAL LOAD RATE function will only activate at the end of the pause.

5.8 DataManager Report window

This window is for entering information about the test and the sample. See Fig. 5-21
Many of the parameters, including the test type, can be customized. See Section 6.3 for instructions. To view the window, click the 'Report' tab.

5.8.1 5.8.2



5.8.4 5.8.5 5.8.6 5.8.7 5.8.3

Fig. 5-21

5.8.1 Test type control

For selecting which type of test is being carried out. The standard options offered with the software are as follows:

- Flexural tests on beams to EN 12390-5 testing standard
- Flexural tests on curbs to EN 1340 testing standard
- Indirect tensile tests on concrete to EN 12390-6
- Indirect tensile tests on block pavers to EN 1338
- Compression tests on concrete to EN 12390-3
- Compression tests on blocks to EN 772-1
- Compression tests on elements to EN 1917
- Cement tests to EN 196-1
- Compression test on concrete to ASTM C39
- ACV test to BS 812-110

Customized test types can also be created – see Section 6.3 for instructions.
To change the test type, click the arrow to the right of the cell, and select an option from the drop-down list. Some of the test parameters will change depending on what test type is selected.

5.8.2 Test parameters

For entering information about the test sample and how the test will be carried out.
Some of the test parameters will change depending on what test type is selected. The parameters are as follows:

- Test organization – the name of the institute carrying out the test.
- Testing machine* – the name of the machine on which the test is being performed.
- Specimen type* – a description of the specimen shape (e.g. cylinder, core, beam, cube)
- Specimen age – the length of time since the sample was made.

This can be entered in any format – however, you want it to appear on the test report e.g. '2 days 8 hours'.

- Load rate – the rate that the sample will be loaded at, in newtons per second or megapascals per second.

Select the units required from the drop-down list adjacent to the cell.
*This cell has the same function as the **Test speed control** (Item 5.7.4) in the **Test settings panel**. If you change the value and/or units in this cell, the **Test speed control** will be automatically updated, and vice versa.*

- Maximum load – the maximum load measured during the current test, in kilonewtons.
- Strength – the strength of the specimen currently being tested, in megapascals, calculated as follows:

$\sigma = P / A$
where:
σ *= strength, MPa (psi)*
P *= maximum load, N (lbf)*
A *= specimen area, mm2 (inch2)*

- Test date – the date that the test is being performed.

*The date can be picked using the **Calendar button**.*

- Operator* – the name of the person carrying out the test.
- Dimensions – for selecting which specimen dimensions will be entered, according to its shape. See Table 5-1.
- Area – the calculated area of the specimen. See Table 5-1 for the calculations for different specimen shapes and test types.

*This cell will not be available if the 'Compression test on elements' is selected for the **Test type control**.*

- Mass – the measured mass of the specimen.
- Density – the bulk density of the specimen.

*This cell is only available when the 'Compression test on concrete' is selected for the **Test type control**.*

- Only for ACV test:
  - M1 Mass
  - M2 Mass
  - Aggregate crushing value
  - A=nominal aggregate size
  - D=piston diameter
  - P=passing
  - R=retained
- Estimated ultimate load (only for **ASTM C39**, when DUAL LOAD RATE is enabled).
- No of upper rollers – the number of upper rollers on the beam testing jig.

*This cell is only available when 'Flexural tests on beams' is selected for the **Test type control**.*

- 'L' distance –the distance between the lower rollers.

*This cell is only available when 'Flexural tests on beams' is selected for the **Test type control**.*

- Failure type – for selecting the failure shape of the specimen after the test.

*This cell is only available when the 'Compression test on concrete' is selected for the **Test type control**. A standard selection of satisfactory and unsatisfactory failure types are provided, for cylindrical and cubic specimens. To select a failure type, click on the failure image and select one from the images shown.*

- K1 factor – the form factor co-efficient (refer to the relevant standard).

*This cell is only available when 'Indirect tensile test on paver blocks', 'Compression test on elements', or 'Compression test on blocks' is selected for the **Test type control**.*

- K2 factor – the conditioning factor co-efficient (refer to the relevant standard).

*This cell is only available when the 'Compression test on blocks' is selected for the **Test type control**.*

- Y-axis unit - for selecting the units to use for the y-axis of the test graph (Item 5.6.2).

Click on the arrow to the right of the select to select from kN or MPa (lbf or psi).
*This control has the same function as the **y-axis unit control** (Item 5.7.7) in the **Test settings panel**. If you change the value in this cell, the **y-axis units control** will be automatically updated, and vice versa. The change will be effective when the next test is started.*

5.8.3 Sample information

- Specimen ID* – the reference number given to the sample to identify it.
- Cement type* – a description of the type of cement being tested
- Preparation date – the date that the sample was made.

*The date can be picked using the **Calendar button**.*

- Cement quantity* – the volume of cement used to prepare the sample, in kilograms per cubic meter.
- Condition when received* – the condition of the sample when it was received by the test organization.
- Sampling location* – the name of the location or site where the sample was taken.
- Preparation method* - a description of the method that was used to prepare the sample, including the test standard, if relevant.
- Condition at the time of test* - the condition of the sample immediately before the test is started.
- Sampling date – the date that the sample was taken

*The date can be picked using the **Calendar button**.*

5.8.4 Report information

- Certificate number –the entry must be made in this field or the test will not save

*When 'Cement test' is selected for the **Test type control**, this field is locked – when a new test is started a window will appear where the certificate number can be entered.*

- Client* – the name of the client who has requested the tests
- Certificate date – the date that the certificate was issued
- Reference – the reference number that has been assigned to the project by the test organization
- Notes – any notes or comments that it is useful to have recorded for the test

** These entries can only be selected from the drop-down lists which are opened by clicking the arrow to the right of the cells. The contents of the lists can be edited using the **Add entry buttons** and **Delete entry buttons** (Items 5.8.5 and 5.8.6).*

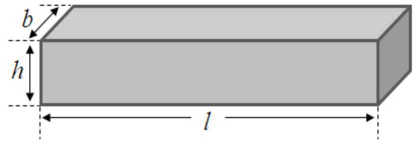
5.8.5 Add entry button

Opens a window allowing a new entry to be added to the drop-down list of the adjacent cell. See Section 6.4.4 for more information and instructions on adding new entries.

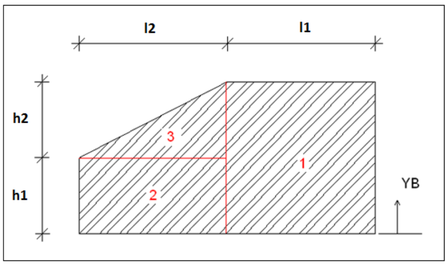
5.8.6 Delete entry button

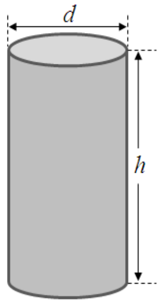
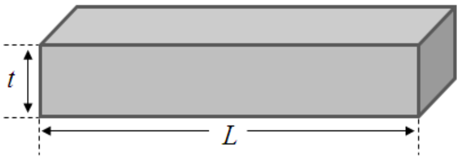
Deletes the entry currently displayed from the drop-down list of the adjacent cell. See Section 6.4.5 for more information and instructions on deleting entries.

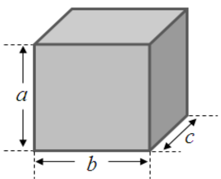
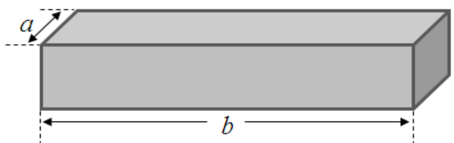
NOTE:
During a test, it is possible to modify all the descriptive fields in the **Report window**. However, when the test is finished, if the **Autosaves** are not ticked, the only field that can be changed is the Failure type (in the case of a 'Compression test on concrete'). It is possible to check **AUTO Save DB** (it automatically saves the test in the archive of the software folder) and/or **AUTO Save TXT** (it automatically saves a txt file named "last_test.txt" in which the last test will be overwritten each time).
If any other fields are changed after the test, the **Save test button** will be disabled and it will not be possible to save the test.

<table>
<tr><th>Test type</th><th>Dimensions</th><th>Diagram</th><th>Area (A)</th></tr>
<tr><td>Flexural tests on beams</td><td rowspan="2">b, h, l</td><td rowspan="2">
</td><td rowspan="2">$A = \frac{2*b*h^2}{3*L}$ 3 points

$A = \frac{b*h^2}{L}$ 4 points

where:
L = distance between rollers (see Item 5.8.2)</td></tr>
<tr><td>Cement tests</td></tr>
<tr><td>Flexural tests on curbs</td><td>h1, h2, l1, l2</td><td>
</td><td>$A = \frac{4 * ITOT}{L * YB}$

where:
ITOT and YB are calculated by the software. See Appendix, Section 8
L = distance between rollers (see Item 5.8.2)</td></tr>
<tr><td>Indirect Tensile tests on concrete</td><td rowspan="2">d, h</td><td rowspan="2">
</td><td>$A = \frac{\pi*h*d}{2}$</td></tr>
<tr><td>Compression tests on concrete</td><td>$A = \pi*(d/2)^2$</td></tr>
<tr><td>Indirect tensile tests on block pavers</td><td>t, L</td><td>
</td><td>$A = \frac{\pi*L*t}{2*k1}$

where:
k1 = form factor (see Item 5.8.2)</td></tr>
<tr><td>Compression tests on concrete</td><td>a, b, c</td><td>
</td><td>A = a*b</td></tr>
<tr><td>Compression tests on blocks</td><td>a, b</td><td>
</td><td>A = a*b

$\sigma = \frac{P}{A} *k1*k2$

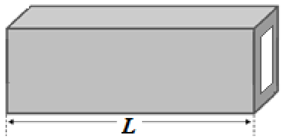
where:
σ = strain
P = load
k1 = form factor
k2 = conditioning factor
(see Item 5.8.2)</td></tr>
<tr><td>Compression tests on elements</td><td>L</td><td>
</td><td>N/A</td></tr>
</table>

Table 5-1

5.8.7 Calendar button

Displays a calendar from which the required date can be selected and entered into the adjacent cell. See Fig. 5-22.

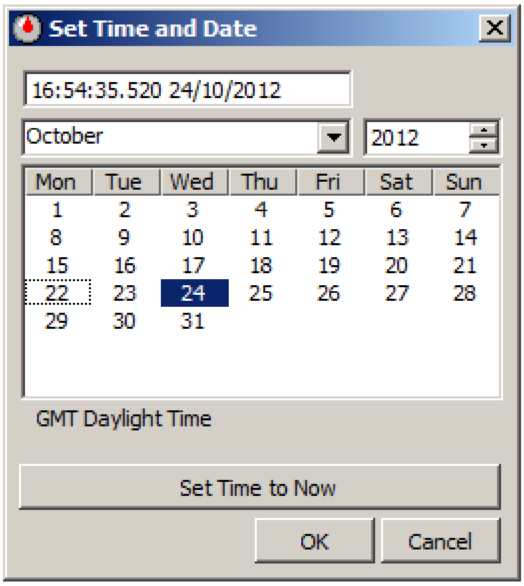


Fig. 5-22

5.9 DataManager Results panel

This panel is for reviewing previous test results and creating reports of single tests or batches of tests. See Fig. 5-23. The tests are saved using an MS Access database.
To open the panel, click the ***Results button*** *(Item 5.5.1) on the* ***DataManager Main panel****.*

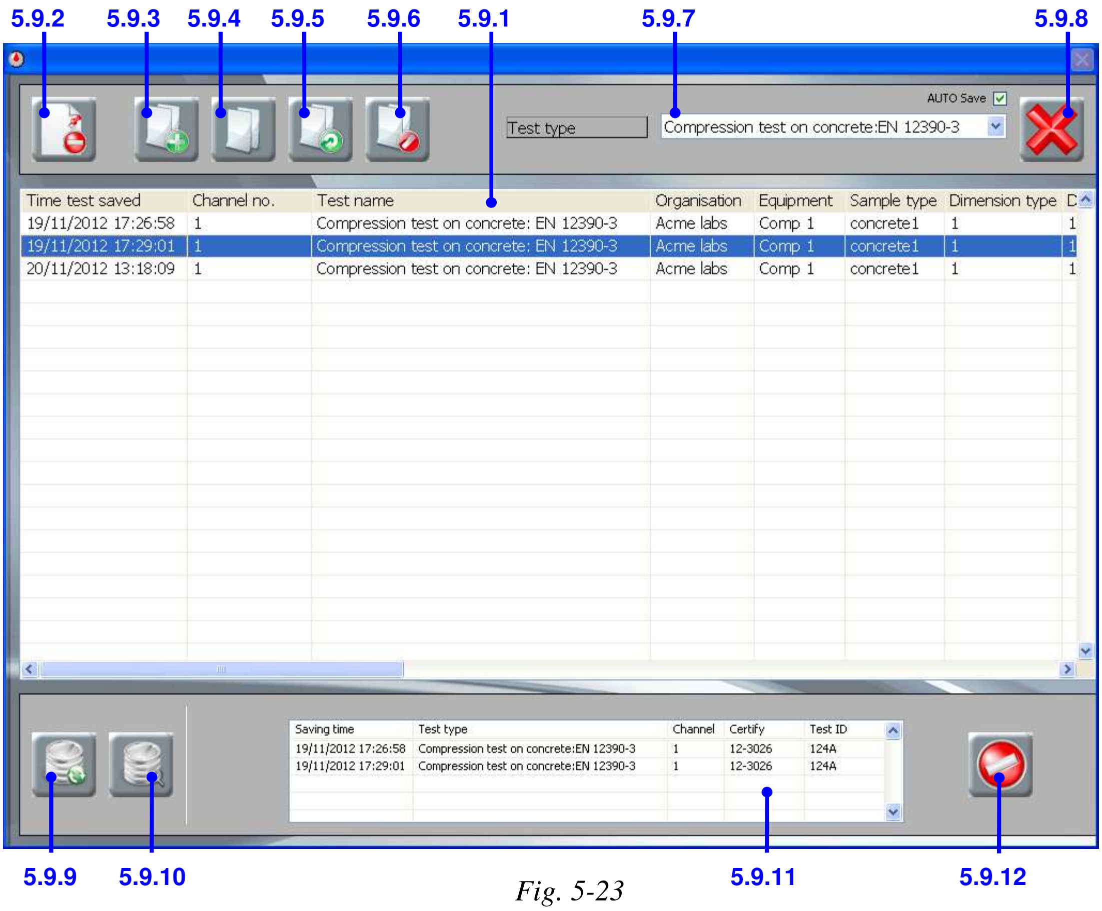


Fig. 5-23

5.9.1 Results table

It contains a list of all the saved tests of the selected test type (Item 5.9.7). Each row represents one test and contains information from the **Report window** as well as the test results.
Use the scroll bars to move vertically and horizontally through the list of records. To open a single test and display it in the ***Review panel****, double-click anywhere on the row.*

5.9.2 Delete record button

Permanently deletes the test in the selected row. See Section 6.6.1 for more information and instructions on deleting test records.

5.9.3 Add record to batch button

Adds the test in the selected row to the **Batch table** (Item 5.9.11).

5.9.4 Create new batch button

Creates a new batch file containing all the tests in the **Batch table** (Item 5.9.11). See Section 6.6.3 for more information and instructions on how to create a batch file.

5.9.5 Open batch button

Opens a window showing all the batch files to allow a file to be selected and displayed in the **Review panel** (Section 5.10). See Section 6.6.6 for more information and instructions on opening a batch file.

5.9.6 Delete batch button

Opens a window showing all the batch files to allow a file to be selected and deleted. See Section 6.6.4 for more information and instructions on deleting a batch file.

5.9.7 Test type control

For selecting which type of test is listed in the **Results table**.
To select a test type, click the arrow to the right of the cell and pick an option from the drop-down list. The contents of the list are the same as those in the ***Test type control*** *(Item 5.8.1) in the* ***Report window****.*

5.9.8 Close button

Closes the **Results panel**.

5.9.9 View all button

Clears the search criteria and re-populates the **Results table** with all the test records.

5.9.10 Search button

Opens a window allowing a search to be carried out to find all saved records that match the search criteria and display them in the **Results table**. The search criteria correspond to the data columns of the table. Up to two criteria may be entered and you can specify whether the records must match both criteria or only one of them. The logic of the search function works in the following way:
Criteria 1 AND Criteria 2 – only tests matching both criteria will be listed;
Criteria 1 OR criteria 2 – any tests matching any of the criteria will be listed;
Criteria 1 [..] – any tests between the two parameters will be listed;
Criteria 1 ---- - only tests matching criteria 1 will be listed.
See Section 6.6.2 for more information and instructions on searching for tests.

5.9.11 Batch table

It contains the test records that have been selected to be added to a batch file using the **Add record to batch button** (Item 5.9.3).
Records are copied from the ***Results table*** *into this table - they are not moved.*

5.9.12 Clear table button

Clears all the records from the **Batch table**.
The actual test records are not deleted, only the contents of the batch table are cleared.

5.10 DataManager Review panel

This panel displays historical test results for single tests or batch files selected from the **Results table** (Item 5.9.1). See Fig. 5-24.

To open a test in the panel, double-click a test record in the ***Results table*** *or open a batch file using the* ***Open batch button*** *(Item 5.9.5).*

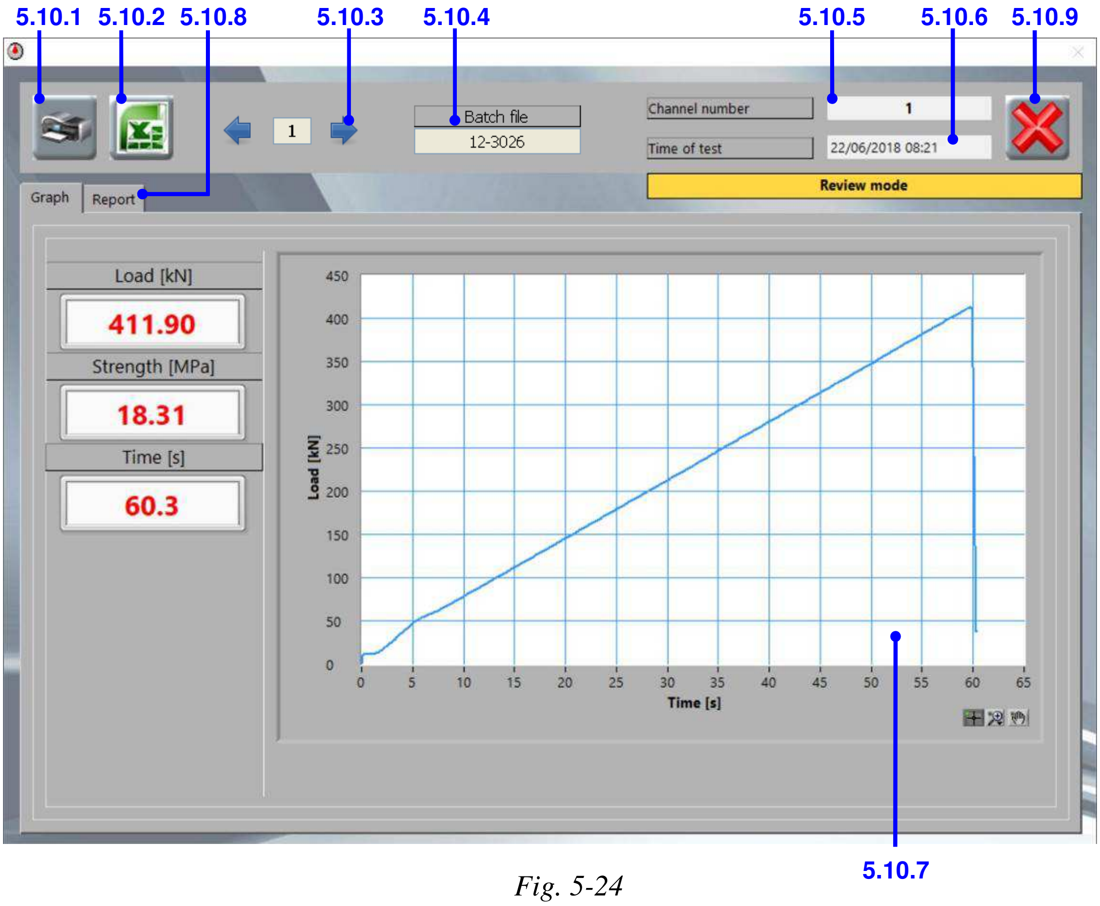


Fig. 5-24

5.10.1 Print button

Prints test report(s) for the test(s) being viewed in the panel on the default printer.

The report contains all the information from the ***Report window*** *and a copy of the graph from the* ***Graph window*** *and is automatically printed on the PC's default printer. In the case of a batch file, a separate report for each test will be printed along with a summary report containing a summary table of all the test results and average values for breaking load and strength. An example of a printed test report is included in the Appendix, Section 8.*

5.10.2 Save to Excel button

Creates report(s) for the test(s) in Excel and automatically saves each one as a workbook.

The report presents the test information from the ***Report window*** *and a copy of the graph on the* ***Graph window****. The second sheet of the Excel workbook contains the graph data, with time (in seconds) in the first column and strength or load in the second column. In the case of a batch file, one Excel workbook will be generated for each test, with the filename* 'batch name_record number*.xls'. A summary report will also be generated containing a summary table of all the test results and average values for breaking load and strength. This will have the filename* 'batch name*_batch.xls'. An example Excel report is included in the Appendix, Section 8.*

NOTE:
The system PC must have MS Excel installed and closed, to use the Save to Excel function.

5.10.3 Record number

Displays the record number within the batch file of the test being displayed.
To view the next or previous test in the batch file, click the arrow buttons on either side of the number. If a single test has been opened, this will not be displayed.

5.10.4 Batch file name

Displays the name of the batch file that is currently open.
If a single test has been opened, this will not be displayed.

5.10.5 Channel number

Displays the channel number of the testing machine that the test was performed on.

5.10.6 Time of test

Displays the time and date that the test was carried out.

5.10.7 Graph panel

The items on this panel are the same as for the **DataManager Graph window** (Item 5.6).
The readings shown are the peak load and strength and the elapsed time at the end of the test.

5.10.8 Report panel

The functions on this panel are the same as for the **DataManager Report window** (Item 5.8) except that the data is read-only; the values cannot be changed.

5.10.9 Close button

Closes the panel.

6. USE OF THE SOFTWARE

6.1 Managing the language

6.1.1 Changing the language of the software

The software comes with English and Italian languages and a custom option (see Section 6.1.3). To change the language used in the software, follow the steps below:

1.  Click the **Options button** on the **MULTITEST Main panel**.

   If you have Datamanager open you will have to close it to access the Main panel.

2. Enter the level 1 password ("controls1") to access the **System options panel**.

3. From the **Language selection cell**, select the language you require (see Fig. 6-1).

Fig. 6-1

4.  Click the **Save button** to save the change.

5.  Click the **Close button** to close the panel.

6. The new language will be effective immediately.

6.1.2 Editing the text of the software

The text used in the software can be edited as follows:

1. Follow steps 1 and 2 of Section 6.1.1.
2.  Click the **Dictionary button** to open the Language panel (see Fig. 6-2).

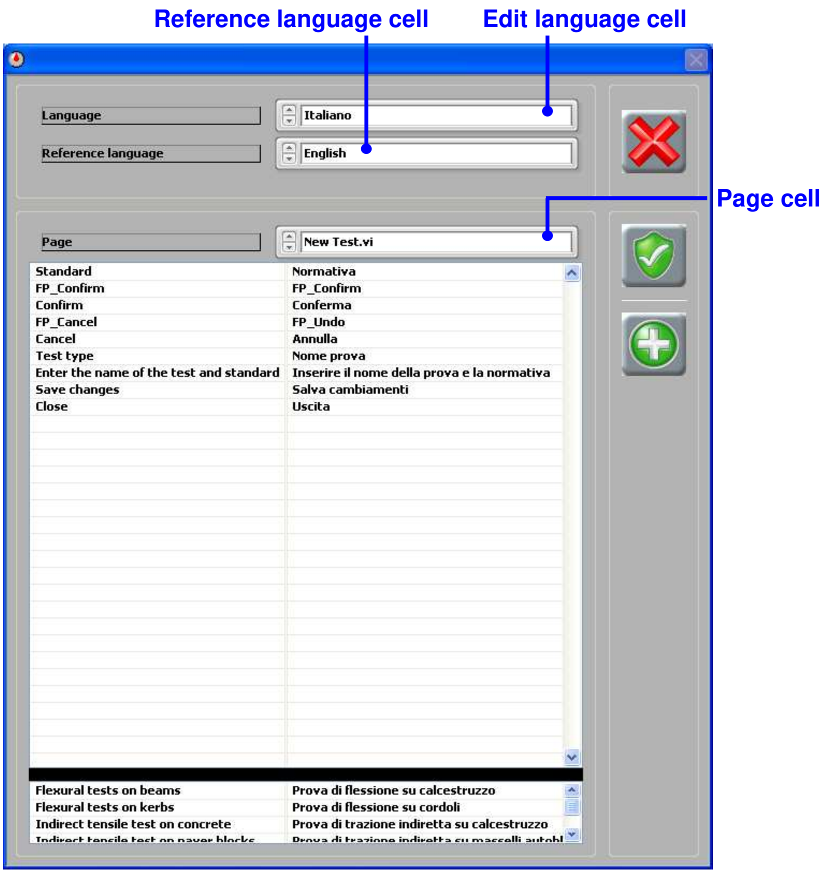


Fig. 6-2

3. Select the language you want to edit from the **Edit language cell** and, if required, the language you want to use as a reference from the **Reference language cell**.

   Use the arrows to the left of the cell to scroll through the options or click on the cell to select from a list showing all the options.

4. Select a section of the software from the **Page cell** and then edit the text in the right-hand column of the table. Repeat for other sections of software as necessary.

   The left-hand column displays the selected reference language.

5.  Click the **Save button** to save the changes.

    *To close the panel without saving the changes, click the **Close** button and then click the **Confirm** button on the message.*

6.  Close the panel by clicking the **Close button**.

6.1.3 Adding a new language

A new language can be added as follows:

1. Follow steps 1 and 2 of Section 6.1.1.
2.  Click the **Dictionary button** to open the Language panel (see Fig. 6-3).
3.  Click the **Add language** button.

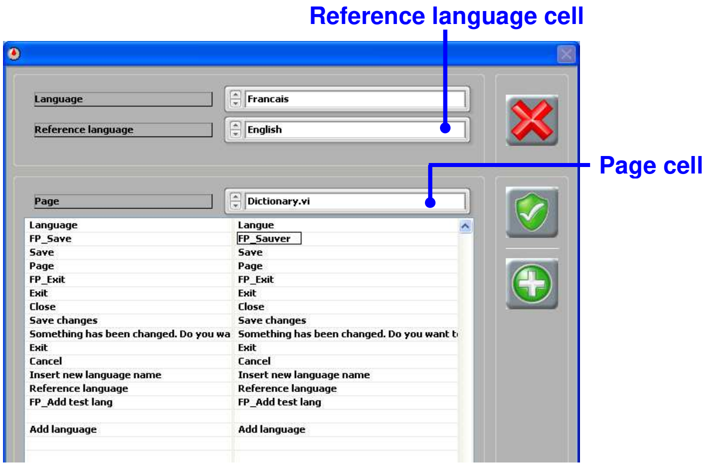


Fig. 6-3

4. Select the language you want to use as a reference from the **Reference language cell** (see Fig. 6-3).

   Use the arrows to the left of the cell to scroll through the options or click on the cell to select from a list showing all the options.

5. Select the first section of the software from the **Page cell** and then edit the text in the right-hand column of the table. Repeat for all the other sections of the software.

   The left-hand column displays the selected reference language.

6.  Click the **Save button** to save the changes.

    *To close the panel without saving the changes, click the **Close** button and then click the **Confirm** button on the message.*

7.  Close the panel by clicking the **Close button**.
8. The new language will now be available in the **Language selection cell** of the **System options panel** (see Fig. 6-4).

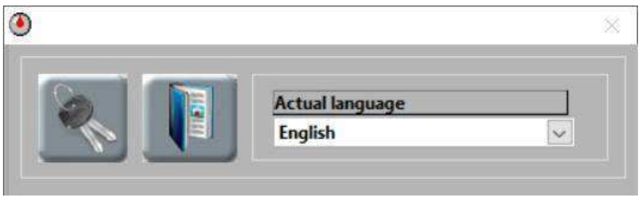


Fig. 6-4

6.2 Setting the password

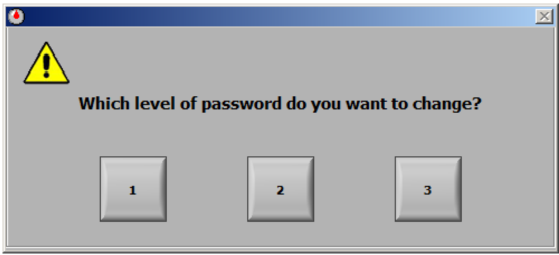
1.  Click the **Options button** on the **MULTITEST Main panel**.

   If you have Datamanager open you will have to close it to access the Main panel.

2. Enter the level 1 password ("controls1") to access the **System options panel**.

3.  Click the **Password button**.

4. Click the relevant button for the level of password that you want to change, 1, 2, or 3, and in the window that appears, enter the current password for the level you selected.



5. In the next window (Fig. 6-5), enter the new password in both cells.

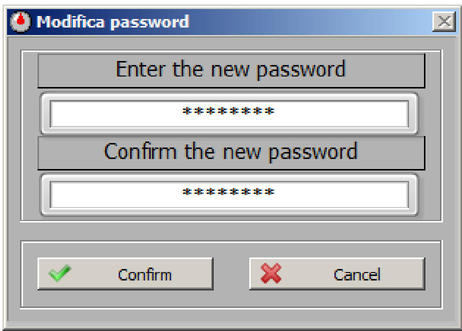


Fig. 6-5

6. Click the **Confirm** button to save the new password.

   If you decide not to save the new password, click the ***Cancel*** *button.*

NOTE:
If you lose or forget the new password it cannot be recovered. It is recommended to keep a list of the passwords in a safe place.

7.  Click the **Close button** on the **System options panel** to close it.

6.3 Calibration

Calibrations of up to ten points can be carried out using the software; the data is then uploaded to the control unit. Refer to the relevant hardware manual for information about controlling the test frame during the calibration procedure.

6.3.1 Calibrating a transducer

Before starting the calibration, decide on the measurement range that needs to be calibrated (normally between 10% and 100% of the full scale of the test frame), and which calibration points within the range will be recorded. This can depend on the linearity of the system, the capacity of the reference load cell, and which part of the range must have class 1 calibration. See the Calibration section of the relevant test frame instruction manual for more information.

A reference load cell of appropriate capacity and accuracy is required for this procedure.

NOTE:
The loads applied during the calibration must not exceed the measurement range of the test frame.

WARNING:
The **Calibration panel** should only be accessed when a calibration needs to be changed. The panel should not be unlocked for any other reason. Before accessing the panel, the following procedure must be carefully read and understood. If any doubts remain, the CONTROLS service department should be contacted for clarification.
CONTROLS will not accept responsibility for any problems caused by incorrect calibration.

1. Place the reference load cell in the test frame.
2. Click the **Channel configuration button** on the **MULTITEST Main panel**.
3. In the channel configuration list that opens (Fig. 6-6), select the channel you want to calibrate and double-click anywhere on the row to open the **Calibration panel**.

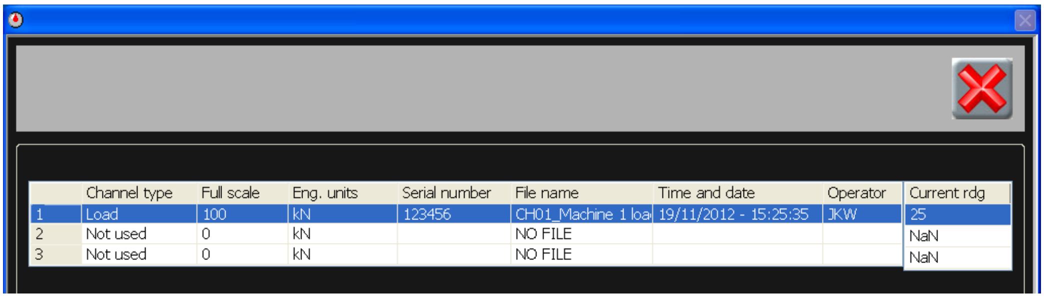


Fig. 6-6

4. Click the **Edit calibration button** to and enter the level 1 password (“controls1”) to unlock the panel.
5. Click the pump on/off button to turn the pump on.
6. Run the test frame up to the maximum load using the **Pump controls** (see Fig. 6-7)

   The function of the controls is described in Item 5.4.23.

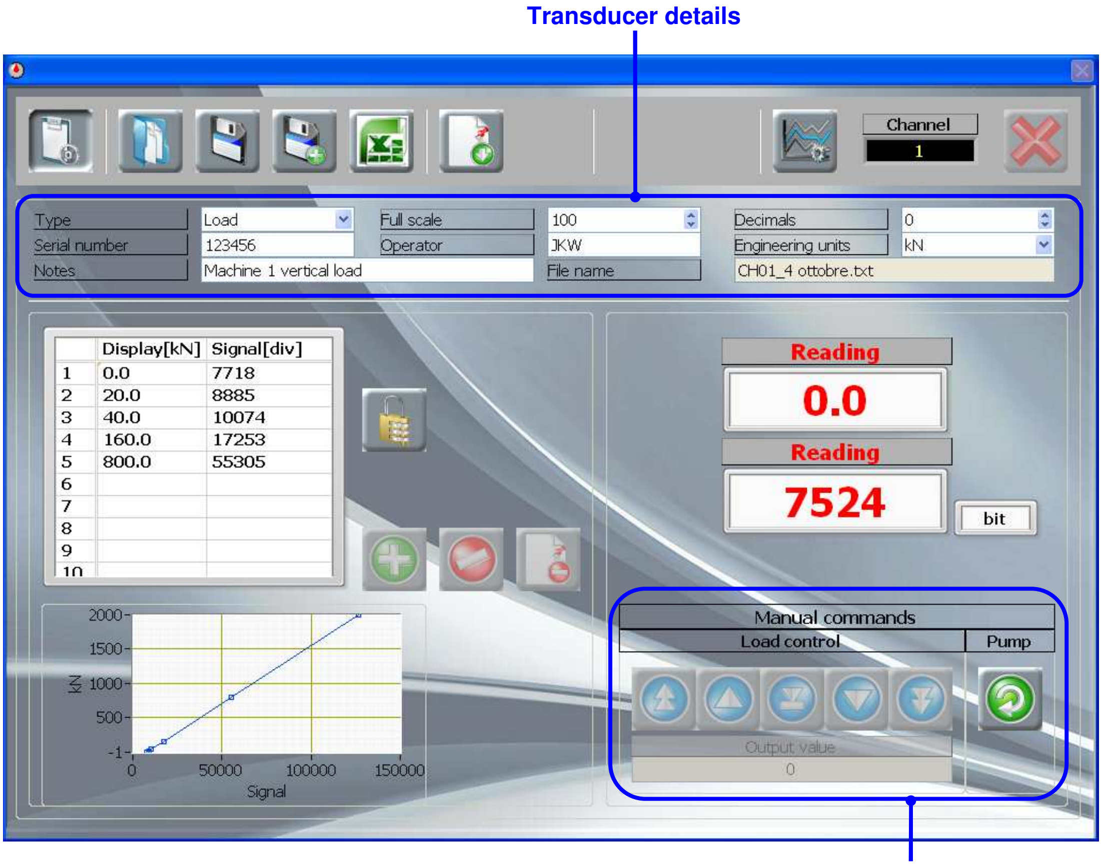


Fig. 6-7

7.  Stop the pump by clicking the pump on/off button.

   You will have to wait for a few seconds for the pressure to fully discharge before it will be possible to start loading again. The **System status** will say 'READY' when the pump is available.

8. Repeat steps 5 to 7 two more times.

9. Enter information about the transducer in the **Transducer details** section (Fig. 6-7).

10.  Click the **Unlock table button** to unlock the calibration table for editing.

11.  Click the **Clear table button** to remove the existing data.

12.  Click the pump on/off button to turn the pump on.

13. To acquire the first calibration point - at zero load - carry out the next steps, 14 to 16, before the load cell touches the upper platen of the test frame.

14.  Click the **Add point button**.

15. In the **Reference reading cell**, enter the reading of the reference transducer, in kilonewtons. See Fig. 6-8.

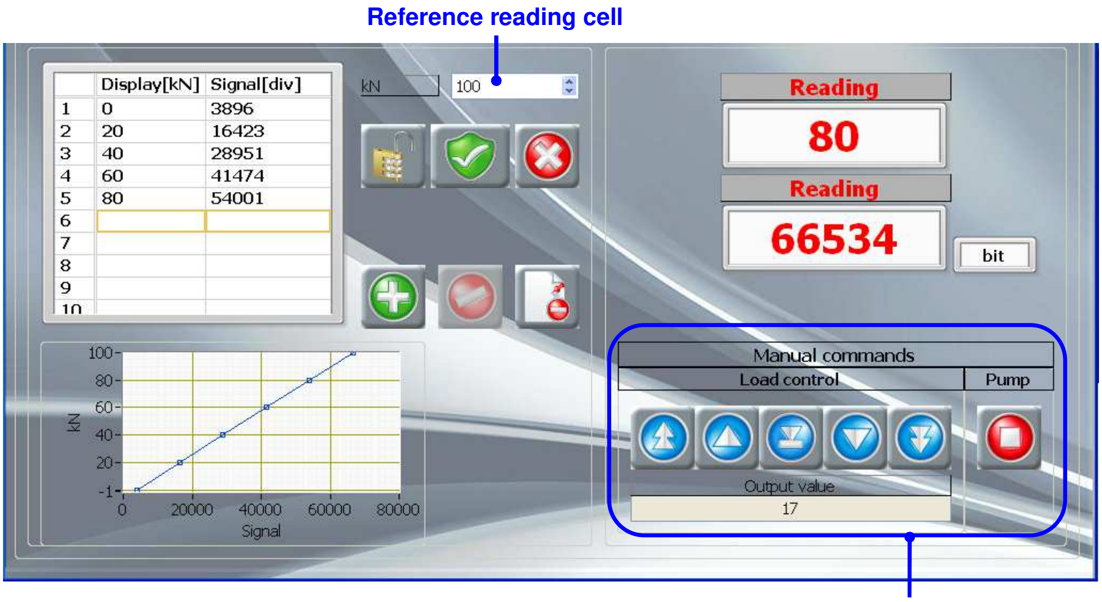


Fig. 6-8

16. When the load cell reading is stable, click the **Acquire point button** to take the reading and put it into the next empty row of the **Calibration table**, along with the reading of the reference transducer.

    *If you decide not to use the point, click the **Cancel point button**.*

17. Run the test frame up to the next required load using the **Pump control**

18. Repeat steps 14 to 17 for all the calibration points.

    *If you need to delete a point from the table, click on the row of the point you want to delete and then click the **Delete point button**.*

19. When all the calibration points have been acquired, stop the pump by clicking the pump on/off button.

20. Click the **Unlock table button** to lock the calibration table.

21. To save the new calibration and overwrite the existing calibration using the same filename, click the **Save button**.

    *If you don't want to overwrite the existing calibration, click the **Save as button** and save the new calibration with a different filename.*

    *The software will automatically insert the channel number ('CH**_') at the beginning of the filename so it is easy to identify which channel the calibration belongs to.*

NOTE:
The default location for the saved calibration files is C:\Users\...\Documents\MULTITEST\data\Channels.

22. If you want to create an Excel report of the calibration data, click the **Save to Excel button**.

    A standard Windows save file dialogue box will open to allow you to choose a location for the file.

    *The report contains all the **Transducer details** and the data from the **Calibration table**.*

NOTE:
The system PC must have MS Excel installed and closed, to use the Save to Excel function.

23. Send the calibration data to the device by clicking the **Send to device button**.
    There will be a slight delay and a message will be shown on the screen while the data is sent.

24. When you've finished, click the **Edit calibration button** to lock the panel.

25. Click the **Close button** to close the panel.

26. If you need to perform another calibration on a different channel, select the channel from the configuration list, and repeat from step 4.

27. Click the **Close button** to close the channel configuration list.

6.3.2 Loading a previous calibration

Historical calibration data is stored for each channel, as long as a new filename is used each time a transducer is calibrated. Historical data can be loaded in as follows:

1. Follow steps 2 to 4 from Section 6.3.1 to access the **Calibration panel** for the channel number you want to work with.
2. Click the **Load calibration button**.

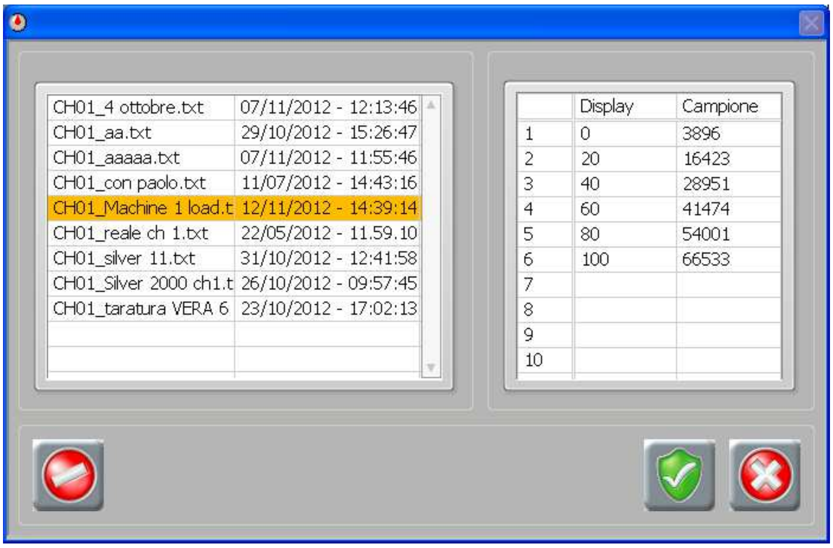


Fig. 6-9

3. In the window that appears (see Fig. 6-9), select the calibration file that you want to use from the list and then click the **Load** button.

   The calibration data for the selected file is shown in the table to the right of the list.

   *If you decide not to load a file, click the **Close** button.*

   *After clicking the **Load file** button, the window will close and the selected calibration file will be displayed on the **Calibration panel**.*

4. When you've finished, click the **Edit calibration button** to lock the panel.
5. Click the **Close button** to close the panel.
6. Click the **Close** button to close the channel configuration list.

6.3.3 Deleting a previous calibration

1. Follow steps 2 to 4 from Section 6.3.1 to access the **Calibration panel** for the channel number you want to work with.
2. Click the **Load calibration button**.
3. In the window that appears (see Fig. 6-9), select the calibration file that you want to delete from the list and then click the **Delete** button.

   You will be prompted to confirm the command before the file is deleted.

4. Click the **Close** button to close the window.
5. When you've finished, click the **Edit calibration button** to lock the **Calibration panel**.
6. Click the **Close button** to close the panel.
7. Click the **Close button** to close the channel configuration list.

6.4 Customizing test types and parameters

Customized test types, based on any of the eight standard test types, can be created with tailor-made parameters.

6.4.1 Adding a new test type

1. Run the DataManager test module.

2. Click the 'Report' tab to open the DataManager **Report window**.

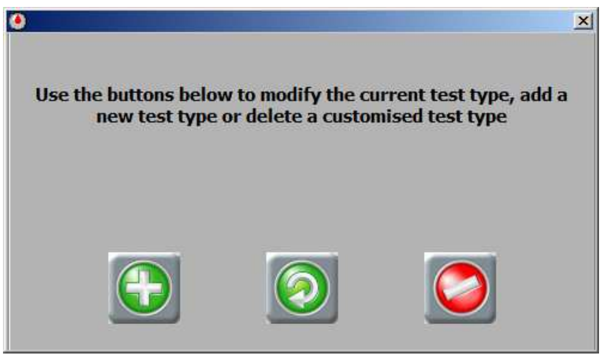
3. Select the type of test you want to work with from the **Test type control** (see Fig. 6-10).

Fig. 6-10

4.  Click the **Customise button**.

5. Enter the level 1 password ("controls1").

6.  In the window that appears, click the **Add test type button**.



7. In the next window, enter a name for the new test type and the standard it is based on and then click the **Confirm** button (see Fig. 6-11).

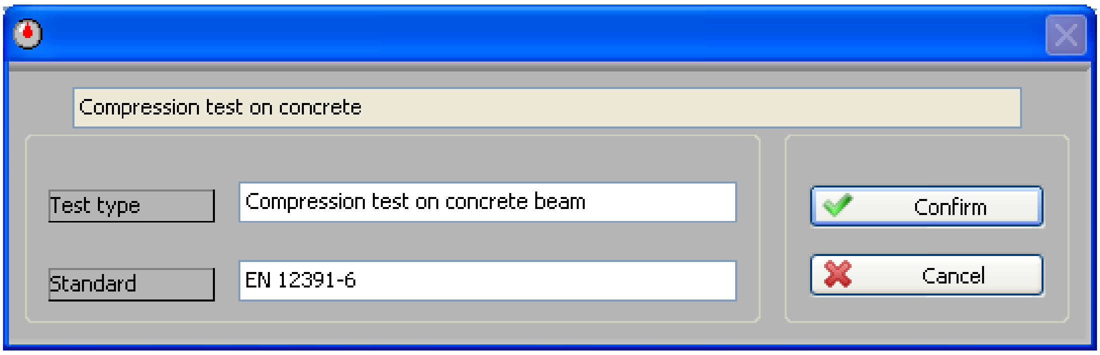


Fig. 6-11

8. All the fields in the **Report window** that are outlined in red can now be edited – change the text as required (see Fig. 6-12).

   *Only the fields outlined in red can be changed. The new descriptions will be used whenever the new test type is selected in this window, in the **Results panel,** and in saved and printed test reports.*

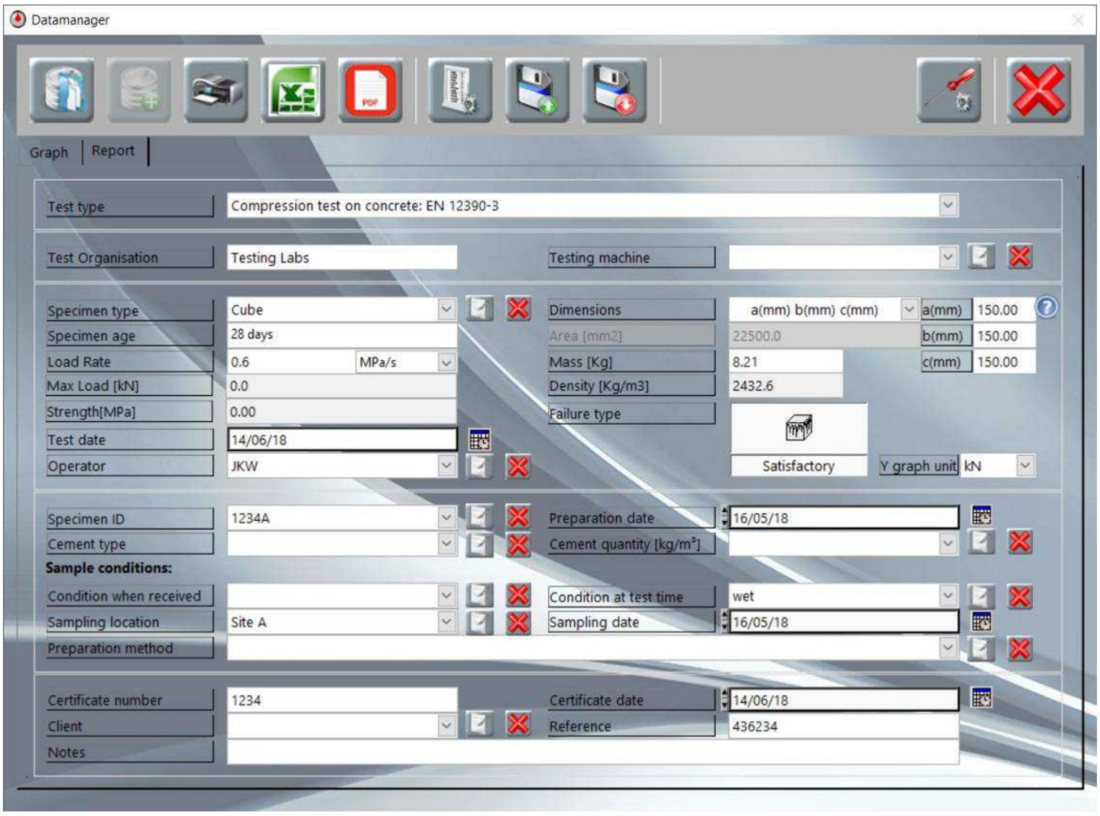


Fig. 6-12

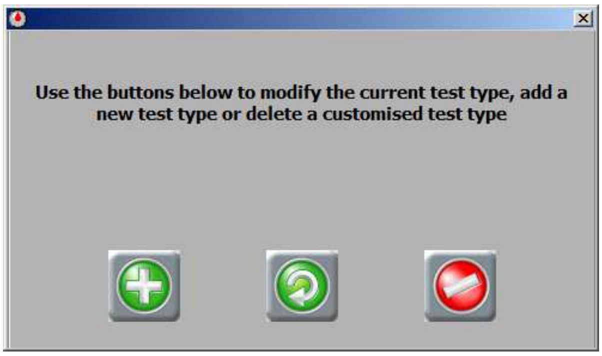
9.  When you have finished, click the **Confirm** button and save the changes.

    *To cancel the customization, click the **Undo** button.*

10. Your new test type can now be selected from the drop-down **Test type control** list. The fields can be edited at any time by following the procedure in Section 6.4.2.

6.4.2 Customizing an existing test type

1. Follow steps 1 to 5 in Section 6.4.1.
2.  In the window that appears, click the **Edit test type button**.



3. All the fields in the **Report window** that are highlighted in red can now be edited – change the text as required (see Fig. 6-12).

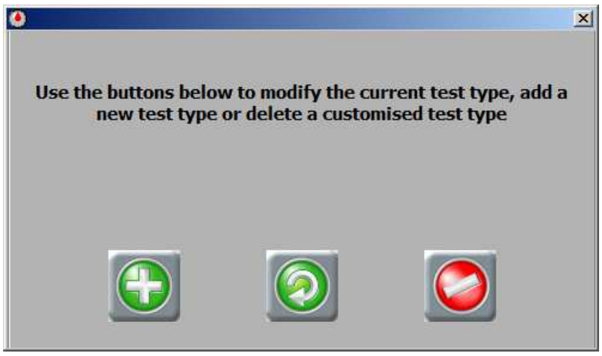
   *Only the fields outlined in red can be changed. The new descriptions will be used whenever the test type is selected in this window, in the **Results panel,** and in saved and printed test reports.*

4.  When you have finished, click the **Confirm** button and save the changes.

    *To cancel the customization, click the **Undo** button.*

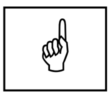
6.4.3 Deleting a test type

1. Follow steps 1 to 5 in Section 6.4.1.
2.  In the window that appears, click the **Delete test type button**.



3.  In the next two windows, click the **Confirm** button to accept the deletion.

    *To cancel the deletion, click the **Cancel** button in the message.*



NOTE:
The selected test type and all tests performed using this test type will be permanently deleted.

6.4.4 Adding a new entry to a field

1. Run the DataManager test module.

2. Click the 'Report' tab to open the DataManager **Report window**.

3.  Click the **Add entry button** adjacent to the cell you want to add an entry to.

4. In the window that appears, type in the new entry.

5.  Click the **Confirm** button

6. The new entry can now be selected from the drop-down list in the cell.

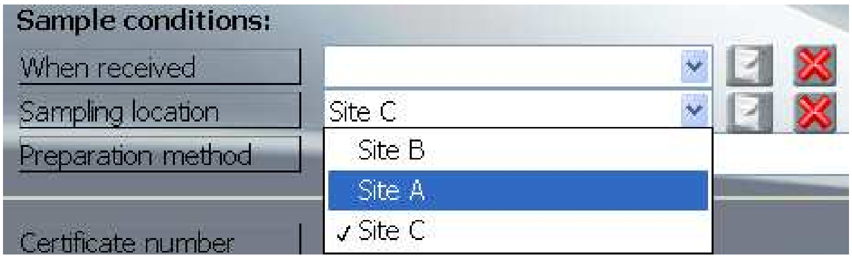


Fig. 6-13

6.4.5 Deleting an entry from a field

1. Select the entry you want to delete from the drop-down list in the cell (see Fig. 6-13).

2.  Click the **Delete Entry button** adjacent to the cell and the item will be deleted.

6.5 Running a test

6.5.1 Compression, flexure or indirect tensile test

This procedure describes how to run a test with the software. Refer also to the instruction manual of the machine that will be used for the test.

1. Run the software.

2. Run the DataManager test module by clicking on the icon on the **MULTITEST Main panel** (see Fig. 6-14).
   It is possible to choose other modules available as an upgrade. For further information refer to the relevant instruction manuals.

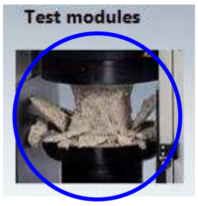


Fig. 6-14

3.  Click the **Test settings button** in the **Graph window** (see Fig. 6-15).

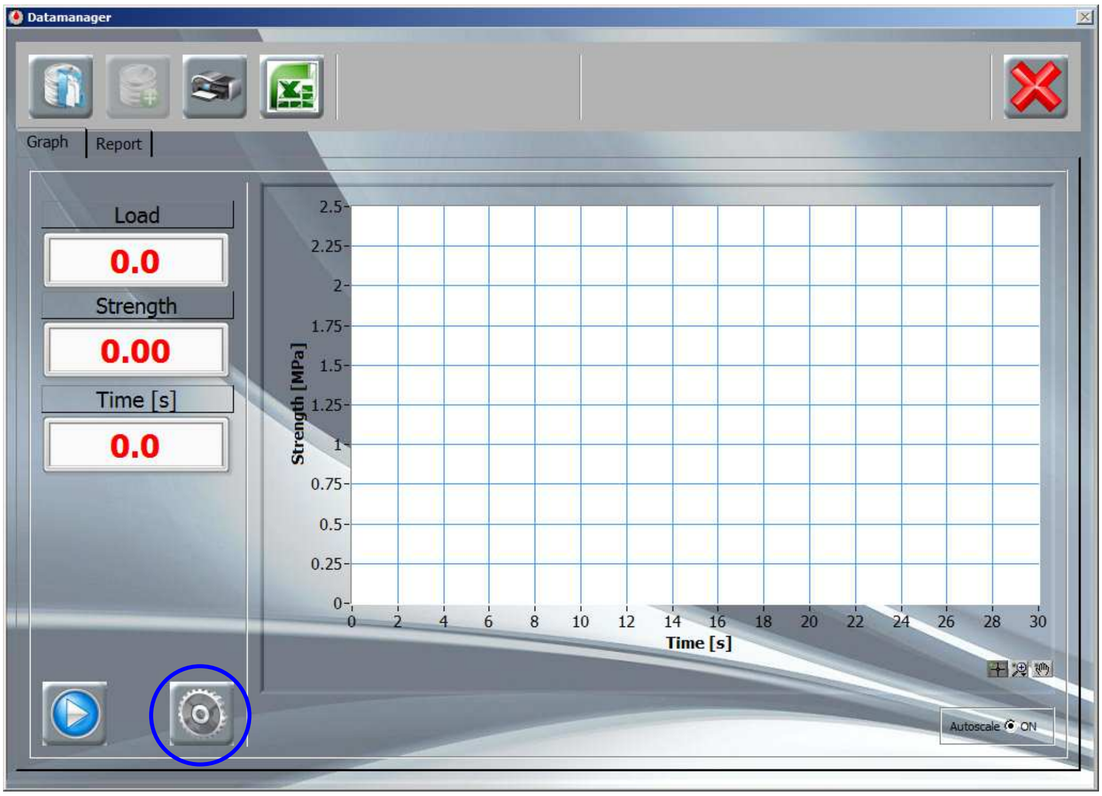


Fig. 6-15

4. In the **Test settings panel** (Fig. 6-16), select the **Channel number** of the machine that will be running the test. Check that suitable values are entered for the **Test speed**, **Peak sensitivity**, **Load limit,** and **Time limit**. If required, change the units that will be used for the y-axis of the graph.

   See Section 5.7 for more information.

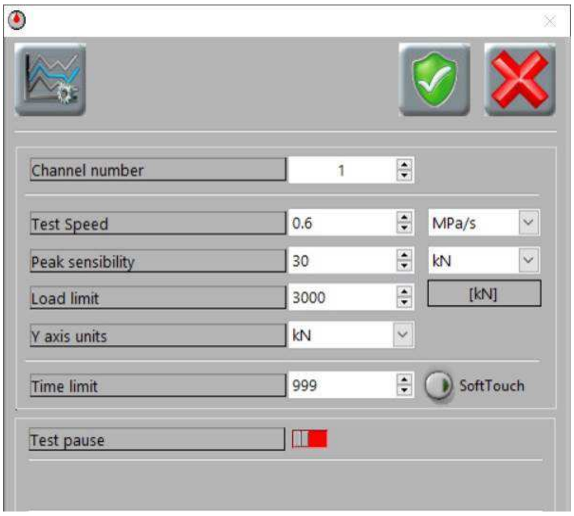


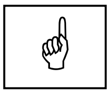
Fig. 6-16

5.  If you made changes, click the **Save** button to save them and close the panel.

    *To close the panel without saving the changes, click the **Close** button and then click the **Confirm** button on the message.*

6.  If you want the tests to be automatically saved, click the **Results button** to open the **Results panel** and make sure that the autosaves are ticked (see Fig. 6-17).

    *Then click the **Close button** to close the panel.*



NOTE:
If you are carrying out a 'Compression test on concrete' and you want to be able to select the failure type after the test has finished, do not tick the **AUTO Save DB** (it automatically saves the test in the archive of the software folder) and/or **AUTO Save TXT** (it automatically saves a txt file named "last_test.txt" in which the last test will be overwritten each time). This will allow you to select the failure and then save the test using the **Save test button**.

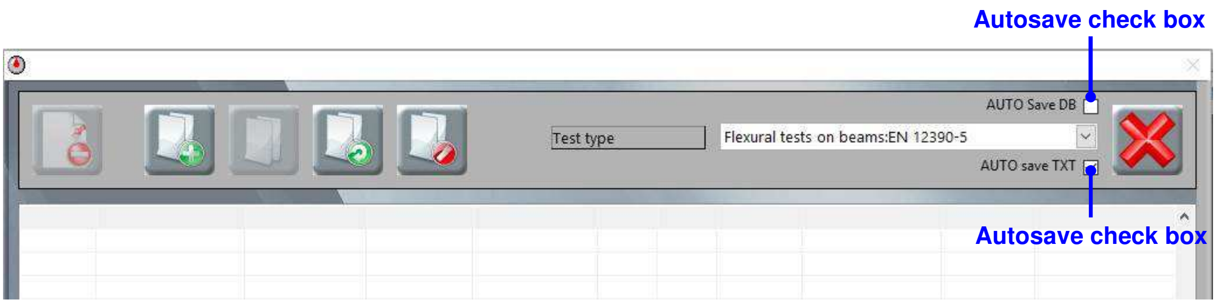


Fig. 6-17

7. Click the 'Report' tab to view the **Report window** (Fig. 6-18).

8. Select the type of test you are performing from the **Test type control**.

9. Enter the sample information and measurements, test information, and client details in the cells. (Refer to Section 5.8 for information.) Make sure that you enter the correct dimensions so that the specimen strength can be calculated correctly.

 *If you saved the report data (see step 10) when carrying out previous tests on the same batch of samples, you can click the **Load report data button** to load the data into the fields.*

You can enter data in as many or as few of the fields as you require, depending on what information you want to see on the test report. The only fields that must have a value are specimen dimensions and certificate number. If you do not enter a certificate number, you will be prompted to enter one when the test finishes before the test can be saved. If you don't enter specimen dimensions, the test will not start.

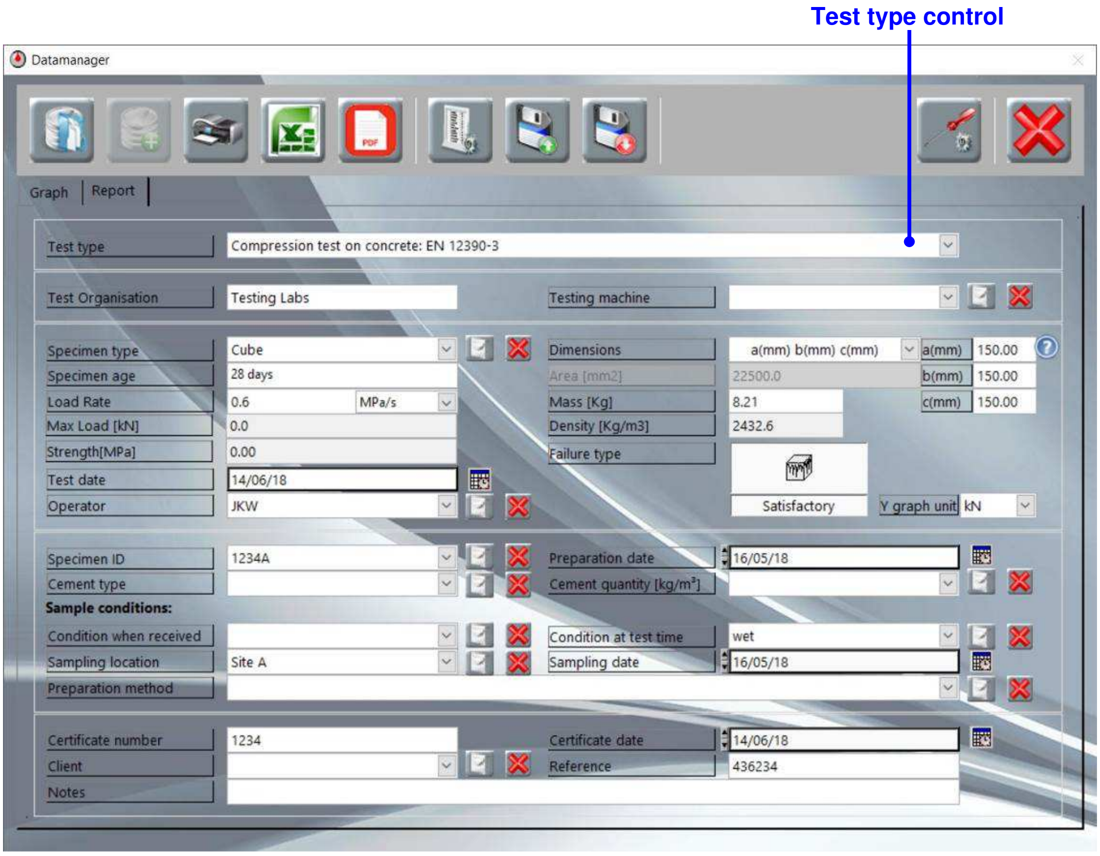


Fig. 6-18

10.  If you want to save the values you have entered so that they can be loaded in quickly another time when you are testing the same batch of samples, click the **Save report data button**.

11. Mount the test sample in the machine and set the machine to the correct position for starting a test (refer to the instruction manual of the machine).

12. Click the 'Graph' tab to view the **Graph window** (Fig. 6-19).

13.  When you're ready to start the test, click the **Start test button**. There will be a short pause while the software communicates with the device – a message will be displayed on the screen telling you to wait. As soon as the message disappears:

    The software will start logging when the preload is reached. Then live load, strength, and elapsed time readings will be displayed in the **Graph window,** and strength or load will be plotted against time on the graph, depending on what you selected for the y-axis units in the **Report window**. The icon of the **Start test button** will change to a stop symbol while the test is running. The system status will display the message 'TEST IN PROGRESS' with a green background.*

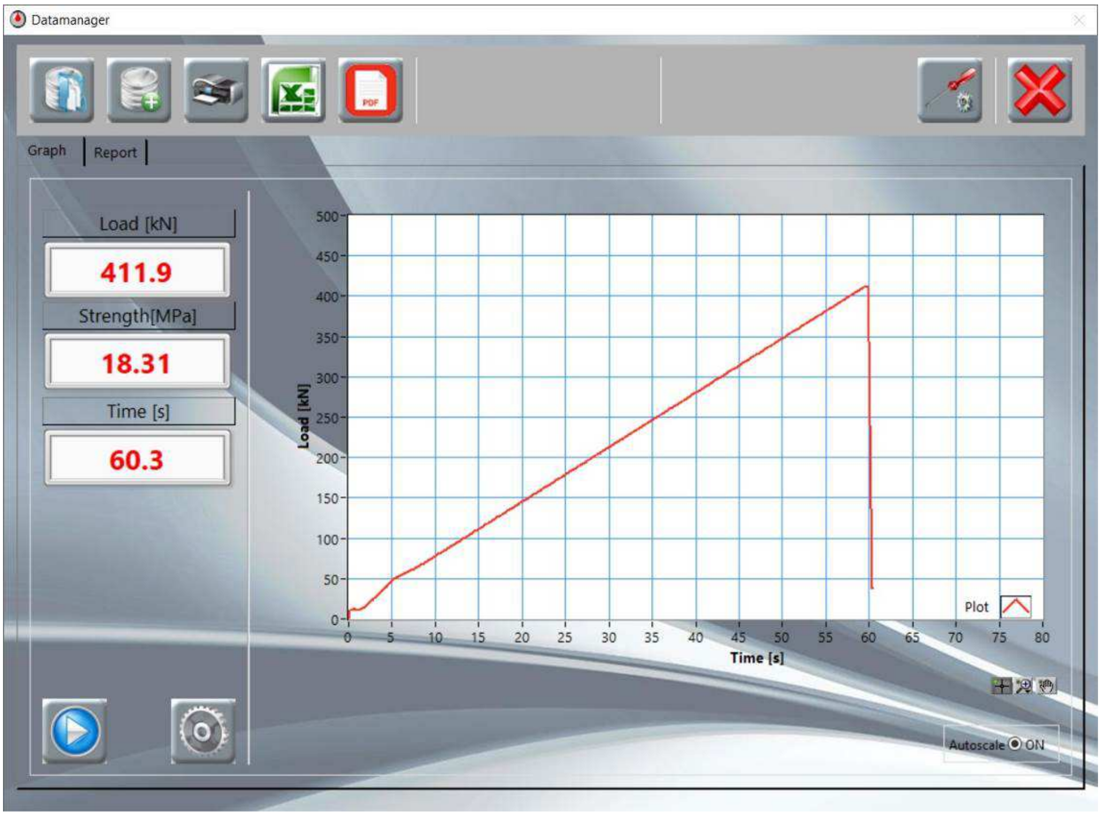


Fig. 6-19

14. The test will stop automatically when specimen failure is detected*, or if the load limit is reached.

    **The preload value and failure detection parameters are specified in the PID settings and should be entered depending on the type of test (see Section 7.3).*

    *The **Peak warning indicator** background will turn green to show that the peak load has been reached (Fig. 6-20).*

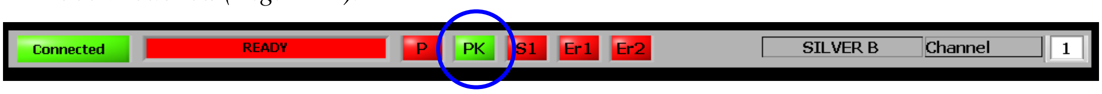


Fig. 6-20

 *If you want to stop the test before this, click the **Start test button** (which will be displaying a stop icon during the test).*

- The AUTOMAX Multitest control console will automatically unload.

15.  When the test is finished, if you didn't turn the **Autosave** functions on (step 6), click the **Save test button**.

16. Remove the test sample from the machine.

17.  If you want to print a test report, click the **Print button**.

    *A report containing all the information from the **Report window** and a copy of the graph from the **Graph window** will be printed on the PC's default printer. An example of a test report is included in the Appendix, Section 8.*

18.  If you want to create an Excel test report, click the **Save to Excel button**. In the standard Windows save file dialogue box that appears, choose a location and enter a name for saving the file and click 'Save'.

NOTE:
The system PC must have MS Excel installed and closed, to use the Save to Excel function.

---

*Excel will open and a report containing all the information from the **Report window** and a copy of the graph from the **Graph window** will be generated and automatically saved using the filename and location that you entered. (The second sheet of the Excel workbook contains the graph data, with time (in seconds) in the first column and strength (in megapascals) or load (in kilonewtons) in the second column.) An example of a test report is included in the Appendix, Section 8.*

19. To perform another test, repeat the above from step 9.

6.5.2 Cement test

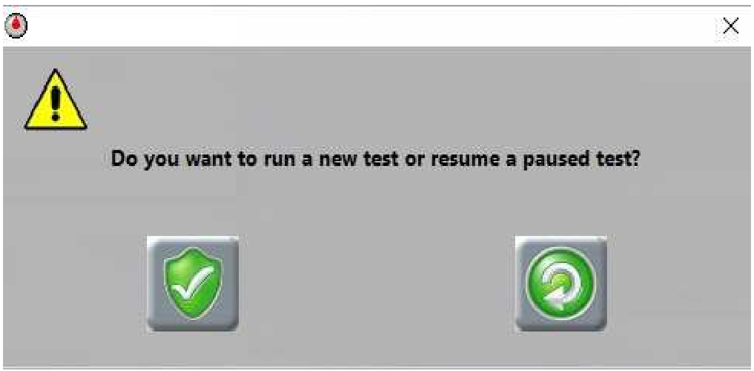
When performing a cement test, the software has extra functions to allow the testing and processing of the three flexure tests on cement prisms and the six subsequent compression tests. The following procedure describes how to run a set of tests.
The cement test program does not allow the management of single tests.

1. Start the software and check the settings as described in steps 1 to 8 of Section 6.5.1.

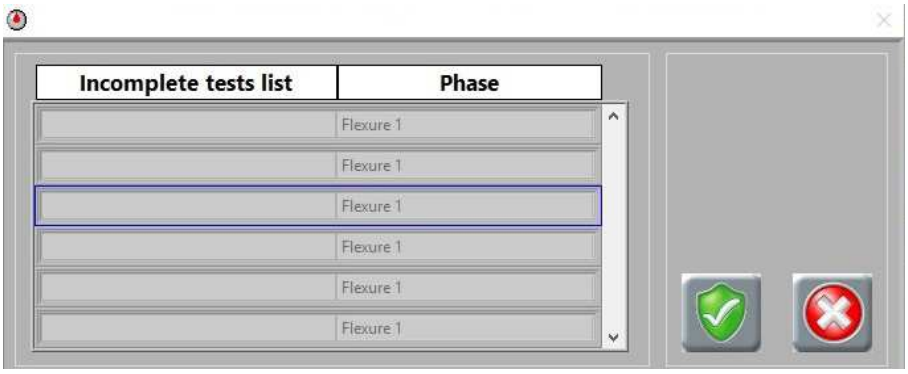
2.  Click the **Cement test** button.

After a test has started, the image and function of this button will change, so it can be used for stopping the test.

3. In the window that appears, it is possible to click the **New Test** button  or click the **Resume Paused Test** button .



- If the operator clicks the **Resume Paused Test** button , the following window appears:



In the window that appears, select the test that you want to continue from the list and click the **Confirm** button .

The first column of the list contains the certificate names of any test sequences that were stopped before they were completed; the second column contains the last test that was performed.

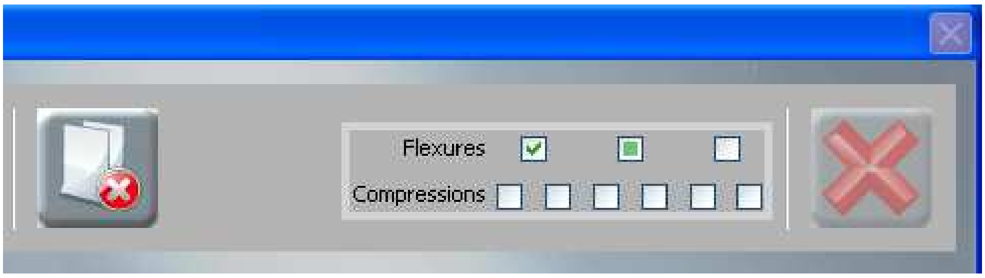
*After clicking the **Confirm** button, the test sequence will be reloaded allowing you to continue with the next test. You will not need to re-enter the information in the **Report window**.*

- If the operator clicks the **New Test** button , in the next window that appears, it is necessary to enter the certificate number, which will be the same for all nine tests and click the **Confirm** button .

4. The **Cement test status indicator** will show that the current test is the first flexure test by displaying a green square in the first flexure box.

5. Run the test following steps 9 to 16 in Section 6.5.1.

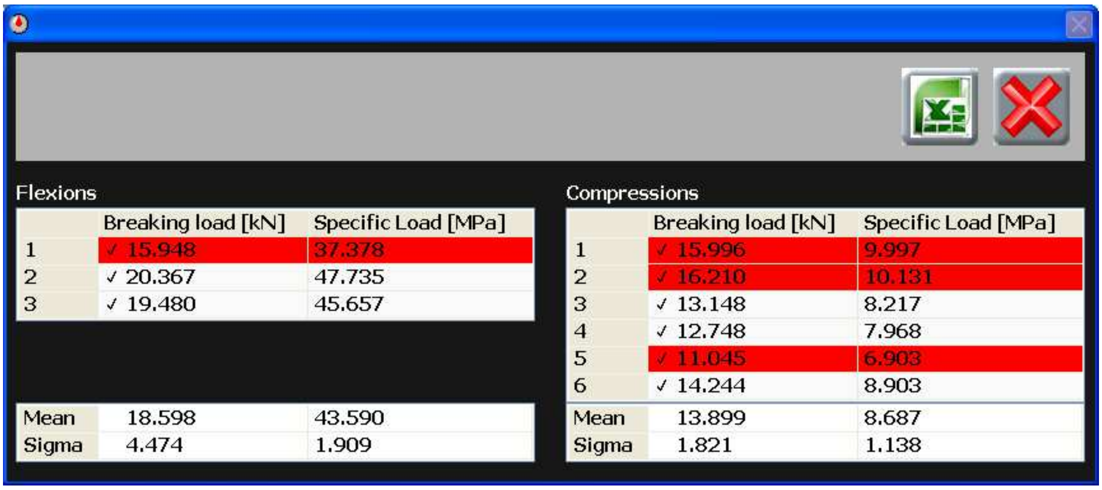
6. After the test is saved, the **Cement test status indicator** will show that the test is complete by displaying a tick in the box and the box for the next test in the sequence will display a green square.



7. To continue with the next test in the sequence, repeat the procedure from step 5.

 *If you need to stop testing before completing all nine tests, click the* ***Cement test button****. The set can be resumed as described previously.*

8.  When the ninth and final test in the sequence has been saved, the button will become available. If required, click the button to display a summary of results for all nine tests (see Fig. 6-21).



Flexions

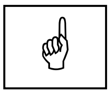
	Breaking load [kN]	Specific Load [MPa]
1	✓ 15.948	37.378
2	✓ 20.367	47.735
3	✓ 19.480	45.657
Mean	18.598	43.590
Sigma	4.474	1.909

Compressions

	Breaking load [kN]	Specific Load [MPa]
1	✓ 15.996	9.997
2	✓ 16.210	10.131
3	✓ 13.148	8.217
4	✓ 12.748	7.968
5	✓ 11.045	6.903
6	✓ 14.244	8.903
Mean	13.899	8.687
Sigma	1.821	1.138

Fig. 6-21

9.  If you want to create an Excel report of the tests, click the **Save to Excel** button on the summary panel.



NOTE:
The system PC must have MS Excel installed and closed to use the Save to Excel function.

- In the window that appears (Fig. 6-22), choose a location for saving the file.

  Do this as follows: the ***file path*** *is shown in the top section – to go up a level, click the* ***back button****; to add a folder to the file path, double-click on the folder in the list. The files will be saved using the certificate number for the file name.*

-  Click the **Confirm** button to create the files.

  Excel will open and a report for each test will be generated and automatically saved in the location that you selected. One Excel workbook will be generated for each test, with the filename 'certificate number_test number.xls'. A summary report will also be generated containing a summary table of all the test results and average values for *breaking load and strength. This will have the filename 'certificate number_batch.xls'. An example of a test report and a summary report are shown in the Appendix, Section 8.*

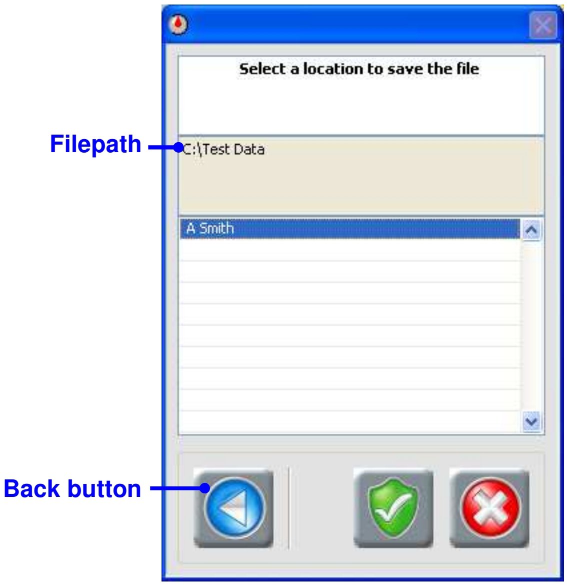


Fig. 6-22

10.  If you want to print a test report, click the **Print button**.

A report for each test containing all the information from the ***Report window*** *and a copy of the graph from the* ***Graph window*** *will be printed on the PC's default printer. A summary report will also be printed containing a summary table of all the test results and average values for breaking load and strength.*

6.6 Managing test results

6.6.1 Deleting a test

1. Run the DataManager test module.
2.  Click the **Results button** to open the **Results panel**.
3. Select the type of test you want to look at from the drop-down list in the **Test type control** (see Fig. 6-23).

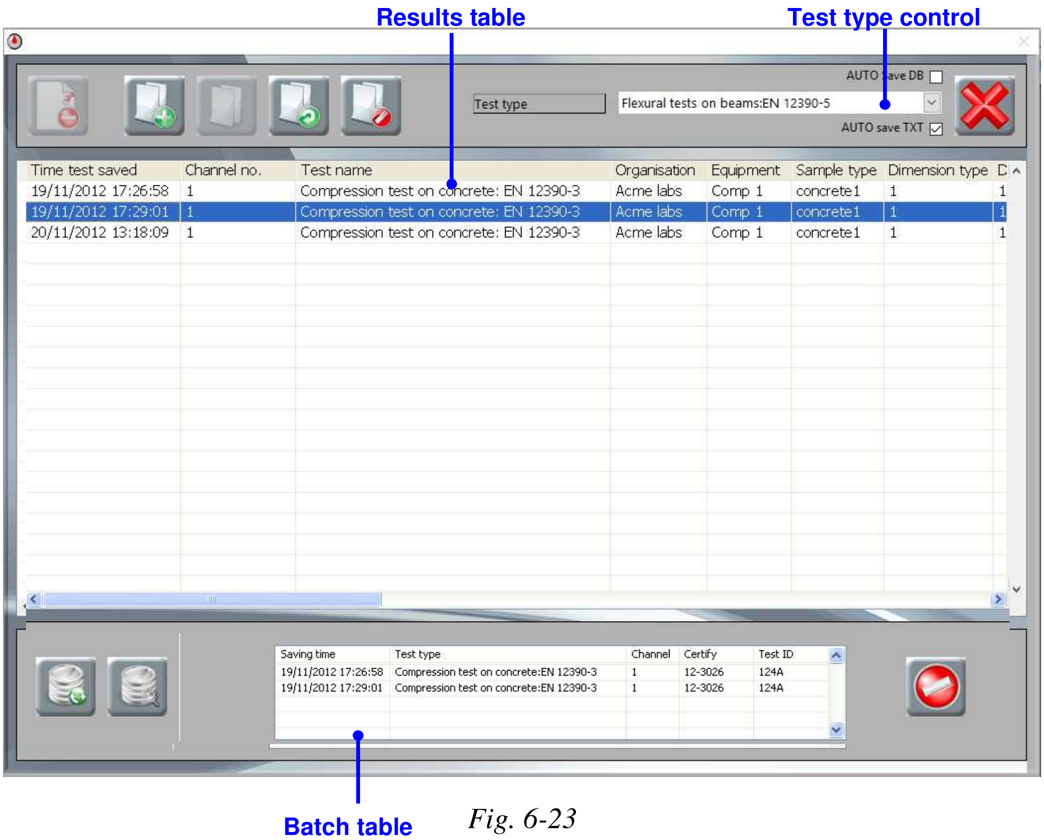
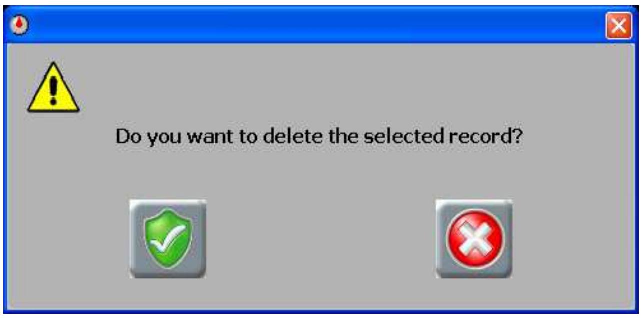


Fig. 6-23

4.  Find the test that you want to delete in the **Results table** (or do a search – see Section 6.6.2) and click anywhere on the row to highlight it. Then click the **Delete record button**.
5.  To delete the test, click the **Confirm** button on the message that appears.



 *If you decide not to delete the test, click the **Cancel** button on the message.*

NOTE:
The test will be permanently deleted.

6.6.2 Searching for a test

This section describes how to search for the test using specific criteria, such as test date, customer, etc.

The search function can only be used to search for one test type.

1. Run the DataManager test module.
2.  Click the **Results button** to open the **Results panel**.
3. Select the type of test you want to add to the batch from the drop-down list in the **Test type control** (see Fig. 6-23).
4.  Click the **Search button**.
5. In the search window (Fig. 6-24), there are several different ways to search:

- If you want to search for tests that match two different criteria, select 'AND' from the middle drop-down list and then select a second criterion (see Fig. 6-24).

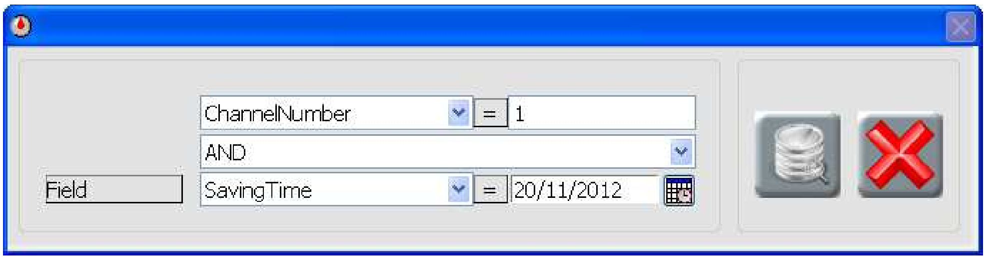


Fig. 6-24

- If you want to search for tests that match at least one of two criteria, select 'OR' from the middle-drop down list and then select a second criterion.

- If you want to search for a test between two parameters, for example, tests that were carried out between two dates, select '[..]' from the middle drop-down list and enter parameters in both the cells (see Fig. 6-25).

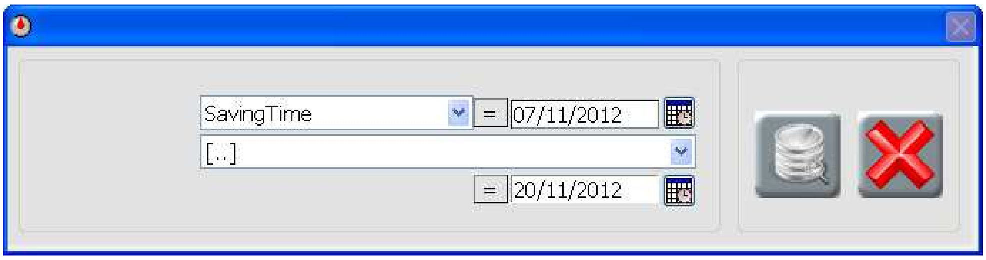


Fig. 6-25

- If you only want to search using one criterion, '---' must be selected from the middle drop-down list (see Fig. 6-26).

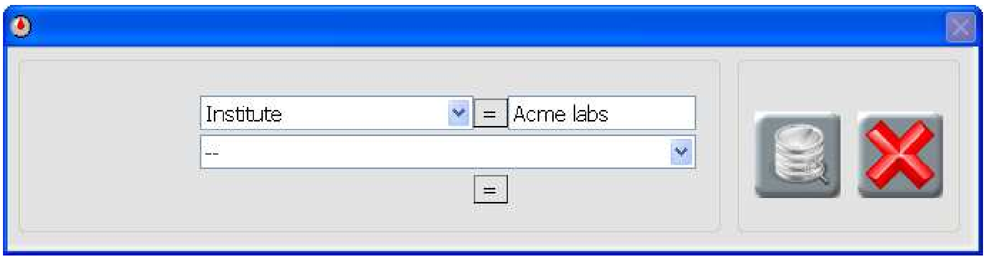


Fig. 6-26

6.  To start the search, click the **Start search** button.

   *When the search has finished, the search window will close and the results will be listed in the **Results table** of the panel.*

7.  To restore the contents of the **Results table**, click the **View all button**.

6.6.3 Creating a batch file

1. Run the DataManager test module.

2.  Click the **Results button** to open the **Results panel**.

3. Select the type of test you want to add to the batch from the drop-down list in the **Test type control** (see Fig. 6-23).

4. Find a test that you want to add in the **Results table** (or do a search – see Section 6.6.2) and click anywhere on the row to highlight it.

5.  Click the **Add record to batch button**.

   *The selected test will be copied into the **Batch table**.*

6. Repeat steps 4 and 5 until all the tests you want are listed in the **Batch table**.

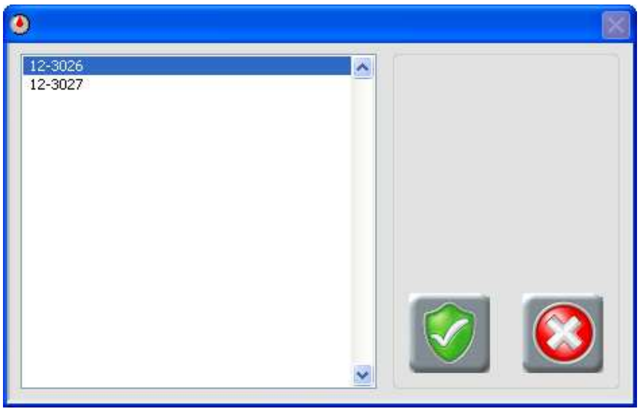
    *If you need to clear the contents of the table, click the **Clear table button**.*

7.  Click the **Create new batch button**.

8.  In the window that appears, enter a name for the batch file and click the confirm button.

9.  To view the file, click the **Open batch button**.

10.  Select the batch file from the list and click the **Confirm** button.



*The **Review panel** will appear (see Fig. 6-27), with the first test of the selected batch displayed. From this panel, the tests can be exported to Excel or printed (see Sections 6.6.5 and 6.6.6). To view the next or previous test in the batch, click the arrows on either side of the **Record number**.*

NOTE:
The system PC must have MS Excel installed and closed, to use the Save to Excel function.

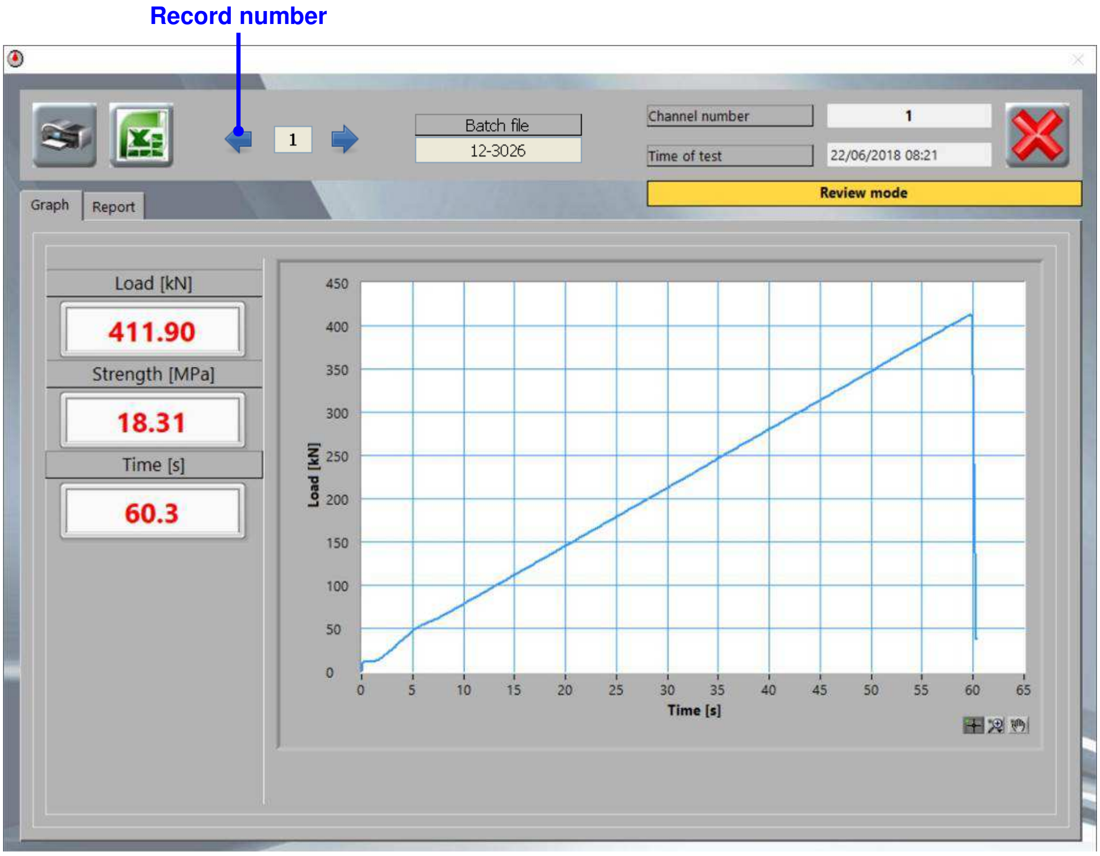


Fig. 6-27

11. When you've finished, click the **Close button** to close the panel.

12. Click the **Close button** on the **Results panel** to return the **DataManager Main panel**.

6.6.4 Deleting a batch file

1. Run the DataManager test module.
2. Click the **Results button** to open the **Results panel** (see Fig. 6-23).
3. Click on the **Delete batch button**.
4. Select the batch file from the list and click the **Confirm** button.

NOTE:
The batch file will be permanently deleted but the actual test records will not be deleted – they will remain in the **Results table**.

*If you decide not to delete the test, click the **Cancel** button.*

6.6.5 Reviewing a single test

1. Run the DataManager test module.
2.  Click the **Results button** to open the **Results panel**.
3. Select the type of test you want to open from the drop-down list in the **Test type control** (see Fig. 6-28).

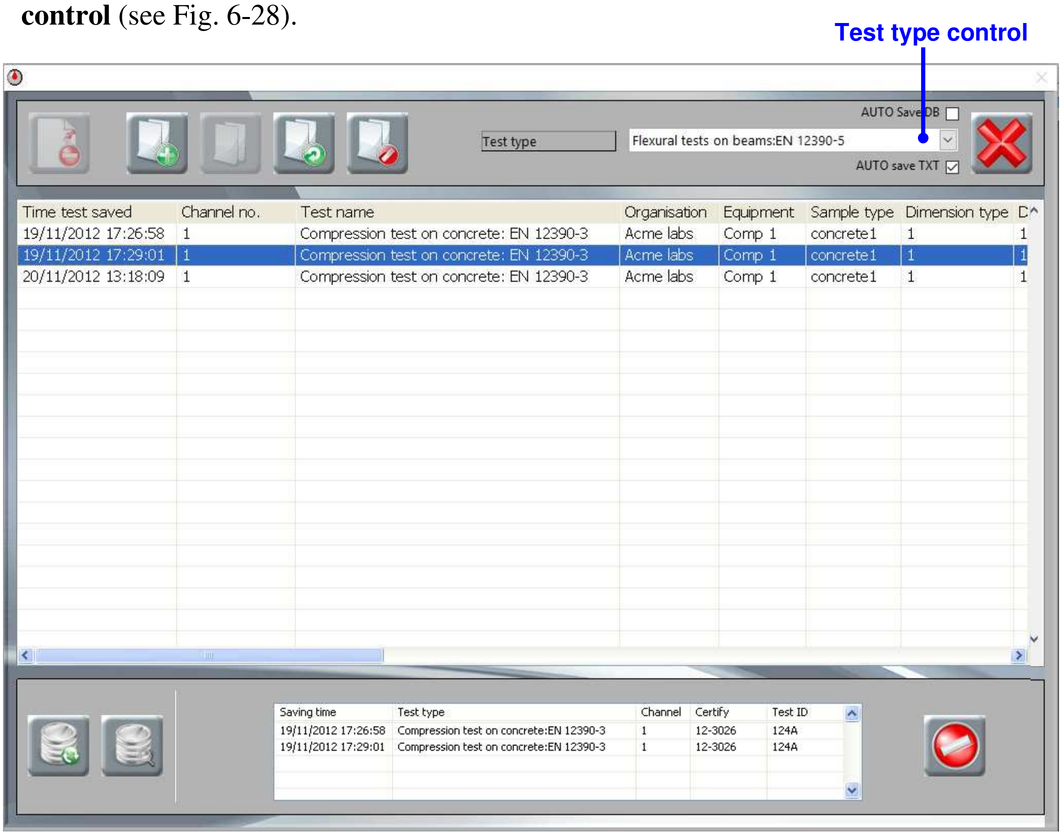


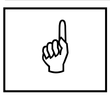
Fig. 6-28

4. Find the test in the **Results table** (or run a search – see Section 6.6.2) and double-click anywhere on the row.

   *The **Review panel** will appear (see Fig. 6-27), with the selected test displayed.*

5.  If you want to create an Excel report of the test, click the **Save to Excel button**. In the standard Windows save file dialogue box, choose a location and enter a name for saving the file and click 'Save'.

   Excel will open and a report for the test will be generated and automatically saved using the filename and location that you entered. An example report is included in the Appendix, Section 8.



NOTE:
The system PC must have MS Excel installed and closed, to use the Save to Excel function.

6.  If you want to print a test report, click the **Print button**.

   A report for the test will be printed on the PC's default printer. An example report is included in the Appendix, Section 8.

7.  When you've finished, click the **Close button** to close the **Review panel**.

8.  Click the **Close button** on the **Results panel** to return the **DataManager Main panel**.

6.6.6 Reviewing a batch of tests

1. Run the DataManager test module.
2.  Click the **Results button** to open the **Results panel**.
3. Select the type of test you want to look at from the drop-down list in the **Test type control** (see Fig. 6-28).
4.  Click the **Open batch button**.

   If you didn't create a batch file yet for the group of tests you want to look at, follow Section 6.6.3 and then come back to this step.
5.  Select the batch file from the list (see Fig. 6-29) and click the **Confirm** button.

   *The **Review panel** will appear (see Fig. 6-27), with the first test of the selected batch displayed. To view the next or previous test in the batch, click the arrows on either side of the **Record number**.*

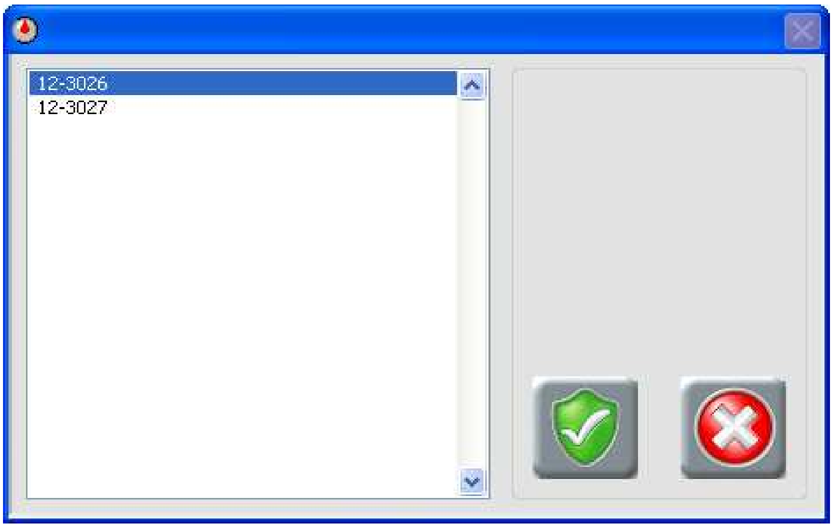
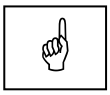


Fig. 6-29

6. If you want to create an Excel report of the test:

-  Click the **Save to Excel button**.

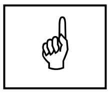


NOTE:
The system PC must have MS Excel installed and closed, to use the Save to Excel function.

- In the window that appears (Fig. 6-30), choose a location for saving the file.

  *Do this as follows: the **file path** is shown in the top section – to go up a level, click the **back button**; to add a folder to the file path, double click on the folder in the list. The files will be saved with the same name as the batch file (this is shown in the **Review panel**).*
-  Click the **Confirm** button to create the files.

Excel will open and a report for each test will be generated and automatically saved in the location that you selected. One Excel workbook will be generated for each test, with the filename 'batch name_record number.xls'. A summary report will also be generated containing a summary table of all the test results and average values for breaking load and strength. This will have the filename 'batch name_batch.xls'. An example of a test report and a summary report are shown in the Appendix, Section 8.



NOTE:
The system PC must have MS Excel installed and closed, to use the Save to Excel function.

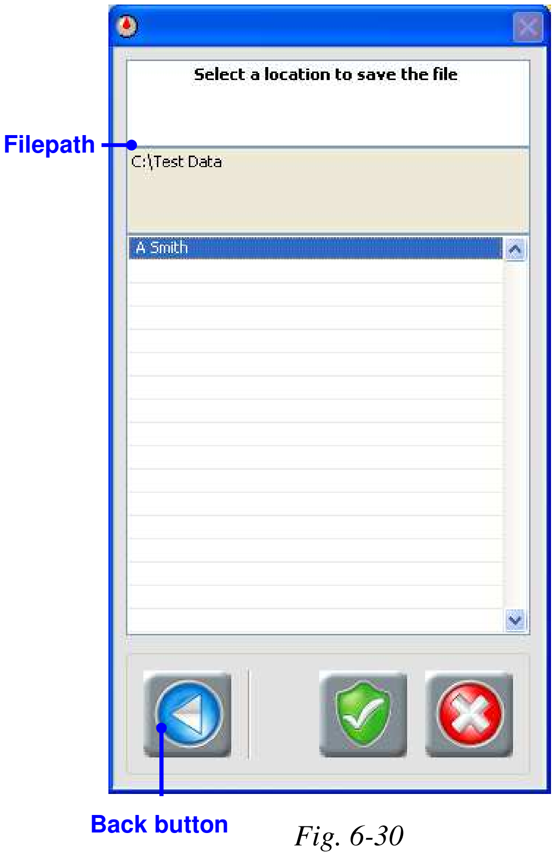


Fig. 6-30

7.  If you want to print a test report, click the **Print button**.

   *A report for each test containing all the information from the **Report window** and a copy of the graph from the **Graph window** will be printed on the PC's default printer. A summary report will also be printed containing a summary table of all the test results and average values for breaking load and strength.*

8.  When you've finished, click the **Close button** to close the **Review panel**.

9.  Click the **Close button** on the **Results panel** to return the **DataManager Main panel**.

6.7 KLAB/E

Introduction

The innovative new approach from CONTROLS now allows AUTOMAX MULTITEST to be a fully integrated and "connected" part of your laboratory infrastructure with the ability to seamlessly take inputs from any number of ancillary measuring systems and devices to further increase efficiency and eliminate transposing errors.

Seamless device integration

Compatible direct-input devices include dimensional measuring stations, calipers, weighing systems and ID barcode readers. Direct acquisition provides a tidier operation eliminating the possibility of data transposition errors.
K-LAB Enterprise is available for new and existing systems controlled by PC via Datamanager Software.

Automatic test results export

It allows direct acquisition, according to the device/s connected, of sample weight, dimensions and identification number.
These data, along with all the relevant test results, are collected and available for:
-Direct use in several formats such as txt, excel, pdf and access (Enterprise version).
-Raw data export to the Laboratory Information Management System (LIMS) or laboratory/ corporate ERP system.
-Full data integration with Prolab.Q Laboratory Information Management System or similar.

Activation of the module

The software must be registered to use it. A trial period of 31 days begins when the software is installed. After the trial period ends, it will not be possible to use the software without applying to CONTROLS for a registration code.
All the functions of the software are fully operational during the trial period.

NOTE:
The software uses the system clock of the PC to work out how many days of a trial period remain. If the PC date or time is changed after the software is installed, it will no longer be possible to register the software on the PC.

During the trial period, the registration window (Fig. 6-31) is displayed every time the software is started. To use the software without registering, click the **Skip** button. The number of days remaining in the trial period is displayed in red at the top of the window.
The software can still be registered even if the trial period has expired.

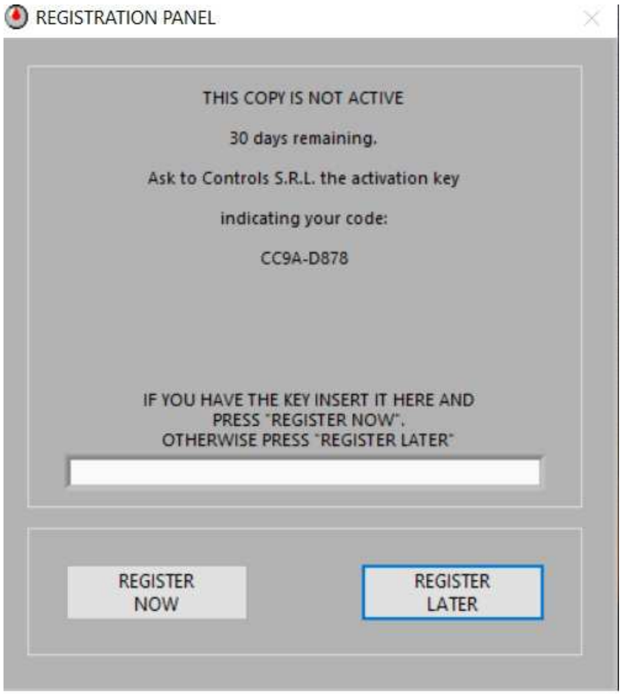
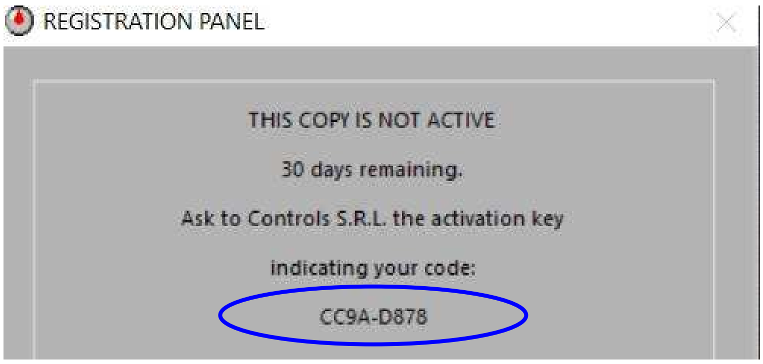


Fig. 6-31

The procedure to register the module is as follows:

1. Open the module Registration Panel.

2. On the Registration Panel that appears when the module starts (Fig. 6-31), you will see a **Registration number**.

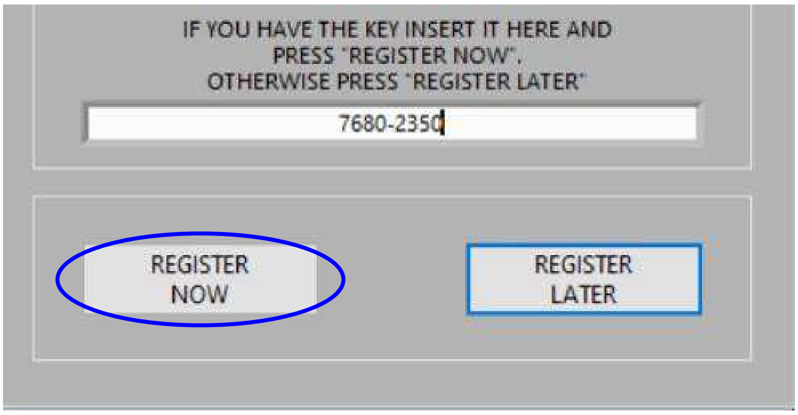


3. Contact CONTROLS After Sales Service (preferably by emailing service@controls.it), give them the **Registration number** and ask for an **Unlock code** for the module KLAB/E.

NOTE:
You must tell CONTROLS your **Registration number** and the name of the module you want to register to receive an **Unlock code**.
You should also include the acknowledgment (ACK) number of your purchase and the name of your company in any correspondence.

*To continue using the software while you wait for the **Unlock code**, click the **Skip** button on the registration window.*

4. When you receive the **Unlock code**, insert the activation code, and click the **Register Now** button on the registration window.



The unlock code should have 9 in the format XXXX-XXXX.

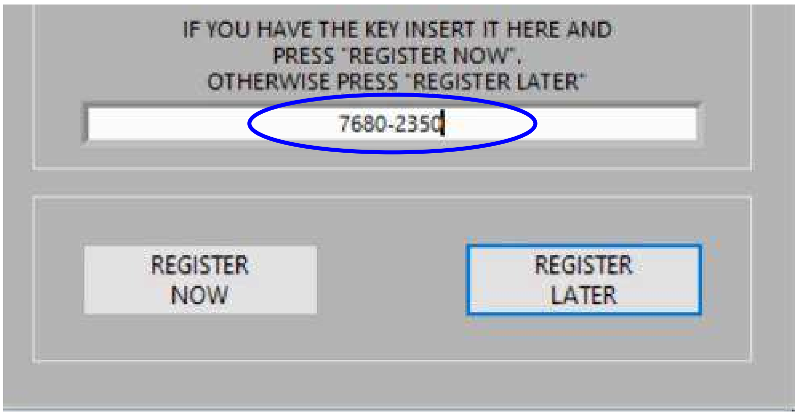


Fig. 6-32

A message will tell you if registration was successful (see Fig. 6-33).

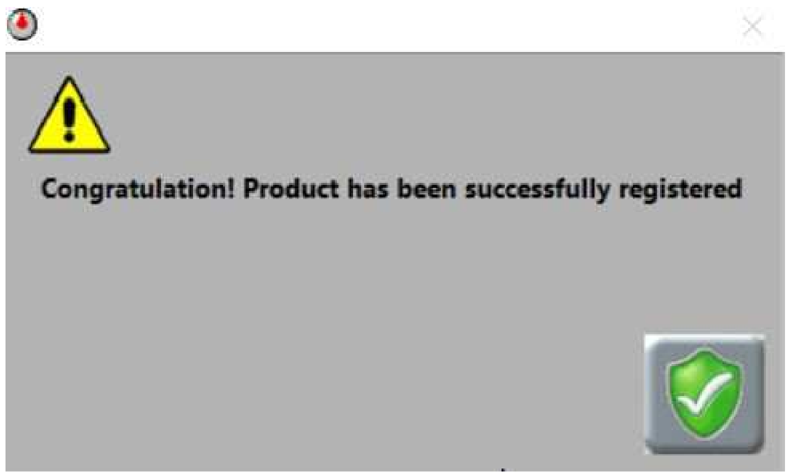


Fig. 6-33

5. Click the **CONFIRM** button to open the module.

Activation troubleshooting

If you followed the steps described before but the module registration failed, check the following things before contacting CONTROLS:

1. Did you provide CONTROLS with the correct **Registration number**?

   *If you made a mistake in the number you gave to CONTROLS, the **Unlock code** will be wrong. Check the number and repeat steps 3 to 5 of the previous Section.*

2. Did you enter the **Unlock code** correctly?

   If you enter the wrong code, make a mistake in the code or enter too few or too many characters, the code will not work and you will see an error message.

3. Is your **Registration number** '0'?

   This means that the User Account Controls have not been set correctly and it is not possible to register the module on the PC.

4. Has the clock or calendar of the PC been changed since the module was installed?

   Changing the time or date of the PC will lock the module and it will not be possible to register the module on the PC.

5. Has someone attempted to bypass the registration procedure?

   Attempting to bypass or hack the registration procedure will lock the software and it will not be possible to register the module on the PC.

List of devices compatible with KLAB/E:

The following devices are compatible with KLAB/E:

- 11-D0630/24K, 24000 g cap. x 0.1g resolution, digital balance. 230V/50-60Hz/1ph.
- 11-D0630/30K, 30000 g cap. x 0.1g resolution, digital balance. 230V/50-60Hz/1ph.
- 82-D1655/K, Digital caliper 300 x 0,05 mm
- 82-D1950/K, Barcode reader

NOTE: Controls SPA does not guarantee the compatibility of other similar instruments and will not assist in the connection of these instruments unless specifically agreed during purchasing.

Instruments connections

IMPORTANT: Before starting the software, all the following instructions must be followed, otherwise the instruments won't work properly.

- Balance

Make sure the RS232 cable is connected to the instrument and connect the other end of the cable to the PC using a suitable RS232-USB converter (we suggest our Q0800/3 RS232/USB converter). Remember also to install the correct drivers for the device as provided by the converter's manufacturer.
The settings required for the communication between balance and software are:
Baud rate: 9600 baud
Autoprn: ON
To change these settings on the balance please refer to its manual.

- USB caliper

To connect the caliper, connect it to a USB port available on the PC. Remember also to update the connection drivers supplied with the instrument to ensure it communicates properly with the software.

- Bluetooth caliper

To connect the bluetooth caliper, the following procedure must be followed the first time that the caliper is connected:
Download the caliper's own graphic interface. If the PC in use doesn't have a bluetooth board, connect the USB bluetooth device to a USB port available on the PC. Pair the bluetooth caliper to the PC through PC settings. Open the caliper's graphic interface and pair the the caliper to this interface. The caliper is now ready to enter measurements in SW/DM software.

- Barcode reader

To connect the barcode reader, connect it to a USB port available on the PC. When done, a sound should be heard from the instrument meaning that connection is established.

Follow the instructions on the device's instruction manual and enable the following settings:
- USB keyboard
- Automatic "Carriage Return" command

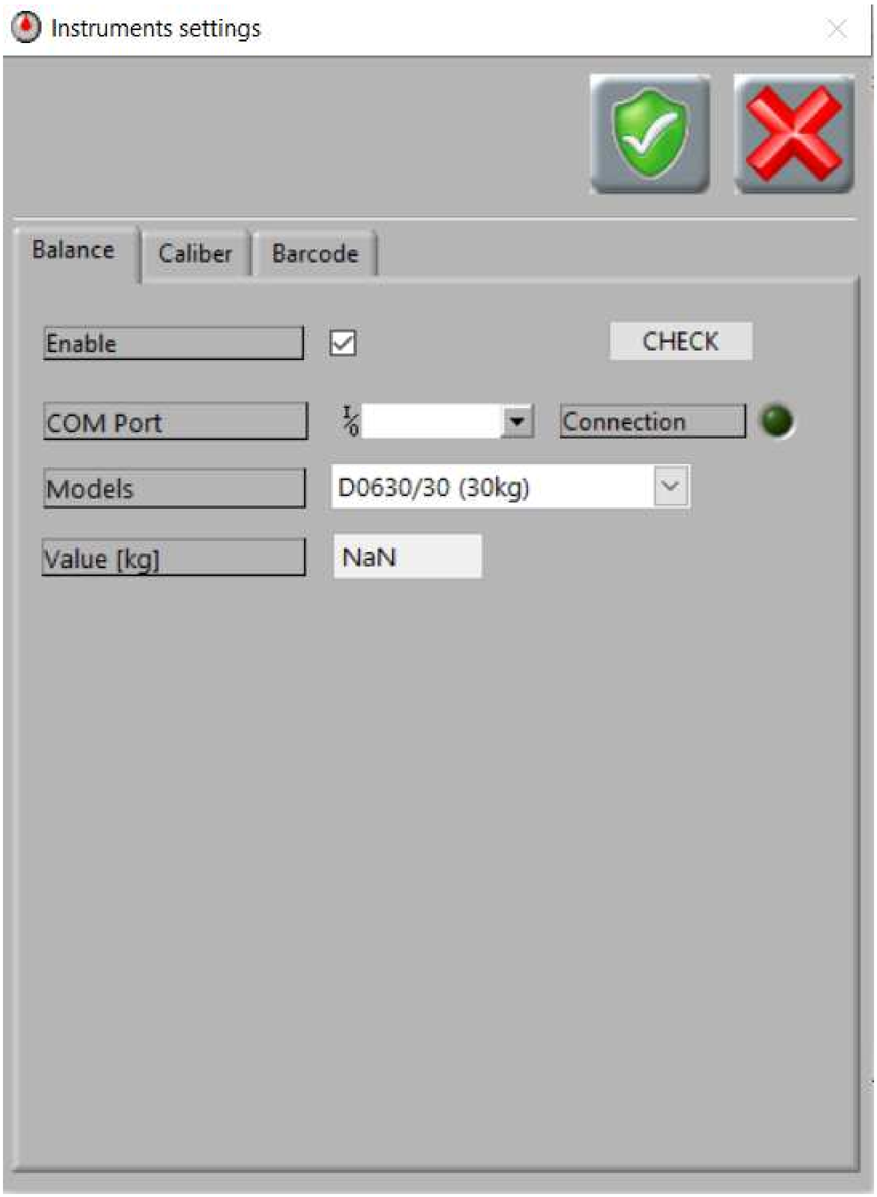
Once all the instruments have been connected, take note of the enumeration of the COM Port used by each device using the Device Manager tab in Windows' Control Panel.

Instruments settings

Before entering the acquisitions in the test menus, the user must first insert the settings of the various devices connected to the machine and check if they do work properly. To do so, click on the Instruments settings button, located in the Options screen on the Multitest main panel.

A new screen will appear, with three tabs dedicated to the three types of instruments.

Balance



Enable: Enable the balance dimensional acquisition and allows the other settings.

CHECK: Click to ping the selected COM Port to check the connection to the balance.

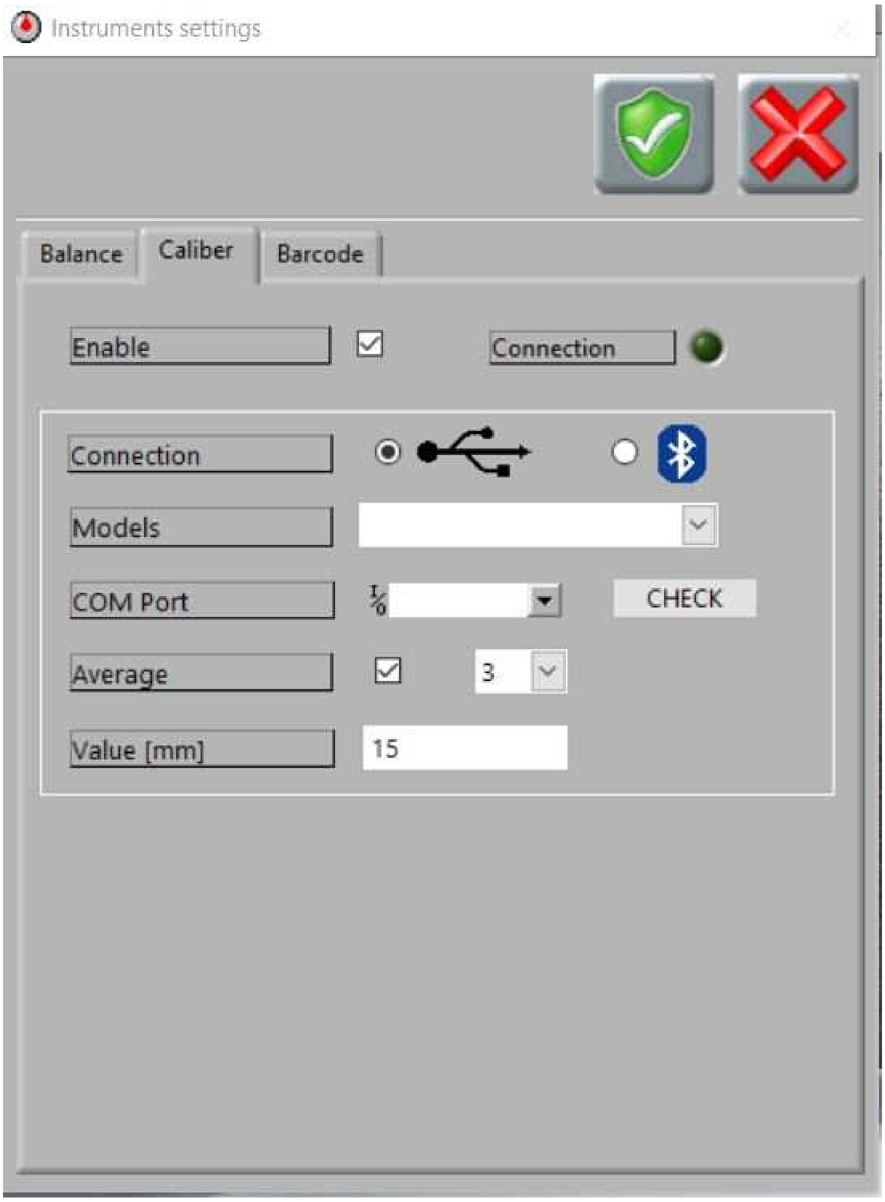
COM Port: Select the COM Port connected to the balance, as in the Device Manager tab in Windows' Control Panel.

Connection: If green, the connection is established with the balance.

Models: Select the balance model in use.

Value: the communication between the balance and the software is continuous, so this field should be filled automatically when the communication is correctly established.

Caliper



Enable: Enable the Caliper dimensional inputs and allows the other settings.

Connection Led: If green, the connection is established with the caliper.

Connection: select if a USB or bluetooth caliper is in use.

Models: Select the caliper model in use.

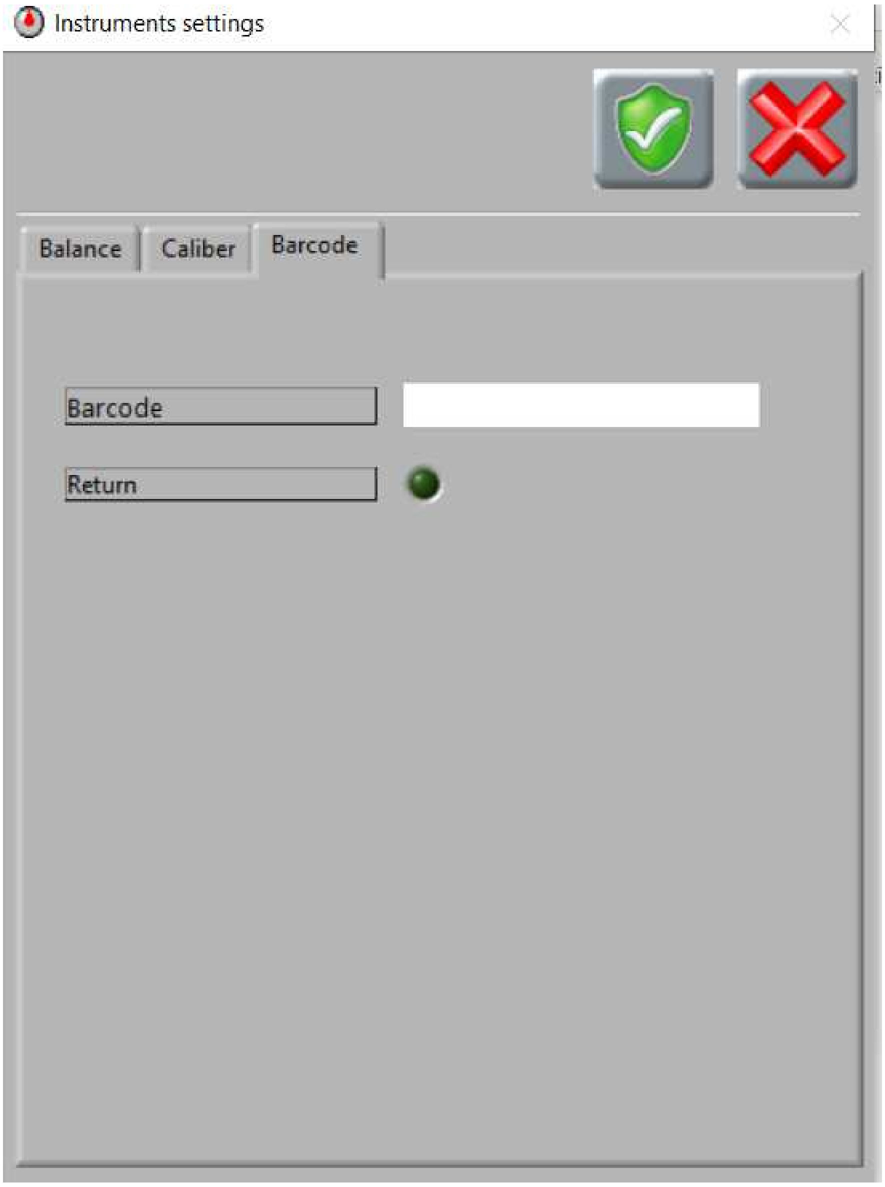
COM Port: Select the COM Port connected to the caliper, as in the Device Manager tab in Windows' Control Panel.

CHECK: Click to ping the selected COM Port to check the connection to the caliper.

Average: check the box to automatically calculate the average of "x" measurements for each dimensional field instead of taking the first measurement, where "x" is a number between 2 and 5 selectable from the drop down menu.
Example: if "3" is selected, the system will take 3 measurements for the same dimension before switching to the next one, calculating the average.

Value: when pressing the “send” button on the caliper, the same number read on the instrument should be shown in this field. If nothing happens, check the connection of the caliper.

Barcode



Barcode: shows the alphanumeric reading taken by the barcode reader each time the trigger is activated

Return: If the connection is correctly established and the barcode reader is set as “return after reading”, the green light shall appear each time the trigger is activated.

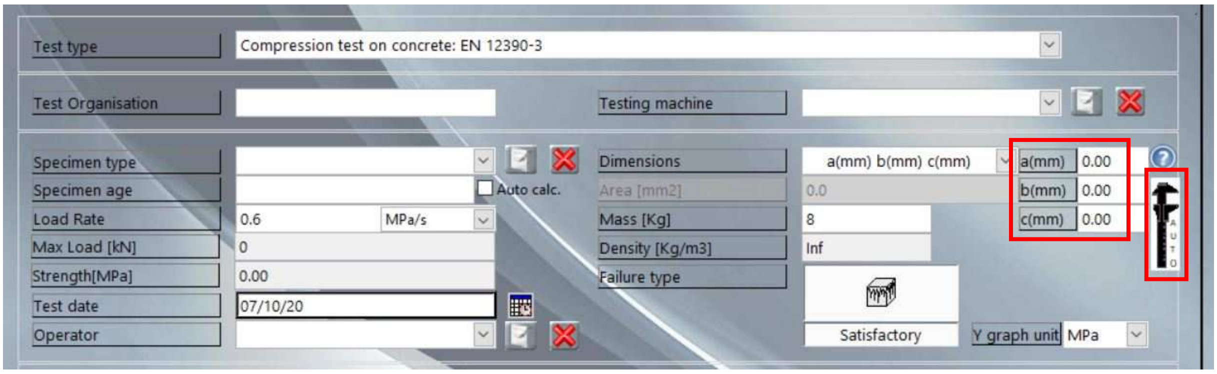
How to input specimen information through connected instruments

This chapter will show how to input specific information through the devices installed.
It's not needed to have all the three devices connected, and it's still possible to insert all data manually.

Caliper

NOTE: Caliper can be used to insert dimensional values only when the selected test is "Compression test on concrete: EN12390-3", "Compression test on concrete: ASTM C39" or "Cement test: EN 196-1".

When the caliper is connected, the specimen dimensional fields can be filled with information coming from the device rather than inserting by hand. An example of the specimen dimensional fields, for compression test on concrete cubes, is in the following picture:

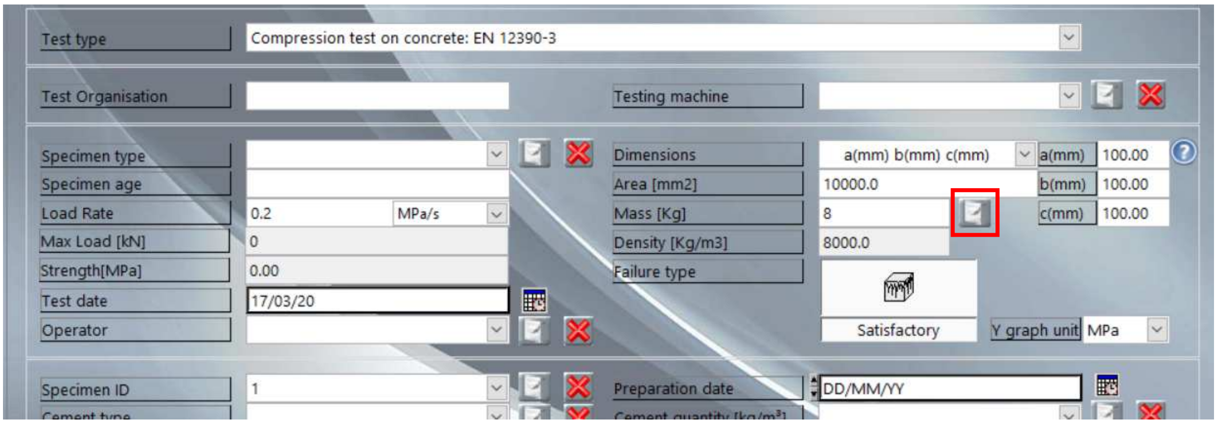


To insert the value, after ensuring the correct communication as seen in the previous chapter, select the field to fill, measure the dimension requested, and press the SEND button on the caliper. The reading should appear in the box.

"Auto switching" measurements: by pressing this button, the user doesn't need to switch manually from one dimensional field to another. The software, after the first measurement (or its average if active) is taken, will switch to the second automatically.

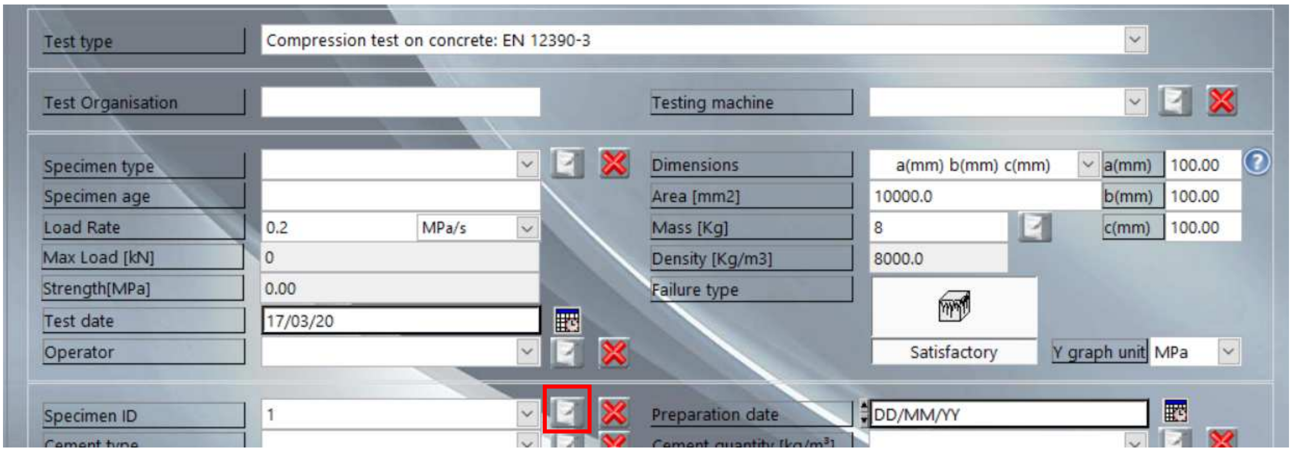
Balance

When the balance is connected, the specimen Mass field can be filled with information coming from the device rather than inserting by hand.
After placing the specimen on the balance, wait for the reading to stabilize itself and press the Balance acquisition button at the right of the field to fill it.

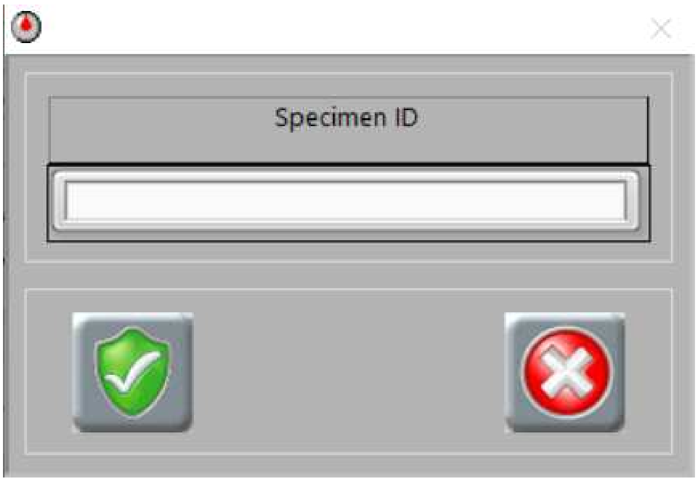


Barcode

When the barcode is connected, the specimen ID field can be filled with information coming from the device rather than inserting it by hand.
To do so, click on the Edit button at the right of the Specimen ID field.



This screen should appear:



Make sure the field is selected (there should be a line at the middle flashing), aim at the barcode on the specimen, and press the trigger of the barcode reader to acquire the data.
Press SAVE to confirm or CLOSE to cancel.

6.8 CMDqc

DataManager – Command Alkon COMMANDqc integration

CONTROLS developed a two-way communication protocol that allows DataManager to communicate with a proprietary database using "*http-PUT*" commands, which allow reading and writing on Command Alkon's COMMANDqc database. These communication functions have been developed for the DataManager software module.

Communication between DataManager and COMMANDqc is possible both using the connection of two different PCs through LAN ports or installing the two software on the same PC.

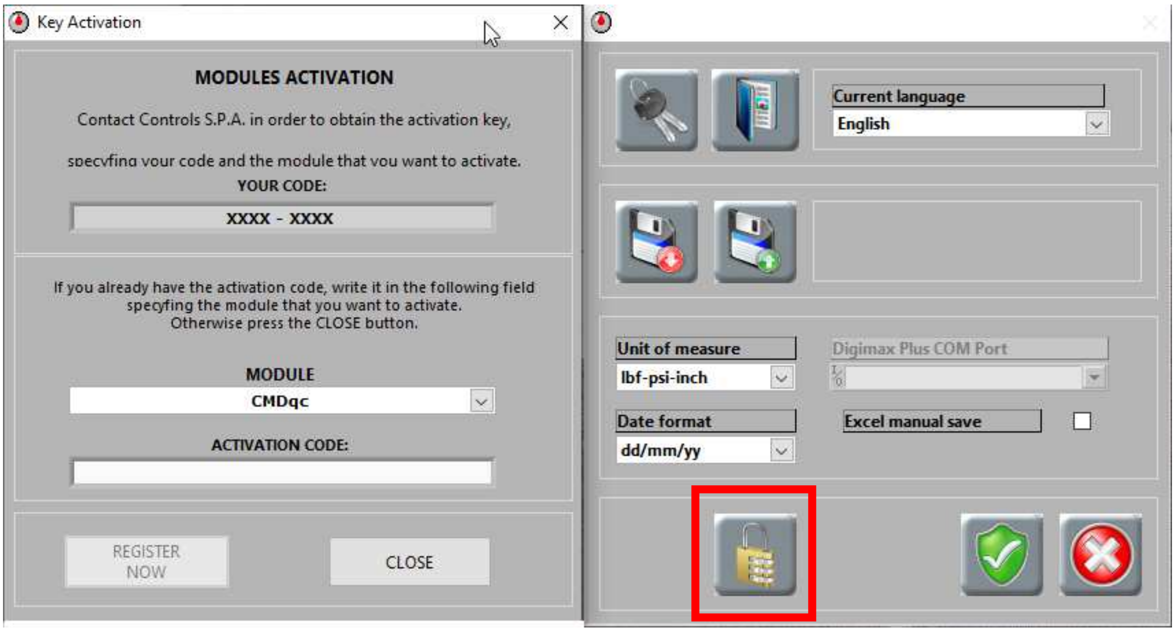
These COMMANDqc integration functions can be found in a submenu of the DataManager module, which needs an activation code to be enabled. To do so, please follow these steps:

- Open the SETTINGS menu on the main MULTITEST software

- Click on the MODULE ACTIVATION key (the one with the lock icon)

- Contact CONTROLS sending the code generated by your PC (YOUR CODE)

- Once the activation code has been received, paste it in the ACTIVATION CODE field, making sure that CMDqc MODULE is selected.



DM.ini file settings

Before launching the software for the first time, it is necessary to verify and modify the settings in the main file DM.ini, to make sure that CMDqc will properly work. The file can be found in MULTITEST path, in the folder C:\Users*yourPCname*\Documents\MULTITEST\data\MODULES\DM, and can be opened with Windows' Notepad.

Scroll down or use the search option to find these sections, verify the parameters, and make changes if needed:

[QC_Info]
Server = "ABCD-EFGH-ILMN\SQLEXPRESS"
Database = "CMDqc_Demo"
User = "conadgpm"
Lab = 1

[QC_Server]
IP_Server = "localhost"
Port_Server = 9010

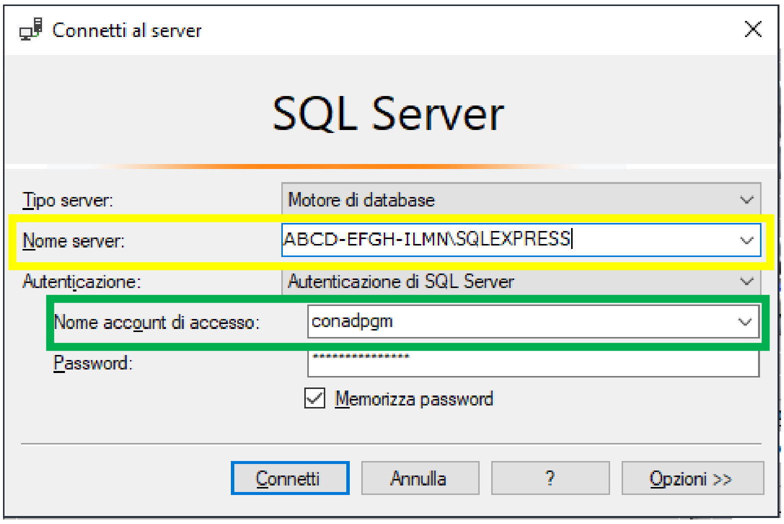
Notes for [QC_Info] Sections:

- Server: it's the server name that targets to the database currently in use. It's used by Microsoft SQLServer as a DB manager. Exchange "ABCD-EFGH-ILMN\SQLEXPRESS" with the information found in the field "Server name" in the starting page of SQLServer (in yellow in the next picture)

- Database: Database name in SQLServer

- User: username to connect to the Database (in green in the next picture)

- LAB: Reference COMMANDqc laboratory.



Notes for [QC_Server] Sections:

- IP_Server: IP address of the server where SQLServer and the database are located. If both Datamanager and COMMANDqc are on the same pc, insert "*localhost*"

- Port_Server: Server connection port (default: *9010*)

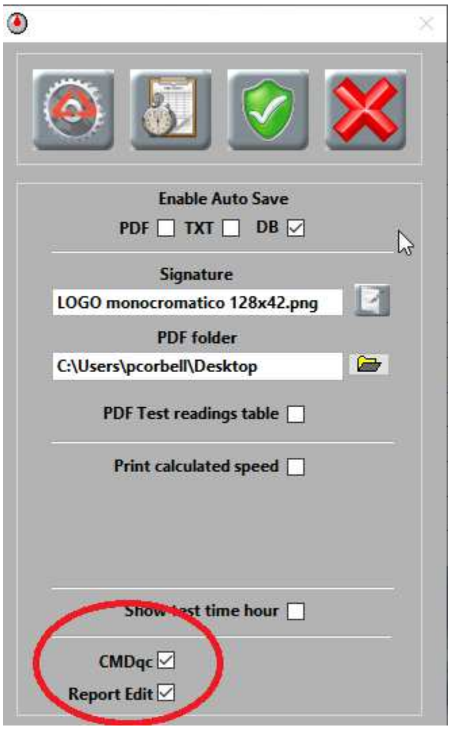
CMDqc activation

Start DataManager and enter the software settings through this key

Tick the "CMDqc" option. When activated, CMDqc mode will update the Datamanager interface with data from COMMANDqc and will block any attempt of modification to them.

If modifications are needed, tick also the "Report Edit" option.

Anytime using Datamanager it will be possible to disable CMDqc mode. This is necessary when testing specimens not managed by the COMMANDqc database.



CMDqc <-> DM fields matching

The combination between the COMMANDqc fields' meaning and the Datamanager ones' is required to obtain all the information correctly in the test screen.

QC_Conversion.ini

This file is created automatically in the "*data\MODULES\DM*" folder at the first Datamanager boot, and at the beginning, it contains only default values. These values are all the possible answers that can be sent from COMMANDqc when Get_List command is issued (see next)

Example:

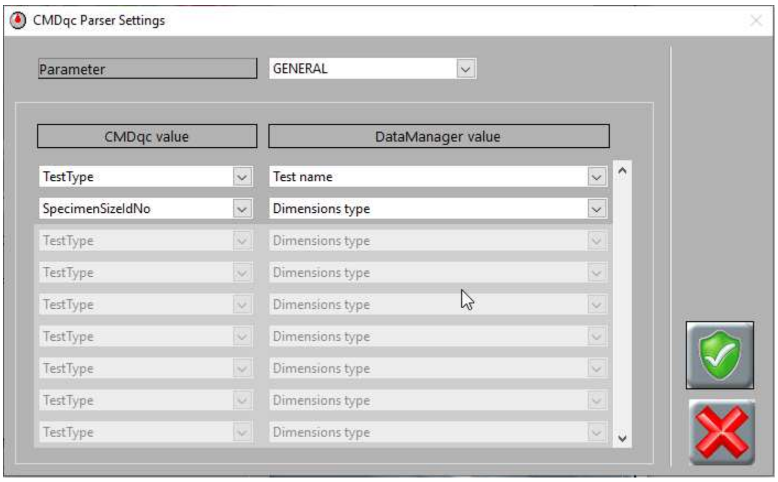
If the "*Indirect tensile test*" is selected as test type in COMMANDqc, the answer to the *Get_List* will have as *TestType* node a value = "I". In this case, the file "*QC_Conversion.ini*" will contain, in the section [*TestType*], the key: *Indirect Tensile Test = "I"*.

CMDqc <-> DM field matching settings

These setting are found using the key 

By pressing the key, Datamanager interrogates *QC_Conversion.ini* and shows in the *Parameter* drop-down menu as many fields as the sections in the uploaded file, plus the *GENERAL* one.

The *GENERAL* section is used to associate the DataManager fields to the COMMANDqc ones, as shown in the picture. In case of errors is possible to delete the undesired fields by pressing the right-click of the mouse and select *Delete element*.

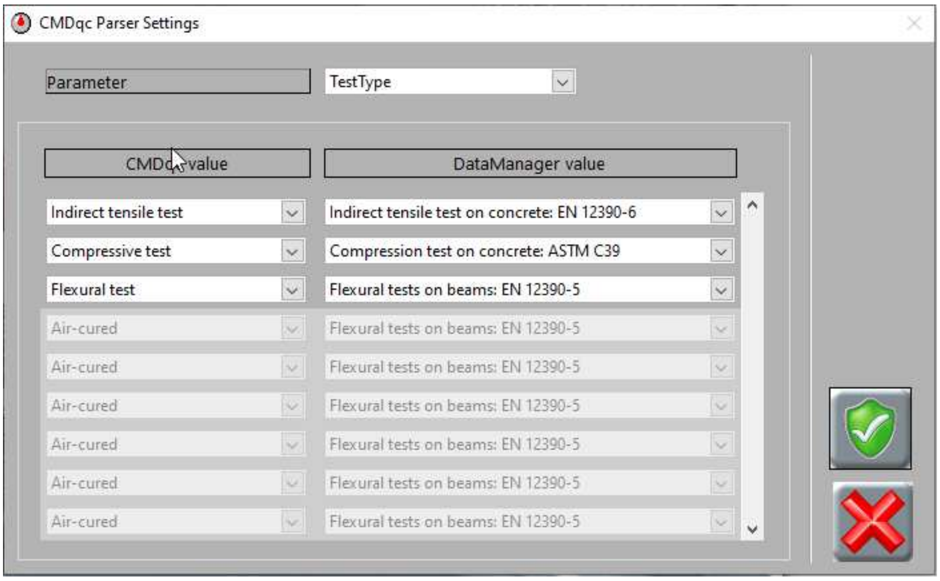


NOTE: these associations are needed only the first time, and before proceeding with the next instructions!

When *GENERAL* is selected, the CMDqc column is populated with the sections' names found in the QC_Conversion.ini file. When a parameter is selected, instead, the column CMDqc will be populated with the keys found in that section.

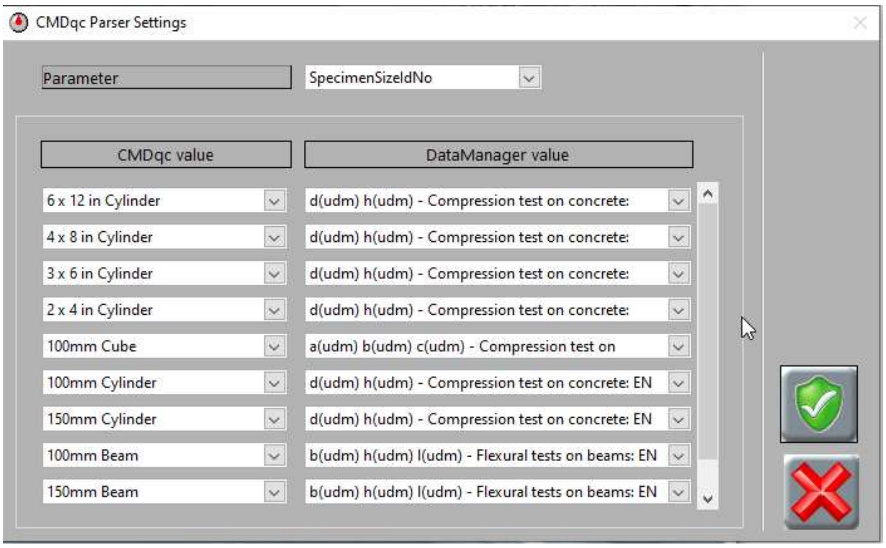
TEST TYPE section associates the test types available on COMMANDqc with those on Datamanager. All the test tyes available on Datamanager can be found in this section, including the user-customized ones. In case of errors is possible to delete the undesired fields by pressing the right-click of the mouse and select *Delete element*.

NOTE: these associations are needed only the first time, and modifications will be possible anytime.



Specimen size ID is used to associate the specimen types available on COMMANDqc with those on Datamanager. All the specimen types available on Datamanager can be found in this section, and each one is associated with one specific test type.

For example, the Cylinder 6”x12” will be associate to the specimen having “*d*” and “*h*” dimensions of a compression test:



In case of errors is possible to delete the undesired fields by pressing the right-click of the mouse and select *Delete element*.

NOTE: these associations are needed only the first time, and modifications will be possible anytime.

When all the combinations are set, press the  confirm key

By doing this, all the combinations will be saved in the DM.ini file, divided by parameters in the section *[QC_Parser]*. Combinations are made by an array of clusters containing two strings:

- *QC_value*, containing the fields' value taken by *QC_Conversion.ini*

- *DM_value*, containing the information selected in the parser.

Download list of specimens to be tested

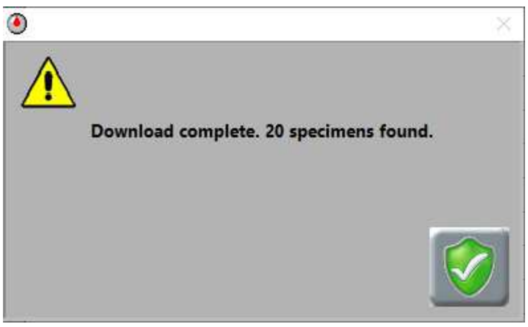
After the operations on 1.1, 1.2, and 1.3 have been performed (only the first time), it will be possible to download from COMMANDqc the list of specimens to test on that day.

- Press the key: 

- Confirm the operation: 

- Wait for the download to finish 

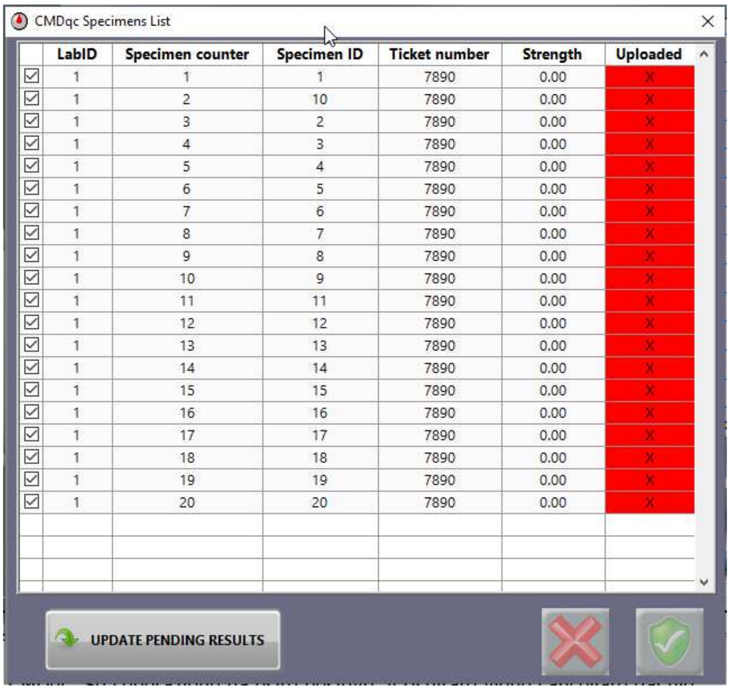
The duration of the download depends on the number of specimens and the network speed. An average of 1 second per specimen is needed. In the end, a message will be shown indicating how many specimens have been downloaded.



Summary table

It's possible to open a summary table of all the specimens downloaded and some of their information.

Press the  key to open the table.



CMDqc Specimens List

	LabID	Specimen counter	Specimen ID	Ticket number	Strength	Uploaded
☑	1	1	1	7890	0.00	X
☑	1	2	10	7890	0.00	X
☑	1	3	2	7890	0.00	X
☑	1	4	3	7890	0.00	X
☑	1	5	4	7890	0.00	X
☑	1	6	5	7890	0.00	X
☑	1	7	6	7890	0.00	X
☑	1	8	7	7890	0.00	X
☑	1	9	8	7890	0.00	X
☑	1	10	9	7890	0.00	X
☑	1	11	11	7890	0.00	X
☑	1	12	12	7890	0.00	X
☑	1	13	13	7890	0.00	X
☑	1	14	14	7890	0.00	X
☑	1	15	15	7890	0.00	X
☑	1	16	16	7890	0.00	X
☑	1	17	17	7890	0.00	X
☑	1	18	18	7890	0.00	X
☑	1	19	19	7890	0.00	X
☑	1	20	20	7890	0.00	X

UPDATE PENDING RESULTS

The table shows the specimens to be tested in the same order of COMMANDqc interface. It's possible to exclude one or more specimens clicking on the flag of the first column. Press the CONFIRM key to update the table.

NOTE: if you access the software settings after downloading the specimen list, the table will be emptied and it will be necessary to download it again.

Test execution

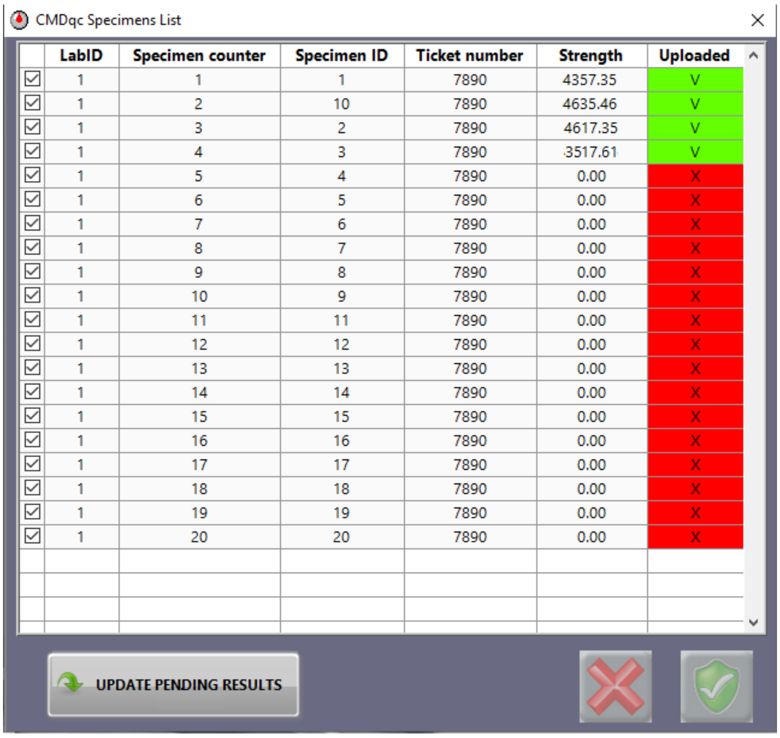
The *Report* page on Datamanager will now show the parameters of the first specimen on the list. This page will not be editable by the user if the "*Report edit*" option (see par. 1.2) is not active.

Perform the test on the specimen and wait for the machine to stop itself automatically at specimen failure.

To save and upload the results on COMMANDqc database press the key 

Alternatively, it's possible to activate the autosave in the local Datamanager Database.
NOTE: it won't be possible to perform the next test if the test is not saved!

Table and counter are updated in real-time:



	LabID	Specimen counter	Specimen ID	Ticket number	Strength	Uploaded
☑	1	1	1	7890	4357.35	V
☑	1	2	10	7890	4635.46	V
☑	1	3	2	7890	4617.35	V
☑	1	4	3	7890	3517.61	V
☑	1	5	4	7890	0.00	X
☑	1	6	5	7890	0.00	X
☑	1	7	6	7890	0.00	X
☑	1	8	7	7890	0.00	X
☑	1	9	8	7890	0.00	X
☑	1	10	9	7890	0.00	X
☑	1	11	11	7890	0.00	X
☑	1	12	12	7890	0.00	X
☑	1	13	13	7890	0.00	X
☑	1	14	14	7890	0.00	X
☑	1	15	15	7890	0.00	X
☑	1	16	16	7890	0.00	X
☑	1	17	17	7890	0.00	X
☑	1	18	18	7890	0.00	X
☑	1	19	19	7890	0.00	X
☑	1	20	20	7890	0.00	X

The test session ends when all the specimens in the list have been tested. At that point, the Start test command automatically deactivates:

Test results upload

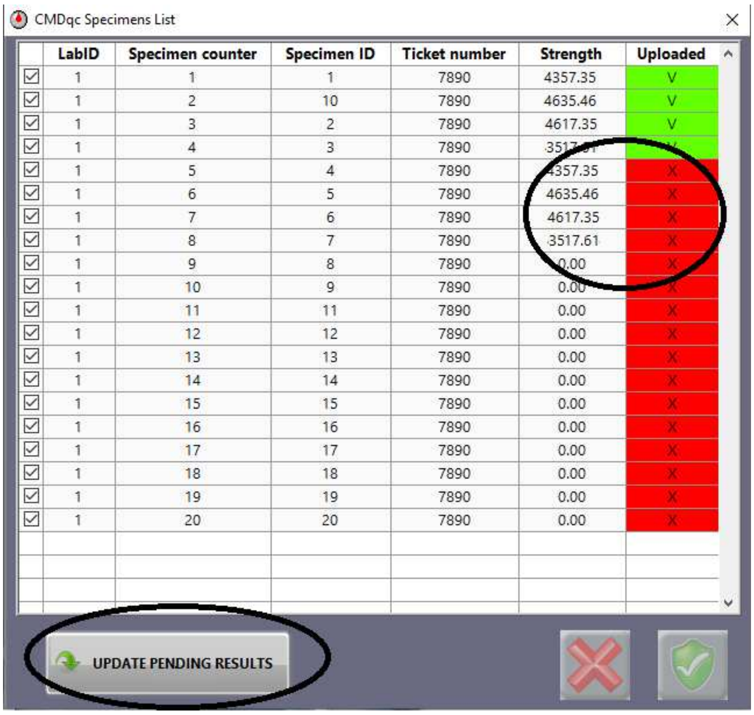
Test results upload on the COMMANDqc database is performed each time that the "save on local Database" key is pressed.

NOTE: This command is available only at the end of a failure test and only if the machine registers the actual failure of the specimen. This command won't be available if the test is stopped by pressing the STOP key!

Test results are saved on a binary file *(QC_data_to_Server.bin*) located in the folder "*data\MODULES\DM*". The file is then read and command "*Set-result"* is issued to upload the results on the COMMANDqc database. If the operation succeeds, the result gets removed on the binary file. If the operation does not succeed, the test result will be kept stored in the binary file and the upload will be performed with the next test result.

In this case, an error message will pop up.

It is possible, once the connection between Datamanager and COMMANDqc is reestablished, to manually upload the results through the “*UPLOAD PENDING RESULTS*” key.



Communication http-PUT

Communication to COMMANDqc is performed by sending a string in JSON through the command *http-PUT*.

COMMANDqc communication url is:

http://***IP_Server***:***Port_Server***/COMMANDqcBroker/service/processmsg

where IP_Server and Port_Server are the values set in DM.ini

- Specimen receiving command (Get_List)

The command sent as a JSON string is as follows:

```
{"cmdqc":
    {"request":
        {"Server":"Server",
        "Database":"Database",
        "context":"sampleManager",
        "group":"Lookups",
        "cmd":"getTestSpecimens",
        "User":"User",
        "Lab":"Lab",
        "SearchBySampleDate":"0",
        "FromDate":"2020-02-18T00:00:00",
        "ToDate":"2020-02-18T23:59:59"
        }
    }
}
```

Where:

- *Server, Database, User* and *Lab* are values found on *DM.ini*,

- *FromDate* and *ToDate* is the time span of the specimens to be tested, the 24 hours of the day.

The answer received by CMDqc containing data of the specimens to be tested is as follows:

{"cmdqc":
{"request":
{"Server":"*Server*",
"Database":"*DataBase*",
"context":"sampleManager",
"group":"Lookups",
"cmd":"getTestSpecimens",
"User":"*User*",
"Lab":"*Lab*",
"SearchBySampleDate":"0",
"FromDate":"2020-02-18T00:00:00",
"ToDate":"2020-02-18T23:59:59"},
a
"reply":
{"SpecimenList":
{"Item":[
{"LabId":"1",
"SampleNumber":"1",
"Comment":"",
"SpecimenID":"1",
"MouldNumber":"",
"TesterID":"",
"StrengthClassCode":"3500",
"SpecimenSize":"9",
"SpecimenType":"4 X 8 in Cylinder",
"GradeStrength":"3.500000000000000e+003",
"SpecimenTypeID":"2",
"SpecimenTypeDescription":"Cylinder",
"SpecimenSizeIdNo":"9",
"MouldLength":"4.00",
"MouldWidth":"4.00",
"MouldDiameter3":"4.00",
"MouldHeight":"8.00",
"MouldHeight2":"8.00",
"MouldDimensionUnitUID":"10001",
"MouldDimensionUnitAbbrev":"in",
"TestTypeDescription":"Compressive test",
"PlantDescription":"Birmingham Plant   ",
"ProductCodeDescription":"5000 - 3/4   ",
"Length":"4.000000000000000e+000",
"LengthEntry":"4.000000000000000e+000",
"Width":"4.000000000000000e+000",
"WidthEntry":"4.000000000000000e+000",
"Height":"8.000000000000000e+000",
"HeightEntry":"8.000000000000000e+000",
"Diameter3":"4.000000000000000e+000",
"Diameter3Entry":"4.000000000000000e+000",
"Height2":"8.000000000000000e+000",
"Height2Entry":"8.000000000000000e+000",

"Height3":"8.000000000000000e+000",
"Height3Entry":"8.000000000000000e+000",
"CapType":"N","SampleControlUID":"-999",
"ConditionPrior":"N",
"FailureMode":"X",
"Crushinglab":"1",
"TestType":"C",
"DocketNumber":"1",
"DocketDate":"2020-02-18T00:00:00",
"CompactionMethod":"R",
"ProductCodeID":"5002",
"PlantID":"1",
"SampleDate":"2020-02-18T00:00:00",
"TestingMachine":"",
"TestAge":"1",
"TestAgeUnitUID":"82004",
"TestAgeUnitAbbrev":"hr",
"TestAgeGroup":"Inter #1",
"CrushingTech":"",
"MeasuringTech":"",
"TestDate":"2020-02-18T00:00:00",
"CrushTime":"2020-02-18T00:00:00",
"TimeIntoTank":"2020-02-18T00:00:00",
"DimensionUnitUID":"10001",
"DimensionUnitAbbrev":"in",
"DimensionUnitUIDEntry":"10001",
"DimensionUnitAbbrevEntry":"in",
"MassUnitUID":"60002",
"MassUnitAbbrev":"lb",
"MassUnitUIDEntry":"60002",
"MassUnitAbbrevEntry":"lb",
"DensityUnitUID":"61001",
"DensityUnitAbbrev":"lb/ft3",
"LoadUnitUID":"71001",
"LoadUnitAbbrev":"lbf",
"LoadUnitUIDEntry":"71001",
"LoadUnitAbbrevEntry":"lbf",
"StrengthUnitUID":"70001",
"StrengthUnitAbbrev":"psi",
"SpecimenUID":"4291",
"ReportNumber":"",
"ReportInvoice":"",
"NotTestedReason":"",
"Exclude":"N",
"IsStrengthAccepted":"0",
"LabStdCuringDays":"-999",
"EquivalentUnits":"",
"ChargeForTest":"",
"PrintTestCert":"Y",
"StandardNormUID":"2",
"MoistureAtTestCode":"",
"CalcUsedAttributeUID":"-1",

"UseAlternativeUnits":"0",
"QCDocketUID":"1225"},
{…}
]
},"RowCount":"2",
"logUID":"11488"
},
"error":"",
"errParseStart":"09:14:57.639",
"errParseEnd":"09:14:57.639",
"brkEnd":"09:14:57.641"
}
}

Specimens are translated from JSON string to Typedef "*DM_Result_control.ctl*" through the research of the nodes. The array of the specimens already tested and to be tested of the requested time span is saved in a Functional Global Variable.

NOTE: in case a specimen has already been tested, COMMANDqc answer will be slightly different and will have also the node "*Strength*" that usually is not present. This node has the Strength result as value and, if available and different from 0, it means that the specimen has already been tested.

Saving results command (Set_Result)

The command, sent as JSON string for saving results, is as follows:

{"cmdqc":
{"request":
{"context":"sampleManager",
"group":"Lookups",
"cmd":"setSpecimenStrength",
"Server":"***Server***",
"Database":"***DataBase***",
"SpecimenUID":"***SpecimenUID***",
"EntryMode":"803",
"Load":"***Load_Result***",
"LoadUnitUID":"70001",
"Strength":"***Strength_Result***",
"StrengthUnitUID":"70001",
"MORBreakDepth":"",
"MORBreakDistance":"",
"MORBreakWidth":"",
"FracturePointUnitUID":"70001",
"TestDate":"***2020-02-18T13:12:11***"
}
}
}

where:

- *Server* and *Database* are values found in DM.ini,

- *SpecimenUID* is the univocal reference of the specimen (obtained when *Get_List* is launched),

- *Load_Result* and *Strength_Result* are the test results (Max load and stress),

- *TestDate* is the date and hour of test execution.

The answer received by CMDqc is as follows:


```
{"cmdqc":
    {"request":
        {"context":"sampleManager",
        "group":"Lookups",
        "cmd":"setSpecimenStrength",
        "Server":"Server",
        "Database":"DataBase",
        "SpecimenUID":"SpecimenUID",
        "EntryMode":"803",
        "Load":"Load_Result",
        "LoadUnitUID":"71011",
        "Strength":"Strength_Result",
        "StrengthUnitUID":"70011",
        "MORBreakDepth":"",
        "MORBreakDistance":"",
        "MORBreakWidth":"",
        "FracturePointUnitUID":"10011",
        "TestDate":"2020-02-18T13:12:11"
        },
    "brkStart":"17:52:50.900",
    "reply":{
        "success":"T",
        "logUID":"11100"
        },
    "error":"",
    "errParseStart":"17:52:50.910",
    "errParseEnd":"17:52:50.910",
    "brkEnd":"17:52:50.915"
    }
}
```


If the results uploading operation succeeded, the *success* node will have *T* (true) as value, if not *F* (false).

6.9 Closing the software

1.  Click the **Close button** on the **DataManager Main panel** to exit the test module.
2.  Click the **Close button** on the **MULTITEST Main panel** to close the software.

7. MAINTENANCE

As with all electrical equipment, this unit must be used correctly and maintenance and inspections must be performed at regular intervals. Such precautions will guarantee the safe and efficient functioning of the equipment.

Periodic maintenance consists of inspections made directly by the test operator and/or by the authorized service personnel.

Maintenance of the equipment is the responsibility of the purchaser and must be performed as stated by this chapter.

Failing to perform the recommended maintenance actions or maintenance performed by unauthorized people can void the warranty.

WARNING:
Failing to perform the recommended maintenance actions or maintenance performed by unauthorized people can void the warranty.
CONTROLS will not be responsible for maintenance and service actions performed by unauthorized people.

WARNING:
Before opening/removing the covers, disconnect the mains supply to the device, and wait at least 5 minutes.

WARNING:
All safety devices must be functional at all times. Damaged protective covers or devices must be replaced immediately. When safety components are replaced, the protective devices are to be properly attached and tested. Any manipulation of the safety devices endangers the operating personnel.

WARNING:
Covers/doors can only be removed/opened by maintenance people, by using relevant tools/buttons. These tools and buttons are for the use of maintenance people only. Never leave them attached to the unit as this may endanger operator safety.

After performing maintenance/repair, make sure that all covers/doors are properly closed and locked.

WARNING:
Due to the high pressure inside the hydraulic circuit (up to 700 bars/10,150 psi), only tubes approved by CONTROLS can be used. CONTROLS will not be responsible for damages to the equipment or personnel injures in case of the use of improper material and/or operations not carried out by authorized technical personnel.

WARNING:
For continued fire protection, replace fuses with the same type and rating. Also, in case of failure, components may only be replaced by using original spare parts. It is in the responsibility of the purchaser to ensure that fire prevention policies are properly implemented according to the CE provisions.

WARNING:
Avoid pouring water, even accidentally, or other liquids into the device, as this could cause short circuits. Before cleaning the device, disconnect it from the mains line.

WARNING:
To avoid direct contact with the oil of the hydraulic circuit, always wear protective gloves while performing the checking.

7.1 Operator’s preventive maintenance

The inspections made directly by the operator are the following:

Action	Who	When
Check to ensure that there is no external damage to the equipment, which could jeopardize the safety of use	Operator	Before every working session
Clean and dry platens, distance piece and the ram of the frame from the sample debris	Operator	At the end of each working session
General inspection	Operator	Weekly
Check the status of the hydraulic tubes against leakages and damages	Operator	Weekly
Check status of the safety devices: ***EMERGENCY BUTTON***, max. pressure valve, piston over travel switch and relevant cable on the frame, front door and rear transparent guard on the frame (refer to chapter 1.5 for their identification)	Operator	Weekly
Check status and functioning of the operator's commands	Operator	Weekly
Check the level of the oil in the tank (see chapter 7.1.1)	Operator	Monthly
Check that all label and rating plates are intact and properly attached	Operator	Monthly

NOTE:
The previous table indicates also the maintenance actions to be performed on the frames normally connected to the control console.

7.1.1 Checking the oil level in the tank

WARNING:
To avoid direct contact with the oil of the hydraulic circuit, always wear protective gloves while performing the checking.

Check the oil level in the tank via the transparent level indicator (the oil should be max. at half of the indicator); if necessary refill the oil tank.

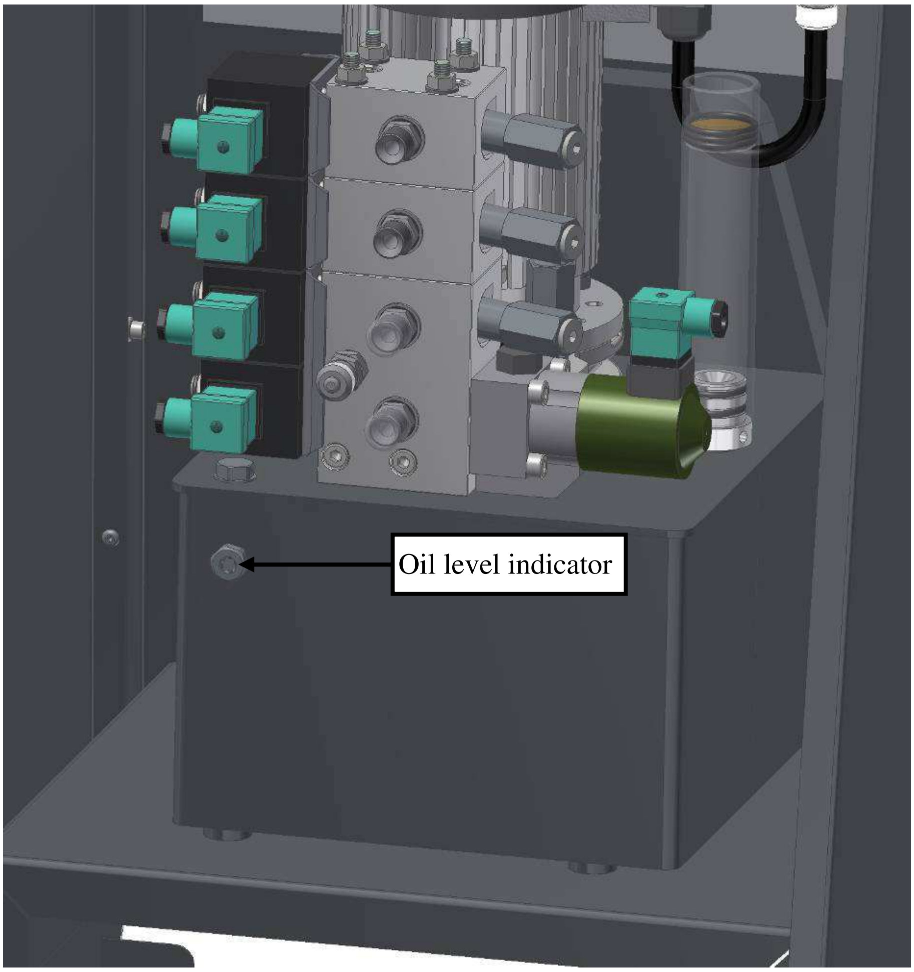


Fig. 7-1

The following type of oil can be used:

Brand	Type
Agip	ATF Dexron
Api	Apilube ATF Dexron
Chevron	Chevron automatic transmission fluid
Elf	Elfmatic G
Exxon	Automatic transmission fluid
Fiat	GI/A
Fina	Dexron ATF
Gulf	Automatic transmission fluid
IP	Dexron fluid
Mobil	ATF 220
Shell	ATF Dexron II
Total	Dexron
Vanguard	ATF fluid DEX II

7.2 Authorized service engineer maintenance actions

In addition to the maintenance actions performed by the operator, the performance of the equipment is checked and, if necessary corrected, during the maintenance activities performed by the authorized service engineer, following the indications provided in the present manual.

The following table lists the maintenance actions and the relevant timing:

Action	Who	When
Checking the centering of the ram assembly	Authorized service engineer	Yearly or before carrying out the calibration of the load channel(s) and the checking of the stability of the frame through the force transfer test with the strain gauged column
Checking the hydraulic circuit	Authorized service engineer	
Checking the status of distance pieces	Authorized service engineer	
Performing the maintenance of the spherical seat	Authorized service engineer	
Checking the calibration of the load channel	Authorized service engineer	Yearly or more frequently depending on the quality procedures of the laboratory
Checking the stability of the frame through the force transfer test with the strain gauged column	Authorized service engineer	Yearly or more frequently depending on the quality procedures of the laboratory
Status of the internal and external cables wear and tear and fastenings	Authorized service engineer	Yearly
Grounding of all the accessible conductive parts	Authorized service engineer	Yearly
Replacing the oil of the hydraulic circuit (see chapter 7.2.1)	Authorized service engineer	Every 3 years or more frequently in very cold or humid environments (where condensation may form in the oil tank)

WARNING:
Refer to qualified service organization authorized by CONTROLS to carry out the maintenance actions described in the chapter "Authorized service engineer maintenance action". CONTROLS has not to be held responsible for damages to the equipment and/or injuries to personnel in case the above is not strictly followed.

NOTE:
The previous table indicates also the maintenance actions to be performed on the frames normally connected to the control console.

7.2.1 Replacing the oil of the hydraulic circuit

WARNING:
To avoid direct contact with the oil of the hydraulic circuit, always wear protective gloves while performing the checking.

Periodically the oil of the hydraulic circuit must be completely replaced.

1. Discharge the exhaust oil via the drainage screw located on the bottom of the oil tank;

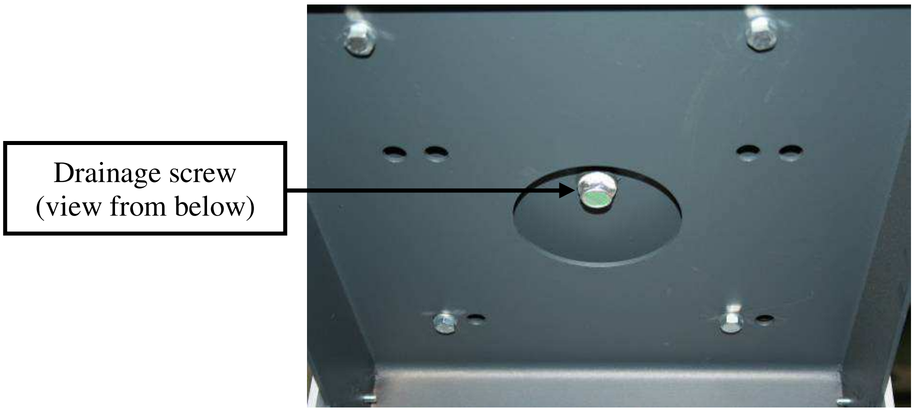


Fig. 7-2

2. Collect the old oil and dispose of it following the local regulations;

3. Refill the tank (9 liters approx.) with the oil type shown in next table;

Brand	Type
Agip	ATF Dexron
Api	Apilube ATF Dexron
Chevron	Chevron automatic transmission fluid
Elf	Elfmatic G
Exxon	Automatic transmission fluid
Fiat	GI/A
Fina	Dexron ATF
Gulf	Automatic transmission fluid
IP	Dexron fluid
Mobil	ATF 220
Shell	ATF Dexron II
Total	Dexron
Vanguard	ATF fluid DEX II

4. Let the pump run and check the oil level in the tank via the transparent level indicator (the oil should be max. at half of the indicator);

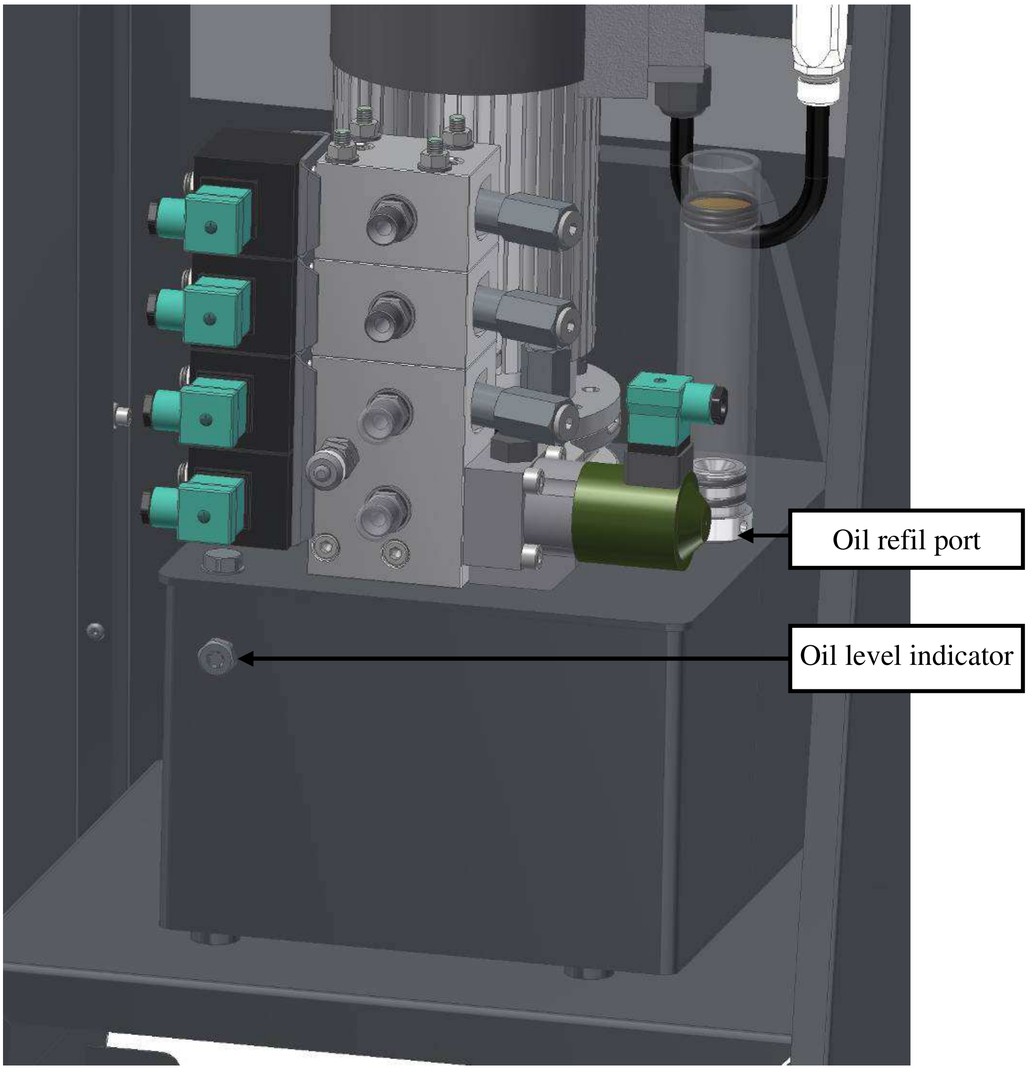


Fig. 7-3

5. Top up if necessary.

7.3 Edit calibration

7.3.1 Syncronyze the calibration data between device and software

1.  Click the **Channel configuration button** on the **MULTITEST Main panel** (section 5.2). Screen will prompt WAIT while calibration data is retrieved by the software from the device.

2. Select an active (used) channel from the channel configuration list (Fig. 7-4). Double-click anywhere on the row to open the **Calibration panel**.

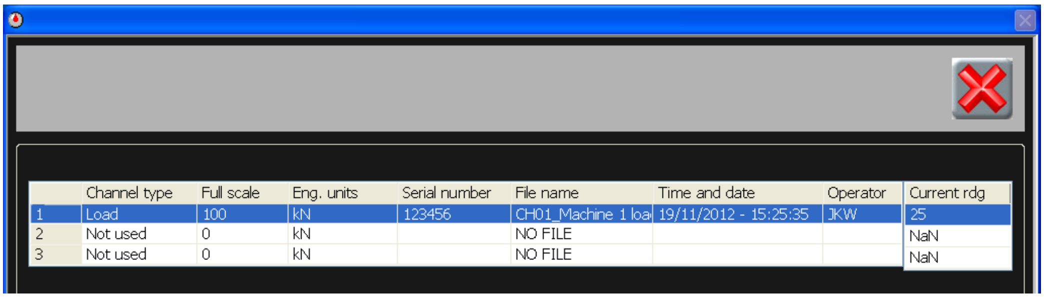


Fig. 7-4

3.  Click the **Edit calibration button** to unlock the panel.

4. Enter the level 1 password (“controls1”).

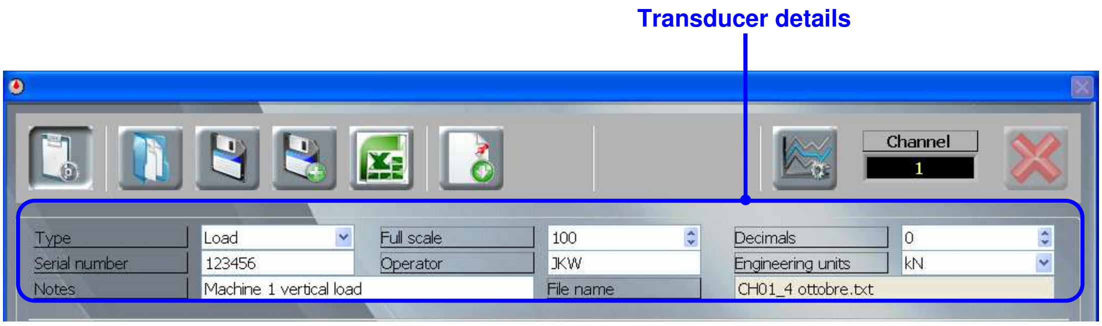
5. The data retrieved from the device and shown in the **Edit Calibration Screen** shown below should be correct. If not, enter the followig information in the **Transducer details**:
   - Type – select ‘Load’
   - Serial number – enter the serial number of the test frame
   - Operator – enter the name or initials of the operator
   - Decimals – select the number of decimal places according to the capacity of the frame:

   1 for 2000kN up to 5000kN

   2 for 200kN up to 1500kN

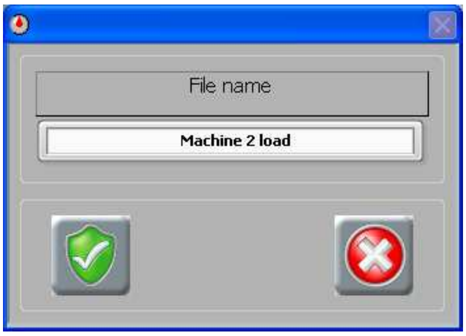
   3 for 15kN up to 150kN

   - Engineering units – select the Engineering Units



6.  Click the **Save button** and enter a name for the file in the window that appears.

   *The software will automatically insert the channel number (‘CH**_’) at the beginning of the filename so it is easy to identify which channel the calibration belongs to.*



7.  Send the calibration data to the machine by clicking the **Send to device button**.
8.  Click the **Edit calibration button** to lock the panel.
9.  Click the **Close button** to close the panel.
10. Repeat the procedure from step 2 for other channels, if required.
11.  Click the **Close** button to close the channel configuration list.

7.3.2 Checking the setting of the ADC and PID parameters

You will need to know the level 2 password ("controls2").

1.  Click the **Channel configuration button** on the **MULTITEST Main panel**.

2. Select an active (used) channel from the channel configuration list (Fig. 7-4). Double-click anywhere on the row to open the **Calibration panel**.

3.  Click the **PID settings button**.

4. Enter the level 2 password ("controls2**"**) to access the PID panel.

5. Select the type of machine that is connected to the selected channel from the drop-down list in the **PID type** cell (see Fig. 7-5).

   Select from 'Compression', 'Flexural' or 'Indirect tensile'.

6. Make sure that all the parameters are set according to the values specified in the instruction manual for the machine (see also the tables below).

Compression tests:

Frame type	K Proportional	K Integrative	K Derivative	Set up time (sec)	Set up delivery (%)	P. Under shoot (%)	Rapid Approach (%)	Preload (kN)	Zeroing time (s)
300 kN (66,000 lbf)	600	200	1	0	10	50	20	1	1
600 kN (135,000 lbf)	600	200	1	0	10	50	20	1	1
1000 kN (225,000 lbf)	500	200	1	0	10	25	20	5	2
2000 kN (450,000 lbf)/110 V	400	150	1	0	10	25	25	10	1
2000 kN (450,000 lbf)	500	200	5	0	10	25	25	10	2
3000 kN (660,000 lbf)	300	100	1	0	10	25	32	10	1
3000 kN (660,000 lbf)/110 V	400	200	1	0	10	25	32	10	1
4000 kN (900,000 lbf)	300	100	1	0	10	25	40	10	1
5000 kN (1,100,000 lbf)	300	100	5	0	10	25	40	10	1

Parameters common to all models/frames:

Parameters	Value
Timeout approach [s]	60
Timeout unload [s]	10

Flexural tests:

Frame type	K Proportional	K Integrative	K Derivative	P. Under shoot (%)	Rapid Approach (%)	Preload (kN)	Zeroing time (s)	Timeout approach [s]	Timeout unload [s]
15 kN (3,500 lbf)	6000	1000	1	50	15	0.1	1	30	5
100 kN (22,500 lbf)-C1201/FR	800	400	5	50	20	0.5	2	40	5
150 kN (30,000 lbf)	500	300	5	50	20	0.5	2	40	5
C0910/FR	400	200	5	50	20	0.5	2	40	10
C15XX/FR	1000	400	1	50	20	0.5	1	60	15
C1601/FR	500	1000	5	50	20	0.5	1	60	15
C17XX/FR	500	100	5	50	20	0.5	1	60	15

Parameters common to all models/frames:

Parameters	Value
Set up time (sec)	0
Set up delivery (%)	10

7.  Click the **Save** button to save the settings.
8.  Click the **Close** button to close the panel.
9.  Close the **Calibration panel** by clicking the **Close button**.
10. Repeat steps 1 to 9 for any other active channels.

NOTE:
The PIDs can be adjusted, if necessary, for each test type ('Compression', 'Flexure', 'Indirect Tensile'), in order to get the optimum performance from the system. It is recommended that the same values are used initially for all test types and then the values are fine-tuned where necessary.

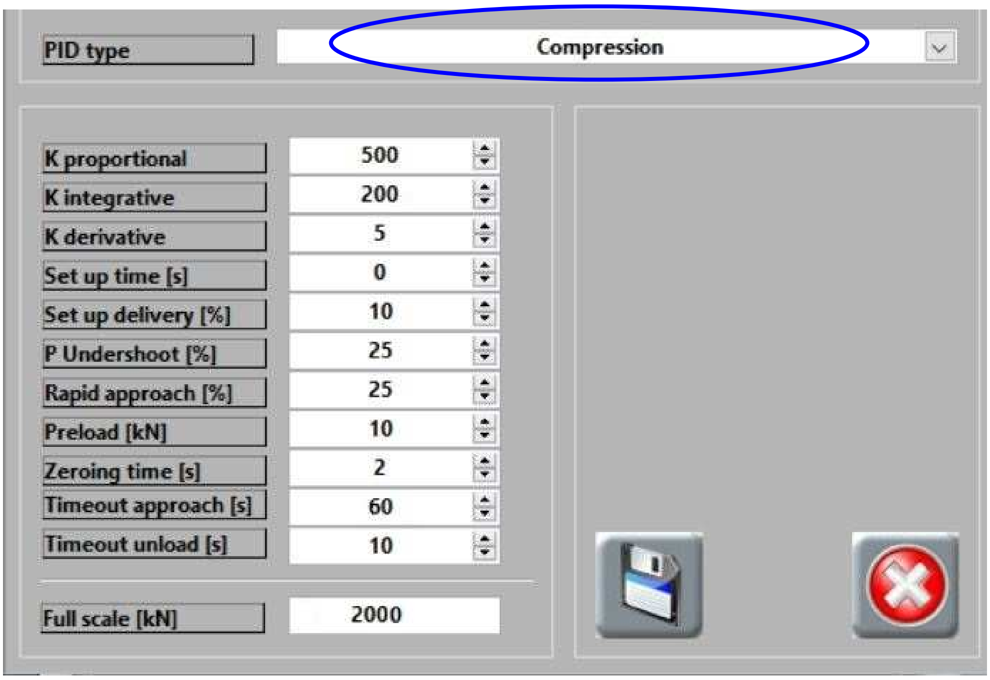


Fig. 7-5

11.  Close the channel configuration list by clicking the **Close** button.

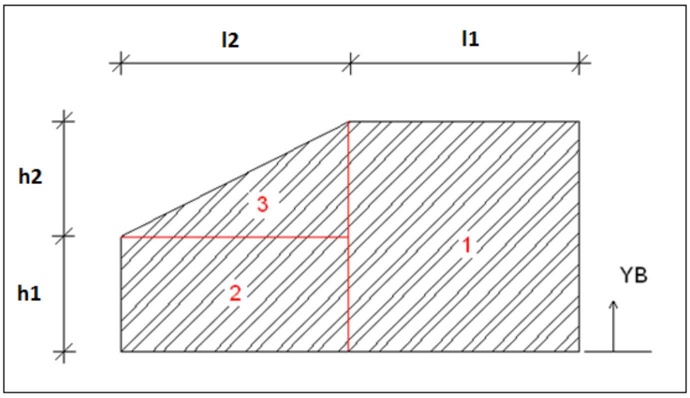
8. APPENDIX

8.1 Calculations used in the software

Where the test type is ‘Flexural test on curbs’, the values used for the area calculation are as follows:

YB = $[h1*h_{tot}*h_{tot} / 2+h1*l2*h1 / 2+h2*l2 / 2*(h1+h2 / 3)] / [l1*h_{tot}+h1*l2+l2*h2 / 2]$
where:
$h_{tot} = h1 + h2$

ITOT = I1+I2+I3
where:
$I1 = (l1 * h_{tot}^3) / 12 + h_{tot} * l1 * (h_{tot} / 2-YB)^2$
$I2 = (l2 * h1^3) / 12 + h1 * l2 * (h1 / 2-YB)^2$
$I3 = (l2 * h2^3) / 36 + l2 * h2 / 2 * (h2 / 3 + h1-YB)^2$



8.2 Example test reports

An Excel batch file summary:

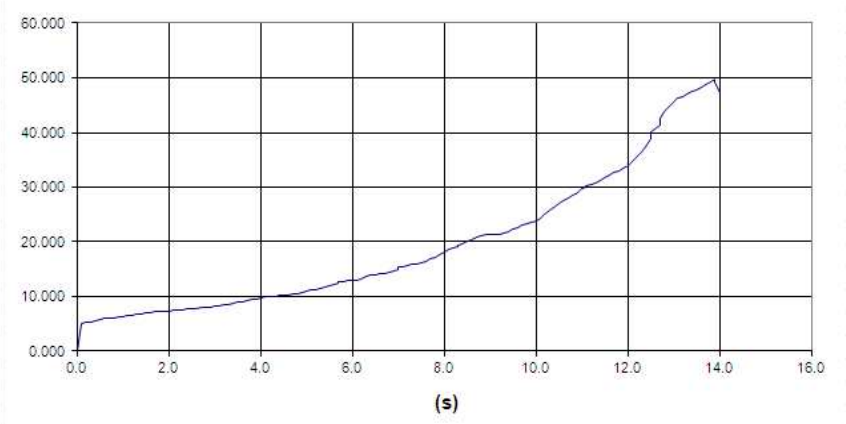
	A	B	C	D	E	F	G	H	I	J	K
1	ACME1										
2											
3	Saving time	Testing machine	Specimen type	Specimen age	Mass [kg]	Test date	Specimen ID	Breaking Load [kN]	Specific Load [MPa]	Time [sec]	
4	12/05/2012 15:48	Comp 1	concrete1	5 days 8 hours	3.5	19/11/2012	124A	50	6.34	14	
5	12/05/2012 15:49	Comp 1	concrete1	5 days 8 hours	3.4	19/11/2012	124B	100	12.7	14.4	
6											
7	Breaking Load [kN]			Specific Load [MPa]							
8	Mean	75		Mean	9.52						
9	Sigma	35.355		Sigma	4.497						

An Excel test report:

Acme labs

Compression test on concrete: EN 12390-3

Certificate number	12-3026	Sampling date	01/11/2012
Testing machine	Comp 1		
Client	A Smith		
Reference	50236		
Specimen type	concrete1	Cement quantity [kg/m³]	5.5
Cement type		Test date	19/11/2012
When received		At time of test	
Sampling location	Site A	Sampling date	01/11/2012
Preparation method			
Specimen ID	124A		
Dimensions	d(mm) 100.0 h(mm) 200.0	Mass [kg]	3.5
	500 0		
Load Rate	0.5		
Area [mm²]	7854.0 Specir 5 days 8 hours	Preparation date	14/11/2012
Load [kN]	50	Strength [MPa]	6.34
Notes			



Operator
JKW

It is forbidden to reproduce this certificate or any part of it

A printed test report:

Acme labs

Compression test on concrete: EN 12390-3

Certificate number	: 12-3026	*Certificate date*	: 19/11/2012
Testing machine	: Comp 1		
Client	: A Smith		
Reference	: 50236		
Specimen type	: concrete1	*Cement quantity [kg/*	: 5.5
Cement type	:	*Preparation date*	: 01/11/2012
When received	:	*At time of test*	:
Sampling location	: Site A	*Sampling date*	: 14/11/2012
Preparation method	:		
Specimen ID	: 124A		

d(mm) h(mm) *d(mm)* 100.0 *h(mm)* 200.0 *Mass [kg]* : 3.500

:

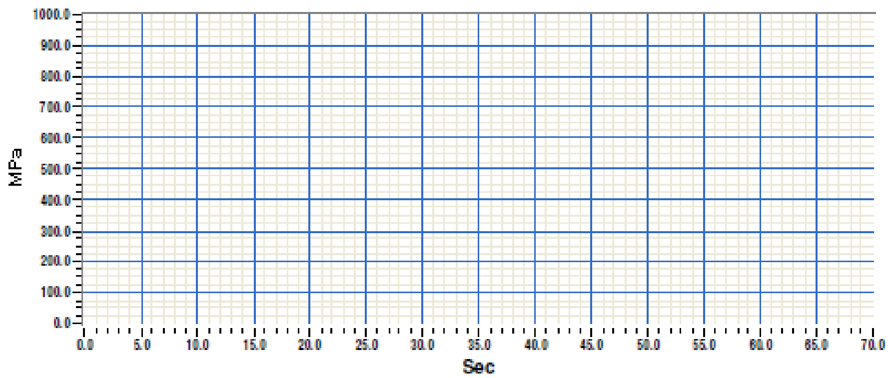
Age *MPa/s* : 0.500

Area [mm²] : 7854.0 *Specimen age* : 5 days 8 *Test date* : 19/11/2012

: 0.00 : 0.00

Failure type :

Notes :



Operator
JKW

8.3 Example calibration report

	A	B	C
1	Channel calibration1		
2			
3	Channel type	Load	
4	Full scale	100	
5	Number of decimals	0	
6	Engineering units	kN	
7	Serial number	123456	
8	User	JKW	
9	Note	Machine 1 vertical load	
10	File name	CH01_Machine 2 load.txt	
11			
12	Sample [bit]	Display [kN]	
13	0	3896	
14	20	16423	
15	40	28951	
16	60	41474	
17	80	54001	
18	100	66533	
19			

8.4 Example report parameters file

	A	B	C	D	E	F
1	[PARAMETERS]					
2	TEST NAME = "Flexural tests on beams:EN 12390-5"					
3	INSTITUTE = ""					
4	EQUIPMENT = ""					
5	SAMPLE TYPE = ""					
6	DIMENSIONS TYPE = 0					
7	DIMENSION 1 = 150.000000					
8	DIMENSION 2 = 150.000000					
9	DIMENSION 3 = 150.000000					
10	AGE = ""					
11	MASS = 0.000000					
12	AREA = 22500.000000					
13	LOAD SPEED = 100.000000					
14	LOAD SPEED UNIT = 0					
15	BREAKING TYPE CUBE = 0					
16	TEST DATE = "06/02/2040"					
17	OPERATOR = ""					
18	TEST ID = ""					
19	SAMPLE DATE = "06/02/2040"					
20	CEMENT TYPE = ""					
21	DOSAGE = ""					
22	START SAMPLE COND = ""					
23	TEST SAMPLE COND = ""					
24	TAKING DATE = "06/02/2040"					
25	TAKING PLACE = ""					
26	SAMPLE PREPARATION = ""					
27	CERTIFY NUMBER = ""					
28	CERTIFY DATE = "06/02/2040"					
29	CUSTOMER = ""					
30	REFERENCE = ""					
31	NOTE = ""					
32	K1 = 1.000000					
33	K2 = 1.000000					
34	L DISTANCE = 100.000000					
35	NO OF CUTTER = 0					
36	TEST TYPE = 0					
37	Y UNIT = 1					
38	DIMENSION 4 = 0.000000					
39	BREAKING TYPE CYL = 0					
40						

Notes:

Notes:

CONTROLS S.p.A.
is certified **ISO 9001:2008**

controls-group.com

wfi.co.uk

ipcglobal.com.au